

- Epidemic keratoconjunctivitis ("shipyard" conjunctivitis: patients having foreign particles removed acquired these viruses from equipment used on the eyes).
- Acute respiratory disease (ARD) and pneumonia:
 - Major problem in young military recruits
 - Serotypes 4, 7, and 21.
 - Three separate vaccines are used by the U.S. military
 - live non-attenuated
 - serotypes 4, 7, and 21
 - administered separately by enteric-coated capsules
- Adenovirus 40 and 41 cause infantile diarrhea.
- Adenovirus serotypes 12, 18, and 31 are common in *normal* human feces and are not known to cause human disease. These same strains cause malignant transformation of hamster cells. The adenoviruses are the standard examples of a **permissive host** (where virus is produced) and **nonpermissive host** (where the virus is not produced but transforms the cells).

Herpesviridae

Enveloped icosahedral nucleocapsids with dsDNA

DNA is synthesized in the nucleus.

Can enter the **latent state** in the host:

- HSV in neurons
- EBV in B lymphocytes

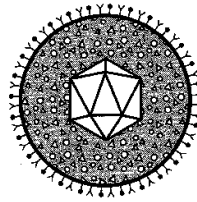


Figure I-5-17. Herpesvirus

Herpes viruses synthesize their own enzymes to synthesize DNA.

Acyclovir (the prodrug form of a nucleoside analog) is effective only in Herpes virus–infected cells because the herpes thymidine kinase is required to activate it, then it inhibits only the herpes polymerase, leaving the host polymerase functioning. No current drug can remove the latent DNA.

Herpes **envelope is from the host nuclear membrane** (virus is assembled in the nucleus).

Herpes form **intranuclear inclusion bodies**.

Herpes produce **distinctive cytopathology**.

Herpes simplex type 1 (HSV-1)

- Gingivostomatitis and recurrent cold sores (skin or lip);
latent in trigeminal root ganglion
- Keratoconjunctivitis generally with lid swelling and vesicles; dendritic ulcers may be seen; untreated repeated attacks may result in visual impairment.
- Meningoencephalitis
 - Characterized by: fever, headache, and confusion (focal temporal lesions; perivascular cuffing)
 - Diagnosis/treatment STAT: PCR on CSF; **RBCs in CSF**; acyclovir: early diagnosis reduces mortality

Herpes simplex type 2 (HSV-2)

- Meningitis: mild self-limiting disease
- Herpes genital infections: painful vesicular lesions of genitals and anal area; **latent in sacral nerve ganglia**
- Neonatal herpes may be one of three presentations:
 1. Disseminated with liver involvement; high mortality;
 2. Encephalitis; high mortality;
 3. Skin, eyes, or mouth.
- Diagnosis:
 - PCR (CSF for encephalitis)
 - Viral culture with fluorescent antibody stain to identify virus
- The **Tzanck smear** (Giemsa stain to show multinucleated giant cells) has **been largely replaced by immunofluorescent staining, which can distinguish HSV-1 from HSV-2.**

Varicella-zoster

- Chickenpox
 - Asynchronous rash
 - **Latent in dorsal root ganglia** → shingles in adults (severe nerve pain)
- Associated with Reye's syndrome
- Disseminated infections in immunocompromised hosts
- Attenuated vaccine
- Passive transfer of immunity with Varicella-Zoster immunoglobulin

Epstein-Barr virus (EBV)

- Selectively **infects B cells binding to CD21 = CR2**
- Many inapparent infections; common, worldwide 90% of the population seropositive
- Infectious mononucleosis, "Kissing disease"
 - **Heterophile positive mononucleosis**, fatigue, fever, sore throat, lymphadenopathy, and splenomegaly
 - Atypical reactive T lymphocytes (Downey Type II cells) may become as high as 70% WBC.
 - Positive for heterophile antibodies that cross-react with Paul-Bunnell antigen on sheep and bovine RBC (only mono that is heterophile antibody positive).
 - Antigens produced by productive cells:
 - EA = early antigen
 - VCA = viral capsid antigen used in diagnostic tests
 - EBNA = Epstein-Barr nuclear antigen
 - MA = membrane antigen
- Burkitt lymphoma, nasopharyngeal cancer, thymus carcinoma: EBNA found in all transformed B lymphoid cells.

Cytomegalovirus (CMV)

- **Herpesviridae** infecting **fibroblasts**; common 80% worldwide.
- **Owl's eyes** = CMV "Sightomegalovirus" basophilic intranuclear inclusion bodies with smaller eosinophilic cytoplasmic inclusion bodies.

Table I-5-9. Acquisition Routes for CMV

Transmission	Patient	Resulting Disease
In utero (most common in utero infection)	Fetuses	Ranges from infected but no obvious defects to severe cytomegalic inclusion disease (jaundice, hepatosplenomegaly, thrombocytic purpura, pneumonitis, and CNS damage [periventricular calcifications] to death)
Birth process, milk	Babies	Serious disease uncommon, heterophile negative mononucleosis
Sex, transfusions	Healthy adults	Serious disease uncommon, heterophile negative mononucleosis
Reactivation in transplanted organ	Transplant patients	Interstitial pneumonitis and systemic disease
Reactivation or new acquisition	AIDS patients	CMV retinitis, pneumonitis, and systemic disease

Human herpesvirus 6

Exanthem subitum (Roseola): common infant disease; fever followed by rash.

Human herpesvirus 8

Probable **cofactor** in **Kaposi sarcoma**

Poxviridae

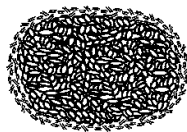


Figure I-5-18. Poxvirus

Complex morphology: Brick-shaped large viruses with ds linear DNA

Virus-coded lipid on surface

Virion-associated DNA-dependent RNA polymerase allows replication in cytoplasm.

Variola

- Smallpox
- Extinct in 1977
- Live attenuated *Vaccinia* vaccine
- Guarnieri bodies found in infected cells (intracytoplasmic)

Vaccinia

Immunogen in smallpox vaccine

Molluscum contagiosum

- Small pink benign wart-like tumors
- Molluscum bodies in central caseous material: oval, **eosinophilic cytoplasmic inclusion bodies**
- Problem in some immunocompromised hosts

Hepadnaviridae

Hepatitis B virus: see earlier notes.

RNA VIRUSES

Generalizations

All RNA viruses are single stranded (ss) except reovirus.

ss (-) RNA viruses carry an RNA-dependent RNA polymerase.

A virion-associated polymerase is also carried by

- Reovirus
- Arenavirus
- Retrovirus (reverse transcriptase)

Most are enveloped; only **naked** ones are

- Picornavirus
- Calicivirus
- Reovirus

Some are **segmented** (different genes on different pieces of RNA)

- Reovirus
- Orthomyxovirus
- Bunyavirus
- Arenavirus
(ROBA sounds like robot, pieces)

Table I-5-10. Positive-Sense RNA Viruses*

Virus Family	RNA Structure	Virion-Associated Polymerase	Envelope	Shape	Multiplies in	Major Viruses
Calicivirus	ss(+)RNA Linear Non-segmented	No polymerase	Naked	Icosahedral	Cytoplasm	Norwalk agent Hepatitis E
Picornavirus	ss(+)RNA Linear Non-segmented	No polymerase	Naked	Icosahedral	Cytoplasm	Polio** ECHO Enteroviruses Rhino Coxsackie Hepatitis A
Flavivirus	ss(+)RNA Linear Non-segmented	No polymerase	Enveloped	Icosahedral	Cytoplasm	Yellow fever Dengue St. Louis encephalitis Hepatitis C
Togavirus	ss(+)RNA Linear Non-segmented	No polymerase	Enveloped	Icosahedral	Cytoplasm	Rubella WEE, EEE Venezuelan encephalitis
Coronavirus	ss(+)RNA Linear Non-segmented	No polymerase	Enveloped	Helical	Cytoplasm	Corona-viruses
Retrovirus-	Diploid ss(+)RNA Linear Non-segmented	RNA dep. DNA polymerase	Enveloped	Icosahedral or truncated conical	Nucleus	HIV HTLV Sarcoma

*Mnemonic: (+) RNA Viruses: Call Pico and Flo To Come Rightaway

**Mnemonic: Picornaviruses PEE Co Rn A Viruses

Polio, Entero, Echo, Coxsackie, Rhino, Hep A

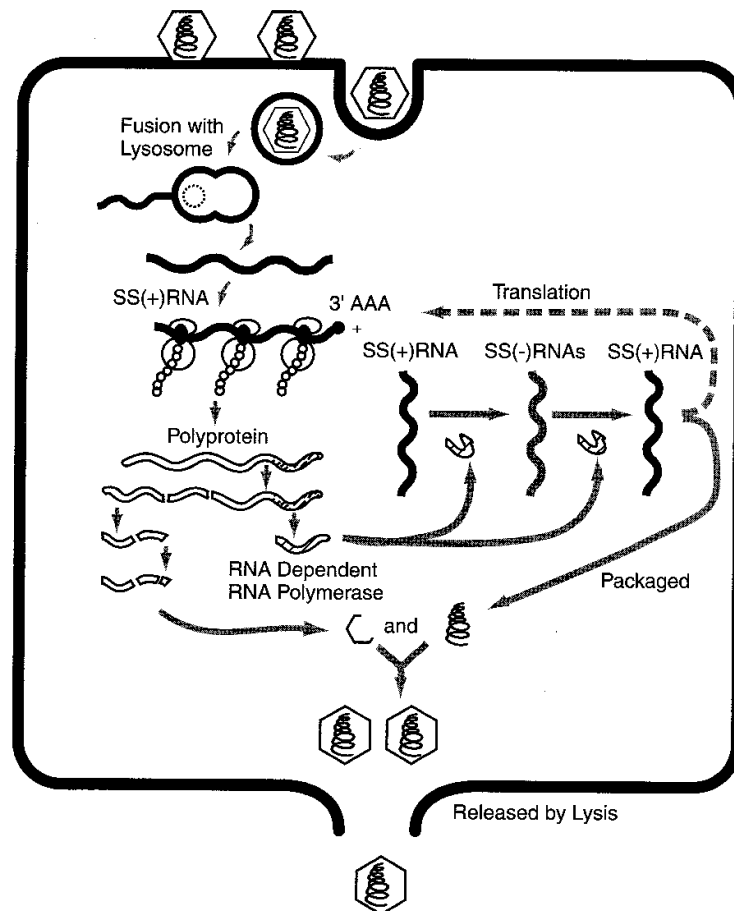


Figure I-5-19. Positive Sense RNA Virus Life Cycle

Abbreviations:

- H = Hemagglutinin—surface glycoproteins that bind to sialic acid (N-acetylneuraminic acid) receptors
- N = Neuraminidase—clips off sialic acids, thus aiding in release of virus
- M = Matrix protein—membrane stabilizing protein underlying the viral envelope
- F = Fusion protein—destabilizes host membrane
- P = Polymerase associated with virion

Caliciviridae

Small (slightly larger than picornaviridae) but similar to picornaviridae

Norwalk agent

- Described as calicivirus-like.
- Naked icosahedral
- Epidemic viral gastroenteritis in school-age kids and adults

Hepatitis E

See hepatitis notes.

Picornaviridae



- Small ss (+) RNA viruses
- Naked

Figure I-5-20. Picornavirus

Enteroviruses (group)

- Summer-fall peak incidence
- Fecal-oral transmission but do not cause diarrhea
- Peak age group <9 years for most
- Stable at pH3
- Resistant to alcohol, detergents because there is no envelope

Polio Virus

- Most infections are asymptomatic; small % cause fever (viremia).
- Smaller % cause aseptic meningitis.
- Poliomyelitis (paralysis) (even smaller %) results from viral damage to **anterior horn motor neurons**.
- **Vaccines:** both are trivalent
 - Sabin (live/oral/best gut immunity)
 - Salk (killed/injectable)

Coxsackie A

- Herpangina (vesicles on soft palate and fauces)
- Hand-foot-and-mouth disease (oral lesions primarily in the anterior buccal mucosa)
- Aseptic meningitis
- Acute lymphoglandular pharyngitis
- Common cold (aachoo's)

Coxsackie B

- Bornholm's disease (a.k.a., pleurodynia or Devil's grip; severe intercostal pain, fever)
- Aseptic meningitis
- Severe systemic illness of newborns
- Possible link to acute-onset, insulin-dependent diabetes in young children
- Myocarditis

Hepatitis A Virus

- ss (+) RNA
- Infectious hepatitis
- Inactivated vaccine
- Hyperimmune serum for post-exposure prophylaxis

Echoviruses and Most Enteroviruses: Aseptic meningitis

Non-Enteroviruses (group)

Rhinoviruses

- The common cold
- Not stable under acidic conditions
- Peaks summer and fall

Flaviviridae

ss (+) RNA icosahedral capsid with envelope

Yellow Fever Virus

- Mosquito-borne (Aedes)
- Liver, kidney, heart, and gastrointestinal mucosa damage
- Attenuated vaccine

St. Louis Encephalitis Virus

- Mosquito-borne (summer)
- Elderly (especially blacks or individuals with hypertension), most likely to have severe disease

Dengue Virus

- Dengue hemorrhagic shock syndrome in previously infected children who are reinfected. Immune enhancement of entry into macrophages.
- Dengue ("break-bone disease"): mild disease with rash and joint or muscle pain
- Mosquito-borne (Aedes)

Hepatitis C

Discussed with hepatitis viruses.



Figure I-5-21. Togavirus

Togaviridae

- ss (+) RNA viruses
- H, no P

Alpha viruses (group)

Equine Encephalitis Viruses: Western, Eastern, and Venezuelan

- All mosquito-borne
- Wild birds are reservoirs.
- Horses are also hosts.

Rubella

- Crosses placenta and is teratogenic.
- Most serious during first 16 weeks gestation
- Congenital rubella = patent ductus arteriosus, pulmonary stenosis, cataracts, microcephaly, deafness
- Attenuated vaccine; single strain—part of MMR

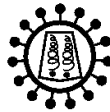
Coronaviridae

- ss (+) RNA, enveloped helical virus
- The large surface glycoprotein spikes give a crown appearance.
- Second most common cause of common cold (peak winter and early spring)

Retroviridae

Diploid ss (+) RNA viruses

Virion-associated reverse transcriptase



Oncovirus group

Human T-cell Leukemia/Lymphotropic (HTLV)

- Adult T-cell leukemia
- C-type particle (most oncoviruses, centrally located electron dense nucleocapsid)
- Japan, Caribbean, southern U.S.

Figure I-5-22. Retrovirus

Lentivirus group

Human Immunodeficiency Virus (HIV); acquired immunodeficiency syndrome (see next section)

HUMAN IMMUNODEFICIENCY VIRUS (HIV)

Structure and Genes of HIV

Positive sense (ss) RNA virus, diploid, non-segmented

Lentivirus in the retrovirus family (not oncogenic)

The HIV virion contains

- Enveloped truncated conical capsid (type D retrovirus)
- Two copies of the ss (+) RNA
- RNA-dependent DNA polymerase (reverse transcriptase)
- Integrase
- Protease

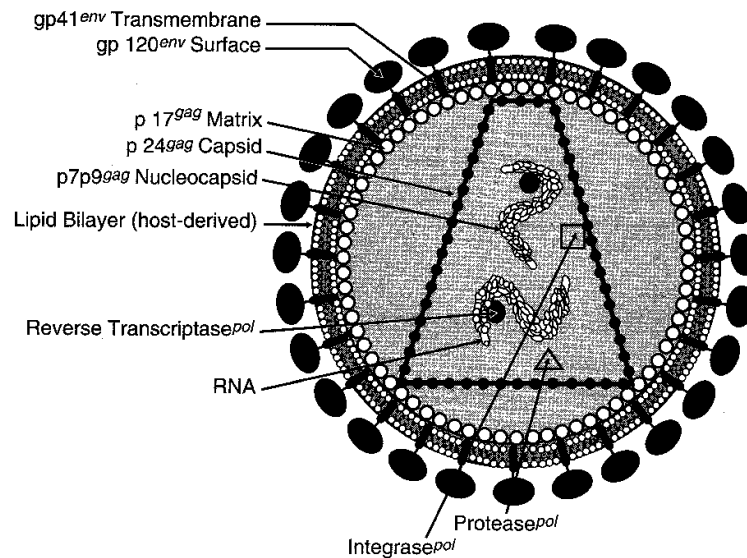


Figure I-5-23. Structure and Genes of HIV

Table I-5-11. Important HIV Genes and Their Functions

Gene	Product(s)	Function
Structural Genes		
Gag	Group specific antigens p24 p7p9 p17	Structural proteins capsid protein core nucleocapsid proteins matrix proteins (stabilizes envelope)
Pol	Reverse transcriptase Integrase Protease	Produces dsDNA provirus Proviral dsDNA integration into host DNA Cleaves polyprotein
Env	gp120 gp41	Surface protein that binds to CD4 on host cell responsible for tropism; genetic drift transmembrane protein for cell fusion
Regulatory Genes		
LTR (U3 U5)	DNA long terminal repeats	Integration and virus gene expression
Tat	Transactivator proteins	Transactivator of transcription; (upregulation); spliced gene
Rev	Regulator virion proteins	Upregulates transport of unspliced and spliced transcripts to the cell cytoplasm; a spliced gene
Nef	Negative factor*	Multiple functions, one of which is to decrease MHCI on infected T cells

*Initially thought to downregulate HIV production; now thought to enhance the pathogenicity of HIV.

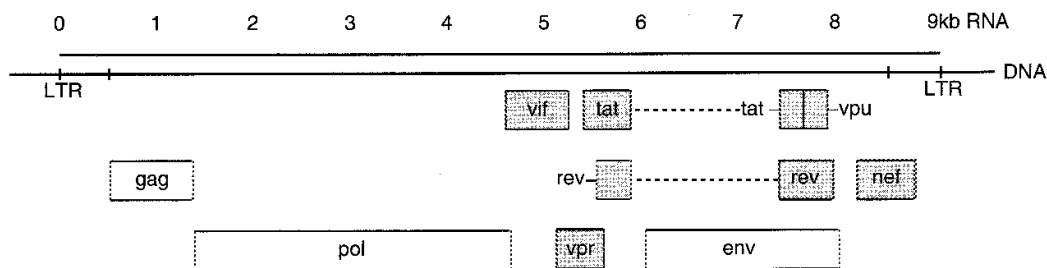
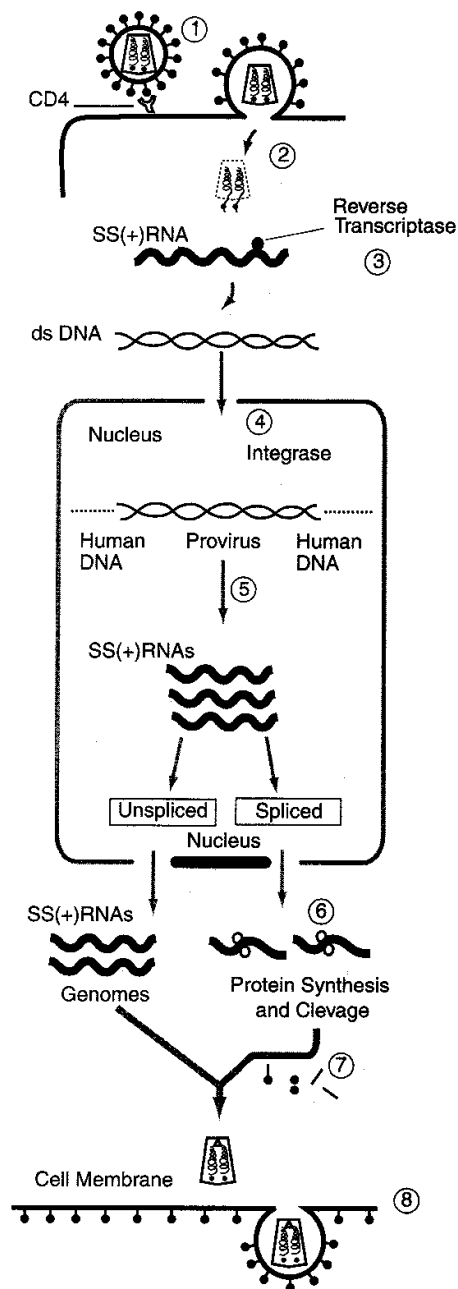


Figure I-5-24. Genetic Map of HIV



1. Surface gp120 of HIV binds to CD4 of T helper cells, macrophages, and microglia.
2. HIV is taken into the cell, losing the envelope; the RNA is uncoated.
3. The RNA is copied using the virion-associated reverse transcriptase; ultimately dsDNA with long terminal repeats is made.
4. The DNA and integrase migrate to nucleus and the DNA is integrated into host DNA forming the **provirus**.

The provirus remains in the host DNA.

The rate of viral replication is regulated by the activity of the regulatory proteins (tat/rev, nef, etc).

Tat upregulates transcription.

Rev regulates transport of RNAs to cytoplasm.

Co-infections (e.g., mycobacterial) stimulate the HIV-infected cells to produce more virus.

5. Transcription produces ss (+) RNAs, some spliced and some remain intact.
 - Spliced RNAs will be used as mRNA.
 - Whole RNA is used as genomic RNA.
6. Translation produces the proteins some of which are polyproteins that are cleaved by the HIV protease.
7. Assembly
8. Maturation/release of virus

Figure I-5-25. Retrovirus Life Cycle: HIV

HIV INFECTION

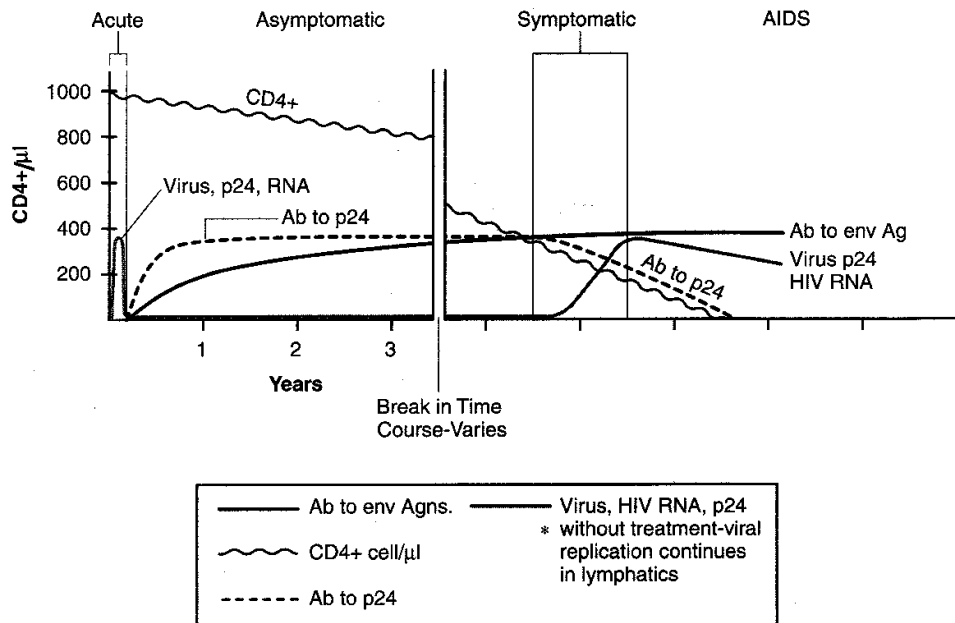


Figure I-5-26. Clinical Stages of HIV Infection

Table I-5-12. CDC Categories*

	Category A	Category B	Category C
CD4+ T cells/ μL	(excludes conditions in B & C) Acute (primary) or asymptomatic HIV infection Persistent generalized lymphadenopathy	Symptomatic but not conditions in C Condition attributed to HIV infection (list below) or are indicative of a defect in cell-mediated immunity	AIDS defining conditions (See following list)
>500	A1	B1	C1
200–499	A2	B2	C2
<200	A3	B3	C3

AIDS = A3, B3, or C1–3

Acute phase has high level of viral production and mononucleosis-like symptoms: fever, sore throat, rash, malaise, lymphadenopathy, diarrhea, etc.

*1993 Revised Classification System for HIV Infection and Expanded Surveillance Case Definition for AIDS among Adolescents and Adults. MMWR December 18, 1992/41(RR-17)

CDC Category B (Symptomatic)

- Bacillary angiomatosis (disseminated Bartonellosis)
- Candidiasis (oral or persistent vulvovaginal)
- Cervical dysplasia or carcinoma *in situ*
- Constitutional sx (fever 38.5°C or diarrhea lasting >1 month)
- Hairy leukoplakia
- Idiopathic thrombocytopenic purpura
- Listeriosis
- Pelvic inflammatory disease (especially with abscess)
- Peripheral neuropathy

AIDS Defining Conditions (C)

- Encephalopathy, HIV-related
- Pneumonia, recurrent (leading cause of death)

Fungal Infections

- Candidiasis of esophagus, bronchi, trachea, or lungs
- Coccidioidomycosis, disseminated, or extrapulmonary
- Cryptococcosis, extrapulmonary
- Histoplasmosis, disseminated, or extrapulmonary
- *Pneumocystis carinii* pneumonia

Carcinomas

- Invasive cervical
- Kaposi's sarcoma; Burkitt's, immunoblastic, or primary CNS lymphoma

Viral Infections

- Cytomegalovirus retinitis (with loss of vision) or disease (other than liver, spleen, or nodes)
- Herpes simplex: chronic ulcer(s) (>1 month); or bronchitis, pneumonitis, or esophagitis
- Progressive multifocal leukoencephalopathy
- Wasting syndrome due to HIV

Parasitic Infections

- Cryptosporidiosis, chronic intestinal (>1 month)
- Isosporiasis, chronic intestinal (>1 month)
- Toxoplasmosis of brain

Bacterial Infections

- *Mycobacterium tuberculosis*, any site (pulmonary or extrapulmonary)
- *Mycobacterium avium* complex or *M. kansasii* or other species or unidentified species, disseminated or extrapulmonary
- *Salmonella* septicemia, recurrent

Diagnosis of HIV Infection

Screening

ELISA (most commonly done) to detect HIV antibodies in patient's serum. (Most tests include the antigens p24, p17, gp160, gp120, and gp41.)

- Anti-p24 is the first reliably detected antibody but declines as viral titers rise in late infection.
- Envelope antibodies rise more slowly but stay high at end. (Env antigens show major antigenic variation.)
- ELISA for p24 antigen useful early.

Confirmation (Using a Second Blood Sample)

- Western blot for antibodies specific for HIV (electrophoretically separated HIV antigens react with the patient's antibody; detection by enzyme-labeled anti-human IgG) or
- Immunofluorescence

HIV DNA PCR

- Qualitative to detect HIV infection in newborns of mothers are HIV+
- Quantitative HIV DNA PCR to determine viral load to assess treatment

Culture for HIV (with Antigen Detection in Culture)

- HIV infection in newborns whose mothers are HIV positive
- To assess drug resistance

Table I-5-13. HIV's Mechanisms of Immunologic Evasion

Characteristic	Function
Multiplication in lymphocytes and macrophages	Eliminates cell-mediated and antibody-mediated immunity
Nef and tat gene products down-regulate class I MHC expression.	Makes infected cells less susceptible to cytotoxic T-cell killing
Destruction of CD4 T cells	Elimination of immune enhancement response
Antigenic drift of the gp120 Heavy glycosylation of gp120	Evade antibody-mediated effector mechanisms

Table I-5-14. Negative Sense RNA Virus

Virus	RNA Structure	Virion-Associated Polymerase	Envelope	Shape	Multiplies in	Major Viruses
Paramyxovirus	ss(-) RNA Linear Non-segmented	Yes	Yes	Helical	Cytoplasm	Mumps Measles Respiratory syncytial Parainfluenza
Rhabdovirus	ss(-) RNA Linear Non-segmented	Yes	Yes	Bullet-shaped helical	Cytoplasm	Rabies Vesicular stomatitis
Filovirus	ss(-) RNA Linear Non-segmented	Yes	Yes	Helical	Cytoplasm	Marburg Ebola
Orthomyxovirus	ss(-) RNA Linear 8 segmented	Yes	Yes	Helical	Cytoplasm & nucleus	Influenza
Bunyavirus	ss(-) RNA Linear → Circular 3 segments Ambisense	Yes	Yes	Helical	Cytoplasm	California encephalitis La Crosse encephalitis Hantavirus
Arenavirus	ss(-) RNA Circular 2 segments 1 (-) sense 1 ambisense	Yes	Yes	Helical	Cytoplasm	Lymphocytic chorio-meningitis Lassa fever

Mnemonic for the ss(-) RNA viruses: **Pairing Rats Fight Over Bunny's Area** or **Pain Results From Our Bunions Always**.
You can remember these are the negative ones since fighting is a negative thing to do.

Note that all are enveloped, all have virion-associated polymerase, and all have helical nucleocapsids.

The oddballs are the last three:

The orthomyxoviruses are linear (ortho) but with eight (ortho/octo) segments, which is one of the reasons they can genetically "mix" it up. The orthomyxoviruses are also odd in that they replicate in both the nucleus and cytoplasm. The bunyaviruses are somewhat contortionists (circular): California playboy bunnies in a ménage à trois?

The arenaviruses have one negative sense and one ambisense strand of RNA.

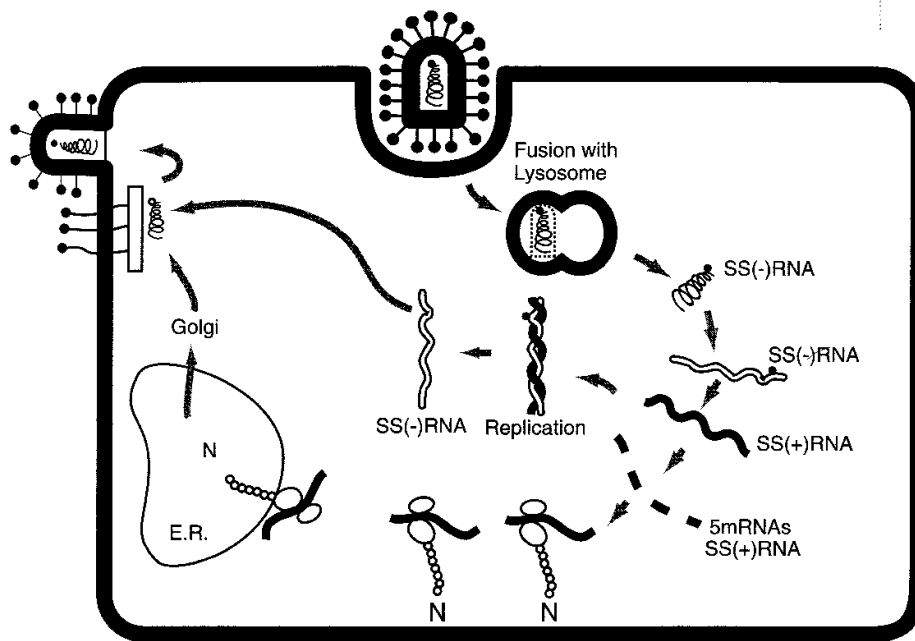


Figure I-5-27. Negative Sense RNA Virus Life Cycle

NEGATIVE SENSE RNA VIRUSES – DISEASES

Paramyxoviridae

- ss(-) RNA strand
- Enveloped helical nucleocapsids

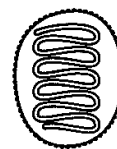


Figure I-5-28. Paramyxovirus

Parainfluenza virus

- Single HN glycoprotein, also fusion protein (F)
- Croup (laryngotracheobronchitis)
- Common cold, bronchitis

Mumps

- ss(-) RNA
- Single HN glycoprotein, also F protein
- Live vaccine
- Parotitis
- Pancreatitis
- Orchitis in adult males
- Meningoencephalitis

Measles (rubeola)

- ss(-) RNA
- H glycoprotein and fusion protein, no neuraminidase
- Measles:

Presentation generally the three C's with photophobia:

Cough, coryza, and conjunctivitis

Koplik spots → maculopapular rash from the ears down → Giant cell

Pneumonia (Warthin-Finkeldy cells)

Rare complication: subacute sclerosing panencephalitis (chronic CNS degeneration)

- Live vaccine (single strain)

Respiratory syncytial virus

- ss(-) RNA
- No H nor N glycoproteins; only F protein
- Major cause of bronchiolitis and pneumonia in infants
- Common cold



Figure I-5-29. Rhabdovirus

Rhabdoviridae

- ss(-) RNA
- Bullet shaped
- Enveloped

Rabies virus

- Negri bodies—intracytoplasmic inclusion bodies
- Inactivated vaccines; passive immunization
- Spread to humans by bites of rabid dogs; contact with bats
- Eastern U.S. reservoirs: foxes & raccoons; western U.S.: skunks

Vesicular stomatitis virus

Foot and mouth disease

Filoviridae

- ss(–) RNA
- Helical capsid with envelope
- Virion-associated RNA-dependent RNA polymerase

Marburg virus

Acute hemorrhagic fever, frequently fatal

Ebola virus

Acute hemorrhagic fever, frequently fatal

Orthomyxoviridae

Influenza

- ss(–) RNA
- Segmented (8)
- Enveloped nucleocapsids
- Separate H and N glycoproteins
- Can lead to Guillain-Barré (A or B) or Reye's syndrome (primarily B)
- Inactivated vaccine, H1N1 and H3N2
- Influenza A & B (and many other viruses, most notably HIV) undergo **genetic drift** = slight changes in antigenicity due to mutations (in influenza responsible for epidemics)
- Influenza A has rare **genetic shift** (genetic reassortment) - major changes from new combinations of RNA segments or recombination between the segments in co-infections causing new pandemics



Figure I-5-30. Orthomyxovirus

Bunyaviridae

- Segmented single (–) RNA
- Enveloped helical capsids (circularize) ambisense
- California group

California encephalitis and LaCrosse encephalitis viruses

- Mosquito-borne
- Young (<15 years) have more severe cases

Hanta virus (Sin Nombre)

- Hanta virus pulmonary syndrome (cough, myalgia, dyspnea, tachycardia, pulmonary edema and effusion, and hypotension [mortality 50%])
- High endemic area: Four Corners region (UT, AZ, NM, CO), but all of North America
- Associated with rodent feces

Arenaviridae

- Segmented RNA (2 pieces)
- ss(–) RNA, ss (ambisense) RNA
- Enveloped
- Helical

Lymphocytic choriomeningitis virus

- Influenza-like with meningeal signs
- Imported from South America

Lassa fever virus

- West Africa
- Hemorrhagic fever
- 50% fatal

DOUBLE-STRANDED RNA VIRUS–DISEASE

Table I-5-15. Double-Stranded RNA Viruses

	RNA Structure	Virion-Associated Polymerase	Envelope	Shape	Major Viruses
Reovirus	Linear, dsRNA 10–11 segments	Yes	Naked	Icosahedral double shelled	Reovirus Rotavirus

Reoviridae

Reoviruses

Upper respiratory tract infections



Figure I-5-31. Reovirus

Rotaviruses

- GI tract infection especially <2 years old, a prolonged diarrhea
- Major cause of infant mortality worldwide

ONCOGENIC VIRUSES

Definitions

Malignant Transformation of Cells

- Dedifferentiation
- Loss of growth control
- Immortalization
- Appearance of new surface antigens ("T" antigens)

Provirus

Viral DNA inserted into host DNA

Oncogenes

Genes with the potential to cause malignant transformation

Cellular Oncogenes (abbreviated c-onc)

These are normal cellular genes whose products control regulation of cell growth and division (e.g., kinases, growth factors and their receptors, G proteins and nuclear regulatory proteins).

Viral Oncogenes (abbreviated v-onc)

Genes carried by certain viruses causing cancer. Viral oncogenes are homologs of cellular oncogenes.

Tumor Suppressor Genes

These genes suppress, or constrain, cell growth and replication.

Major Concepts of Tumorigenesis

Mutation of a c-Oncogene or Tumor Suppressor Gene

- Mutation in one of these control genes may result in unregulated growth of cells.
- Example of mutated oncogene—ras
- Retinoblastoma (Rb) is an example of mutation in tumor suppressor gene.

Dosage Effects

- Oncogenes in amplified DNA—increased number of copies results in overexpression of gene.
- **Translocation**, which links an oncogene with a more active enhancer and/or promoter resulting in overexpression (Burkitt's lymphoma).
- Provirus **insertional mutagenesis**—for example, a retrovirus with its very active transcriptional promoter/enhancer region, the LTR (long terminal repeat) may integrate (insert) near a cellular oncogene. This is one of the mechanisms by which retroviruses that do not have v-onc cause carcinoma.

- Infection with a virus carrying a v-onc: e.g., infection with a retrovirus carrying viral oncogenes such as src. The gene was probably picked up by a provirus inserted near a cellular oncogene picking up copies of c-onc. Viral progeny then contain the new oncogene now called v-onc. When a new cell is infected with the recombinant virus, the oncogene is now under the transcriptional control of the viral enhancer/promoter.
- Interaction between the products of oncogenes and tumor suppressor genes. Proteins E6 and E7 of the human papilloma virus combine with and inactivate the p53 and p110 (Rb), respectively.

Specific Viruses Associated with Human Cancers

EBV

- Burkitt's lymphoma, nasopharyngeal, and thymic carcinoma
- BL occurs only in malarial regions; the plasmodia are thought to produce a slight immunosuppression.
- EBV stimulates B-cell replication and eventually, if a translocation of *c-myc* to the DNA region where genetic rearrangements involved in antibody synthesis occurs, BL develops.

Chronic HBV

Primary hepatocellular carcinoma

Chronic HCV

Primary hepatocellular carcinoma

HPV

- Cervical carcinoma
- Mechanism: inactivation of tumor suppressor gene

HTLV-1

- CD4+ T-cell leukemia/lymphomas
- Provirus insertion or capture

PRION DISEASES

Table I-5-16. Prion Diseases

Disease	Infectious agent	Host	Comments
Kuru	Prion	Human	Subacute Spongiform Encephalopathy (SSE); Fore Tribe - New Guinea; cannibalism
Creutzfeldt-Jakob Disease (and variant)	Prion	Human	SSE Genetic predisposition. Ingestion of infected cow brains
Gerstmann-Straussler	Prion	Human	SSE
Fatal Familial Insomnia	Prion	Human	SSE
Scrapie	Prion	Sheep	SSE—scraping their wool off on fences

Table I-5-17. Slow Conventional Viruses (Viruses)

Disease	Infectious agent	Host	Comments
Measles SSPE	Virus	Human having had measles	Subacute sclerosing panencephalitis
AIDS dementia	HIV	Human	Dementia
PML	JC Virus	Human	Progressive multifocal leukoencephalopathy

VIRAL GENETICS

Phenotypic Mixing

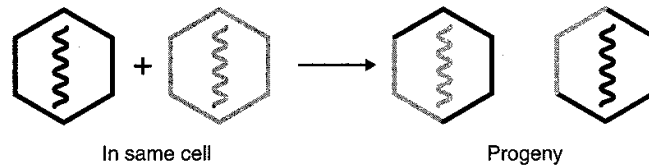


Figure I-5-32. Phenotypic Mixing

- Related viruses coinfect cell (virus A and virus B).
- Resulting proteins on the surface are a mixture capsid of AB around nucleic acid of either A or B.

Phenotypic Masking

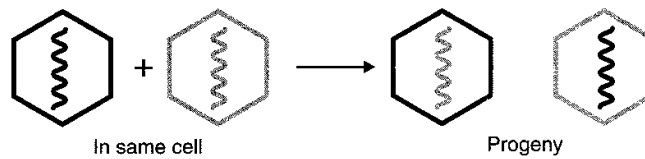


Figure I-5-33. Phenotypic Masking

- Related viruses coinfect cell (virus A and virus B).
- Capsid of proteins of virus A form around nucleic acid of B.

Complementation

- Two related defective viruses infect the same cell. If they are defective in different genes, viral progeny (still with mutated DNA) will be formed.

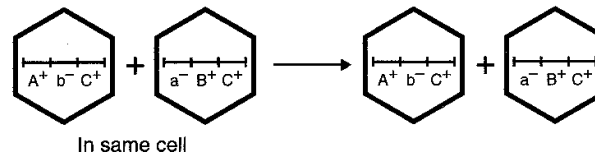


Figure I-5-34. Complementation: Mutations in Different Genes

If they are defective in the same gene, no progeny will be formed.

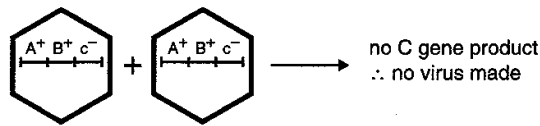


Figure I-5-35. Complementation: Mutations in Same Genes

- Coinfection of hepatitis B and D is a clinical example of complementation where HBV supplies the needed surface antigen for hepatitis D.

Genetic Reassortment = Genetic Shift

- Two different strains of a segmented RNA virus infect the same cell.
- Major new genetic combinations are produced through "shuffling," resulting in stable and dramatic changes.

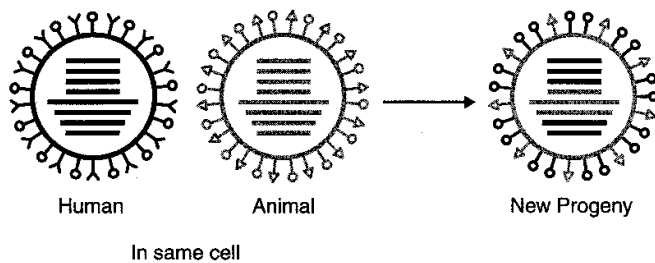


Figure I-5-36. Genetic Reassortment in Influenza A

Genetic Drift

- Minor antigenic changes from mutation
- Occurs in many viruses, particularly RNA ones
- Most noted in HIV and influenza

Viral Vectors

- Recombinant viruses are produced that have combinations of human replacement genes with the defective viral nucleic acid.

Chapter Summary

The virus structure consists of a core of RNA or DNA associated with structural proteins and perhaps key enzymes. This is called the nucleocapsid and constitutes a naked capsid virus. The nucleocapsid plus host membranes, including viral-specified glycoproteins, make up an enveloped virus.

The various viruses have characteristic sizes and shapes, as illustrated in Figure I-5-4.

The first step in viral replication is attachment. This is accomplished by the recognition of specific receptor sites on the host cell. This is followed by replication, release of the nucleic acid core (uncoating), and then macromolecular synthesis. Finally, new virus particles are assembled and released.

The eclipse period is the time it takes to replicate the first viral particles intracellularly. The latent period is the time between infection and the first release of virus from the infected cell.

Viruses are spread like any other pathogen. Arboviruses are those borne by arthropods, most commonly mosquitoes but also ticks, biting midges, and sandflies. Most arboviruses belong to the Togavirus, Flavivirus, or Bunyavirus families.

Virus attachment is mediated by an interaction between specific receptor sites on the host cell and either a specific viral surface glycoprotein on enveloped viruses or a specific surface protein on naked viruses.

Heat, detergents, or organic solvents inactivate enveloped (but not naked) viruses by destroying the lipid envelope.

The distribution of the required host receptors determines the species and tissue specificity for infection by a specific virus. Table I-5-1 summarizes the relationship among four important viruses, their targeted cells, and the cell receptors used for attachment by each of these viruses.

Once attached, the virus may invade the host cell by receptor-mediated endocytosis, by uptake via coated pits, or by the fusion of a viral envelope with the cell membrane.

The initial chore facing the invading virus is to synthesize protein that will replace normal host-cell function with those essential for viral survival and replication. To do so, the virus must use mRNA that will be recognized and used by the host cell's ribosomes.

Double-stranded DNA viruses directly transcribe mRNA using the negative strand as a template.

Retroviruses use their positively stranded RNA genome as templates to reverse transcribe double-stranded DNA using an RNA-dependent DNA polymerase (reverse transcriptase). The negative DNA strand is then used to transcribe mRNA.

Positively stranded RNA viruses can use their genome directly as mRNA.

Negatively stranded RNA viruses make the complimentary positive strand for use as mRNA.

Once reproduced, the naked viruses lyse the host cell. Enveloped viruses acquire host-cell membrane as an outer surface as they leave the cell.

The various patterns of viral infection are shown in Figure I-5-10 and Table I-5-5. They may be abortive, resulting in no infection; cytolytic, generally resulting in an acute infection; or persistent, resulting in a chronic infection, latent with possible later emergence of disease, or transforming, resulting in cancer.

(Continued)

Chapter Summary (continued)

Figure I-5-11 summarizes the ages at which various viral diseases may be expected to occur.

Immune defense against viruses utilizes the innate and adaptive pathways in a fashion similar to that used for protection from bacterial infection. However, a component of the innate system, interferon, is of unique importance as a protective mechanism against viral infection. Interferon is produced and released by virus-infected cells and is transmitted to noninfected cells where it inhibits nascent viral protein synthesis (Figure I-5-12).

Viral vaccines are developed using killed viruses, attenuated viruses (treated to lose virulence), and viral components. The properties and uses of each type of vaccine are summarized.

The responsible viral family, molecular features, transmission mode, disease characteristics, mortality rates, and diagnostic techniques associated with the hepatitis A, B, C, D, and E viruses are summarized in Table I-5-7. The characteristics of infection by the hepatitis B virus are discussed in greater detail on the following page and illustrated in Figure I-5-13.

The medically important DNA viruses include the Parvoviruses, Papovaviruses, Adenoviruses, Herpes viruses, Poxviruses, and the Hepadnaviruses. The properties and associated disease states of important members of each of these families are summarized in Tables I-5-8 and -9, Figures I-5-14, -15, -16, -17, and -18, and the accompanying text.

The medically important RNA viruses include the Calciviruses, Picornaviruses, Flaviviruses, Togaviruses, Coronaviruses, and Retroviruses. The properties and associated disease states of important members of each of these families are summarized in Tables I-5-10, -11, -12, -13, -14, and -15, Figures I-5-19, -20, -21, -22, -23, -24, -25, -26, -27, -28, -29, -30, and -31, and the accompanying text.

Table I-5-16 summarizes the characteristics of Kuru, Creutzfeldt-Jacob disease, Gerstmann-Straussler disease, fatal familial insomnia, and scrapie—all prion-associated diseases.

Table I-5-17 summarizes the slow effects of three conventional virus diseases (not prion). These are subacute sclerosing panencephalitis (SSPE) caused by latent measles virus, HIV-induced dementia, and progressive multifocal leukoencephalopathy (PML) provoked by the JC virus.

Natural genetic modification of viruses can occur by genetic mixing, phenotypic masking, complementation, genetic re-assortment (genetic shift), and genetic drift. In addition, the genetic characteristics of viruses can be altered by recombinant technology.

Review Questions

1. Three genetically distinct and fairly stable serotypes have made this virus's vaccines so successful that worldwide eradication of this virus is anticipated in the next 5 to 10 years.
 - A. Adenovirus
 - B. Herpes virus
 - C. Influenza virus
 - D. Measles
 - E. Poliovirus

2. Roommates of a 19-year-old college student become alarmed when he does not get up to go to swim practice in the morning and they are unable to wake him for his 11 AM class (he had complained of a headache and not feeling well the night before). The rescue squad finds a febrile, comatose young man with a petechial rash. In the emergency room, Kernig and Brudzinski's signs are present. No papilledema is seen so a spinal tap is done. Protein is high, glucose low. CSF WBC count is 9,000 (mainly PMNs) with few RBCs. The characteristics of the most likely causative agent are
 - A. An enveloped dsDNA virus
 - B. A naked (+) ssRNA virus
 - C. A Gram-negative bacillus with a polyribitol capsule
 - D. A Gram-negative, oxidase-positive diplococcus
 - E. A Gram-positive, lancet-shaped alpha-hemolytic diplococcus

3. Serologic test results from a hepatitis patient reveal: anti-HBc positive, HBsAg positive, and anti-HBs negative. The correct interpretation of the patient's status is
 - A. No longer contagious
 - B. Immune to hepatitis B virus
 - C. Evidence of receiving hepatitis B vaccination
 - D. Hepatitis B virus chronic carrier state
 - E. Impossible to have both surface antigen and core antibody positive

4. Two individuals become infected with HIV at approximately the same time. JT progresses rapidly to full-blown AIDS and dies within 3 years. AY is still asymptomatic after 10 years. One possible explanation is that the virus in JT is probably expressing a high amount of activity of which regulatory gene product?
 - A. *vif*
 - B. *vpr*
 - C. *tat*
 - D. *env*
 - E. *pol*

5. The best prospects for treatment and cure of microbial diseases are always those unique factors of a pathogen's life cycle that can be altered without affecting the survival of the host's own cells. In HIV, one such therapeutic target would be the products of the *pol* gene, which codes for the reverse transcriptase unique to the retroviral life cycle. If it were possible to ablate expression of the HIV *pol* gene, what other aspect of the virus's life cycle would be directly altered?
 - A. Transcription of proviral DNA
 - B. Production of viral mRNA
 - C. Integration of proviral DNA
 - D. Nucleocapsid
 - E. Viral maturation
6. Interferons inhibit viral growth primarily by affecting
 - A. Host cytokine production
 - B. Host protein synthesis
 - C. Viral protein synthesis
 - D. Viral transcription process
 - E. Viral assembly and release
7. Transfection with the naked nucleic acid of which virus would result in active viral replication?
 - A. Bunyavirus
 - B. Coxsackie
 - C. Poxvirus
 - D. Retrovirus
 - E. Rhabdovirus
8. A chronic infection of hepatitis B is defined as having demonstrated the presence of
 - A. HBsAg for more than 6 months
 - B. HBsAg for more than 3 months
 - C. Antibody against HBsAg for more than 6 months
 - D. Antibody against HBcAg for less than 3 months
 - E. HBcAg for 3 months
9. Live, attenuated vaccines exist for human diseases caused by
 - A. Hepatitis A and B viruses
 - B. Hepatitis A and polioviruses
 - C. Rubella and rubeola viruses
 - D. Influenza viruses
 - E. Rabies virus and rotavirus

10. A boy with bilateral swelling of his salivary glands is found to have an elevated serum amylase. The most likely viral causative agent
 - A. Has equal amounts of adenine and thymine in its genome
 - B. May cause complications at the time of acute disease or during convalescence
 - C. Has multiple serotypes
 - D. Is characterized by being helical, circular, single-stranded RNA and naked
11. A 5-year-old female presents with a fever and a generalized macular rash that is most dense on the scalp and trunk of the body. Several waves of lesions appear, one after another, and evolve rapidly into vesicles and then pustules over several days. The most likely disease and causative agent is
 - A. Exanthem subitum due to cytomegalovirus
 - B. Chickenpox due to the varicella-zoster virus
 - C. Whitlow's infection due to herpes simplex virus type 1
 - D. Herpetic gingivostomatitis due to the varicella-zoster virus
 - E. Infectious mononucleosis due to the Epstein-Barr virus
12. Infection of appropriate cells with a composite virus made up of Coxsackie virus capsid components and poliovirus RNA would yield progeny which would
 - A. Have the host cell range of Coxsackie virus
 - B. Also be composite viruses
 - C. Show phenotypic mixing
 - D. Have a recombinant genome consisting of both Coxsackie and poliovirus
 - E. Cross-react with Sabin-vaccine-induced antibodies
13. Positive-stranded viral RNA
 - A. Replicates most commonly through a double-stranded DNA intermediate
 - B. Binds to ribosomes and can be translated
 - C. Requires a special polymerase in the virion
 - D. Is not infectious
 - E. Replicates without an intermediate
14. To design a vaccine against HIV infection, a logical goal would be to alter some native molecule or product of the virion in order to make it highly immunogenic. If you wished to prevent the attachment of the virus to helper T lymphocytes, which molecule or family of molecules might best be targeted?
 - A. gp41
 - B. gp120
 - C. nucleocapsid protein
 - D. p17
 - E. p24

15. How do the oncogenes of the human papilloma viruses act?
 - A. They act as tumor suppressor genes
 - B. They function identically to cellular oncogenes
 - C. They kill the infected cell when upregulated
 - D. We don't know how they act because they are present in every cell of every individual and yet only some become cancerous
 - E. They code for early viral proteins, leading to malignant transformation
16. To which of the following viruses is hepatitis A most closely related genetically?
 - A. Hepatitis B
 - B. Poliovirus
 - C. Measles
 - D. Rubella virus
 - E. Influenza
17. A naked protein similar in amino acid sequence to a normal human protein, but which has no nucleic acid associated with it, can cause
 - A. AIDS dementia
 - B. Mumps meningitis
 - C. Progressive multifocal leukoencephalopathy
 - D. Subacute sclerosing panencephalitis
 - E. Subacute spongiform encephalopathy
18. A 37-year-old ambitious executive for a local Health Maintenance Organization comes to your office because he has developed multiple blister-like lesions on his penis over the last 1–2 days. They are somewhat painful, and he is worried that he has AIDS. He denies homosexuality and intravenous drug abuse and had an HIV test prior to his marriage 3 years ago. He reports several similar episodes several years ago when he worked as a photographer in Nepal. He was never told what they were, and they resolved over several days without any treatment. His physical examination is remarkable only for the presence of 6–8 vesicular lesions 3–4 mm in diameter on the glans of the penis. There is no crusting, drainage or bleeding. The lesions are moderately tender and there is mild inguinal adenopathy bilaterally. How does the causative agent produce its messenger RNA?
 - A. By producing a positive sense intermediate
 - B. By direct translation from the genome
 - C. By transcription from proviral DNA
 - D. By producing a negative sense intermediate
 - E. By transcribing the genomic DNA
 - F. By producing a double-stranded DNA intermediate
 - G. The genomic RNA is used directly on the ribosomes

19. The Tzanck test, which aids in the diagnosis of herpes simplex infection, is a search for
 - A. Antibodies to herpes simplex 1 or 2
 - B. Intracytoplasmic inclusion bodies
 - C. Virus shedding from pustular lesions
 - D. Multinucleated giant cells
 - E. Immunofluorescence of infected cells
 - F. Detection of viral RNA
 - G. Koilocytic cells
20. In the U.S., a baby has the greatest chance of acquiring which virus *in utero*?
 - A. Cytomegalovirus
 - B. Hepatitis B virus
 - C. Herpes simplex virus
 - D. Respiratory syncytial virus
 - E. Rubella virus
21. Which of these viruses has RNA for both its genome and replicative intermediate?
 - A. Cytomegalovirus
 - B. Hepadnavirus
 - C. Retroviruses
 - D. Togaviruses
 - E. Poxvirus
22. What is the most common lab testing method for diagnosing infectious mononucleosis?
 - A. The Monospot test to detect EBV-specific antibody
 - B. An assay for Epstein-Barr nuclear antigen
 - C. The presence of atypical lymphocytes in the blood establishes the etiology
 - D. A test for heterophile antibody, which cross-reacts with antigens found on a variety of animal red blood cells
 - E. A simple procedure is done to isolate EBV from saliva, blood, or lymphoid tissue
23. What virus is noted for genetic reassortment, which leads to major pandemics about once every 10 to 11 years?
 - A. Adenovirus
 - B. Herpes virus
 - C. Human immunodeficiency virus (HIV)
 - D. Influenza virus
 - E. Poliovirus

24. What virus is noted for such a high incidence of genetic drift that more than one antigenic variant can be isolated from most infected individuals who have high viral titers?
 - A. Adenovirus
 - B. Herpes virus
 - C. Human immunodeficiency virus (HIV)
 - D. Influenza virus
 - E. Poliovirus
25. A 19-year-old male college student reports sore throat and extreme fatigue following even normal non-taxing tasks like getting dressed and going down to breakfast. He tells you that he has been sick for several weeks, that he has been feverish and that his girlfriend now appears to be getting the same thing. His tonsils are inflamed with a white exudate adhering; cervical lymphadenopathy is prominent, as is splenomegaly. The most likely causative agent is
 - A. ss DNA, naked icosahedral virus
 - B. ds DNA, naked icosahedral virus
 - C. ds DNA, enveloped complex virus
 - D. ds DNA, enveloped icosahedral virus
 - E. ds RNA, naked segmented virus
 - F. ss RNA, segmented enveloped and helical virus
 - G. (–) ss RNA, bullet-shaped helical virus
 - H. (–) ss RNA, naked helical virus
 - I. (+) ss RNA, naked icosahedral virus
 - J. (+) ss RNA, enveloped icosahedral virus
 - K. (+) ss RNA, enveloped diploid virus
26. Cataracts and patent ductus arteriosus in a newborn suggest *in utero* infection with what viral family?
 - A. Adenovirus
 - B. Paramyxovirus
 - C. Parvovirus
 - D. Picornavirus
 - E. Reovirus
 - F. Togavirus
27. What is the most dominant method of spread for measles?
 - A. Animal bite
 - B. Fecal-oral
 - C. Fomite spread
 - D. Respiratory droplet spread
 - E. Sexual contact
 - F. Transfusion or intravenous drug abuse
 - G. Tick bite

28. How are human papilloma virus type 4 warts spread?
- A. Animal bite
 - B. Fecal-oral
 - C. Fomite spread
 - D. Respiratory droplet spread
 - E. Sexual contact
29. A 15-year-old member of the high school swim team notices painless, umbilicated cutaneous lesions on the toes. Large eosinophilic cytoplasmic inclusions are present in the affected epithelia. What is the most likely causative agent?
- A. Adenovirus
 - B. B-19 virus
 - C. Cytomegalovirus
 - D. Herpes simplex virus
 - E. Human papilloma virus
 - F. Molluscum contagiosum virus
 - G. Varicella-Zoster virus
30. A bone marrow transplant recipient becomes febrile and hypoxic and chest films demonstrate diffuse interstitial pneumonia. What is the most likely causative agent?
- A. BK virus
 - B. Cytomegalovirus
 - C. Herpes simplex virus
 - D. Molluscum contagiosum virus
 - E. Paramyxovirus
 - F. Varicella-Zoster virus
31. A 6-month-old infant presents with painless verrucous growths on the laryngeal folds. What is the most likely causative agent?
- A. B-19 virus
 - B. Cytomegalovirus
 - C. Herpes simplex virus
 - D. Human papilloma virus
 - E. Molluscum contagiosum virus

Answers

1. **Answer: E.** Poliovirus has three serotypes, which do not change significantly from year to year or within a person. The World Health Organization (WHO) hopes to have it eradicated within the next five years. (Note the clue of multiple vaccines.)
2. **Answer: D.** The most likely causative agent here is a bacterium. Viral meningitis is usually mild and would not fit the CSF values. Both the age of the patient and the petechial rash suggest it is most likely to be *Neisseria meningitidis*, which is a Gram-negative diplococcus that is oxidase-positive. The overproduction of outer-membrane fragments is what leads to the petechial rash, even prior to antibiotic treatment.
3. **Answer: D.** The positive test for hepatitis B surface antigen indicates that the patient still has the hepatitis B virus and hence is still contagious. The presence of hepatitis B surface antigen and the absence of the surface antibody (anti-HBs) indicate either an acute HBV infection (if patient has had the disease for only a short time) or a chronic carrier state (if the hepatitis has been going on for many months). Because acute HBV is not a choice, choice D then becomes the correct answer. Choice B would be a right answer if HBsAg is negative. Core antibodies would not be present if the person is only vaccinated. Also, HBs antigen should not be present in a detectable amount from vaccination.
4. **Answer: C.** Products of *tat* and *rev* genes increase viral maturation and act to decrease the latent period (J.T. probably had a high expression of those genes). The *vif* gene product affects viral infectivity, and *vpr* gene products increase the efficiency of budding. Another possible difference is that A.Y. may lack the invasion gene, which codes a cellular co-receptor for HIV. *Env* codes for envelope antigen and *pol* codes for reverse transcriptase.
5. **Answer: C.** The *pol* gene codes both for reverse transcriptase and the integrase, which allow the linear proviral DNA to be integrated, apparently at a random site, into a chromosome in the host cell. Of the distractors, both choices A and B are accomplished using the host cell's RNA polymerase. Choice D is a function of the *gag* gene, and choice E is controlled by *tat* and *rev* genes.
6. **Answer: C.** Interferons interfere with virus multiplication by blocking translation of viral protein (by inhibiting viral mRNA and hence inhibiting viral protein synthesis). They do not inhibit transcription, assembly, or release. They also do not inhibit host-protein synthesis.
7. **Answer: B.** Two of the viruses can be eliminated because they are negative RNA viruses and require polymerase accompanying the RNA to replicate. These two are the *Bunyavirus* and the rhabdoviruses. Since the retrovirus RNA requires the reverse transcriptase to replicate, it also cannot start an infection with just the RNA. Because of the locale of replication of the one DNA virus in the list (Poxvirus replicates in the cytoplasm), it must also bring in an enzyme to replicate: a transcribing enzyme.
8. **Answer: A.** Presence of HBsAg (surface antigen to hepatitis B virus) for 6 months (not 3 months) is considered chronic. Students need to know the abbreviations used for hepatitis serology.

9. **Answer: C.** Hepatitis A vaccine is a killed vaccine, and B is a component vaccine. The three important, live viral vaccines are MMR (mumps, measles, or rubeola, and rubella or German measles). The adenovirus vaccine (not in the question) is a live vaccine that is NOT attenuated but is enteric-coated and given orally to reduce respiratory infections in military personnel. Other live viral vaccines are attenuated and include Sabin's polio, varicella, and yellow fever.
10. **Answer: B.** Complications do occur. Choice A is wrong because the agent is the mumps virus, which is RNA (not DNA), and thereby should have uracil instead of thymine. Only one serotype exists, explaining why one gets mumps only once. Choice D is wrong because the genome of the mumps virus is linear and not circular.
11. **Answer: B.** The clinical presentation is consistent with chickenpox caused by VZV. Exanthem subitum is caused by human herpes virus 6, not by CMV. Herpetic gingivostomatitis refers to herpes simplex type 1, not VZV. Students need to know the typical clinical presentations of Whitlow's and infectious mononucleosis.
12. **Answer: E.** This question requires you to understand viral replication and viral genetics quite well. The only nucleic acid in the composite parental virus is the RNA belonging to poliovirus. Thus, only poliovirus is made. The only role the Coxsackie virus would play in the infection is to bind to the host cell and stimulate the uptake of the composite virus. Once uncoating takes place, the Coxsackie components play no further role. A perfect poliovirus will have been made.

So **choice A** is incorrect—the progeny will have the host-cell range of polio because that is what they'll be.

Choice B is incorrect because there is no genetic material coding for the Coxsackie components, so you cannot get a composite.

Choice C is incorrect because no capsid components of Coxsackie will be made; there can be no mixing.

Choice D is incorrect because there was only one type of RNA; there can never be recombination.

Choice E is correct because Sabin is a polio-specific vaccine, and poliovirus will be produced.
13. **Answer: B.** Positive-stranded viral RNA is infectious because it can bind to ribosomes and be translated and therefore can make its polymerase (a RNA-dependent RNA polymerase). It does not need to carry it in its virion. It replicates by making a negative template of RNA, and only HIV replicates through a DNA intermediate. All single-stranded RNA viruses require a replicative intermediate, making choice E false.
14. **Answer: B.** Gp120 is the surface antigen of HIV that mediates its attachment to CD4 lymphocytes. Gp41 is a transmembrane glycoprotein, and p24, p17, and nucleocapsid protein are all internal molecules, which would rarely be accessible to the immune response.
15. **Answer: E.** Our normal tumor-suppressor genes are "anti-oncogenes," which negatively regulate cell growth. The oncogenes of human papilloma viruses code for two early proteins: E6 and E7, which inactivate the tumor-suppressor genes of the infected cells, resulting in malignant transformation. Viral oncogenes are similar but *not identical* to cellular oncogenes. Expression of viral oncogenes may lead to malignant transformation but will not kill the infected cells. HPV oncogenes are not present in every cell and certainly not in every individual.

16. **Answer: B.** Hepatitis A is a picornavirus so that is the most likely choice. If you do not know the family but know that it is naked and positive RNA, you can eliminate hepatitis B because it is DNA. Measles, rubella, and influenza are all enveloped.
17. **Answer: E.** The definition fits a prion. Prions cause subacute spongiform encephalitis (e.g., Kuru, Creutzfeldt-Jakob disease, Gerstmann-Straussler disease, etc.). PML is caused by the JC virus. AIDS dementia is HIV. SSPE follows measles and perhaps is caused by hypermutated, defective forms of the virus.
18. **Answer: E.** The virus is HSV II, a herpesvirus, which are dsDNA viruses that use the mechanisms of our own cells to transcribe an RNA strand from their genomic DNA and use the transcribed RNA as a messenger RNA. Of the distractors: choice A is the technique used by the negative sense RNA viruses; choices B and G are used by the positive sense RNA viruses; choice C is used by the retroviruses; choice D is used during the genomic duplication of negative sense RNA viruses; and choice E is used in genomic replication by the retroviruses. Choice F would not produce RNA.
19. **Answer: D.** In the Tzanck test, a swab is taken from the exposed base of a lesion and observed microscopically by Giemsa staining for the presence of characteristic multinucleated giant cells. The Tzanck test is not an immunofluorescence test (**choice E**) or a test for patient antibodies (**choice A**) or a viral culture. (Both direct immunofluorescent staining of cells from lesions and viral culture with DFA confirmation are commonly used for diagnosis in large hospitals because the Tzanck test is not specific.) Herpes simplex virus produces intranuclear infection (thus **choice B** is not true), and the lesions are generally not pustular (**choice C**). **Choice F** is not true because herpesviruses are DNA viruses, and koilocytic cells (**choice G**) are hallmarks of human papilloma virus infection in PAP smears.
20. **Answer: A.** CMV is an extremely common virus and crosses the placenta oftentimes without causing obvious symptomology. Fortunately, Rubella, which is highly teratogenic particularly in early pregnancy, is generally prevented by routine vaccination in childhood or at least 16 weeks prior to pregnancy. Less than 5% of hepatitis B could possibly be *in utero*. HSV 2 will only cross the placenta if mom acquires Herpes for the first time during her pregnancy. RSV and other respiratory viruses will not. Other viruses that can cross the placenta include Coxsackie B and HIV.
21. **Answer: D.** Cytomegalovirus, Hepadna, and Poxviruses are all dsDNA viruses and not RNA. Retrovirus is an RNA virus but replicates through a dsDNA, so it also is not the correct answer. Toga is a positive RNA virus, which replicates through a negative RNA intermediate and has no DNA; therefore, it's the correct answer.
22. **Answer: D.** Monospot is the most commonly used test for the diagnosis of infectious mononucleosis caused by EBV. However, it does not detect EBV-specific antibody. It instead detects heterophile antibody, which is nonspecific in that it may be present in different organisms and individuals and it cross-reacts with many animal RBCs. Epstein-Barr nuclear antigen test is routinely run in the diagnosis of mononucleosis. Atypical lymphocytes are found in mononucleosis caused by EBV and CMV, but CMV is heterophile antibody-negative. Isolation of EBV, of course, can establish the diagnosis. However, the procedure is cumbersome and laborious, and would identify asymptomatic infected cases as well.
23. **Answer: D.** The segmented influenza viruses may undergo recombination with a similar animal virus. This leads to genetic change that negates everyone's prior immunity, leading to severe pandemics.

24. **Answer: C.** HIV. It is this genetic drift that makes it difficult for the body to fight off HIV and has complicated the development of an effective vaccine.
25. **Answer: D.** Both the symptomology, length of infection, and the epidemiological clues (college student, age 19, has given it to his girlfriend) strongly suggest that this is EBV, which is a herpesvirus.
- Choice A** = parvo; **choice B** = adeno; **choice C** = pox; **choice D** = papova/hepadna because there's no distinction as to circular or partial ds DNA; **choice E** = reovirus; **choice F** = arena virus, bunya, and orthomyxo; **choice G** = rabies; **choice H** = none; **choice I** = calici or picorna; **choice J** = flavi and toga; **choice K** = retro.
26. **Answer: F.** The description fits congenital rubella, a togavirus, which is an enveloped positive RNA virus that is not segmented.
27. **Answer: D.** If you have any trouble, think about which of these viruses has respiratory symptoms (in this case, pneumonia).
28. **Answer: C.** Remember that type 4 strains cause plantar warts, and these are largely transmitted by shower room floors, towels, etc.
29. **Answer: F.** This describes the typical presentation of molluscum contagiosum, which is commonly acquired through small breaks in the skin in environments where moisture keeps the virus viable (swimming pools, showers).
30. **Answer: B.** CMV is the most common viral cause of death in bone-marrow transplant patients, causing an interstitial pneumonia.
31. **Answer: D.** Perinatal infection with human papilloma virus can cause infantile laryngeal warts.

Matching

A. Match disease with causative agent. Remember that choices may be used once, more than once, or not at all.

- A. Adenovirus
- B. Coronavirus
- C. Coxsackie virus
- D. Echovirus
- E. Gerstmann-Straussler
- F. Herpes 1
- G. Marburg agent
- H. Measles
- I. Norwalk agent
- J. Parvovirus
- K. Rabies
- L. Rotavirus
- M. Rubella
- N. Vaccinia
- O. Varicella-Zoster

- ___ 1. A common cause of acute coryza, a usually afebrile upper respiratory infection causing inflammation of the nose, paranasal sinuses, throat, larynx, and trachea.
- ___ 2. The patient has experienced a low-grade fever and stiff neck for the past day. This morning he awakened with difficulty and his wife noted distinct personality changes. He experienced several seizures and was brought into the emergency department. A non-traumatic spinal tap yielded CSF fluid with lymphocytes and RBCs. A diagnosis of encephalitis was made.
- ___ 3. An epidemic of diarrheal disease has occurred at Lincoln Grade School. Twenty percent of the children have been ill enough to stay at home and three teachers and a custodian have also called in sick.
- ___ 4. Several children in a daycare center have developed papulovesicular erythematous lesions over the buccal mucosa and palate. The children had experienced fever, headache, and sore throat a few days preceding the appearance of the vesicular lesions. Similar lesions are noted on the hands and feet and in the diaper area in the very young.
- ___ 5. Over fifty percent of the boys in Scout Troop #27 have developed a febrile disease during the first week after their camp out on the lake. They all complain of sore throat and itchy eyes; their conjunctiva are inflamed. A few other members of the troop have conjunctivitis but no signs of pharyngitis.
- ___ 6. This 17-year-old child has been bothered with insomnia for the past several weeks. Her schoolwork has suffered and she has been having hallucinations. The onset of seizures has prompted a neurologic examination. The presence of elevated levels of gamma globulin in the spinal fluid prompts the neurologist to order an EEG. A diagnosis of subacute sclerosing panencephalitis is made. What earlier viral infection has she had?

- ___7. The patient complains of easy fatigability and insomnia. He is somewhat apathetic and disoriented. Aphasia and other signs of abnormal higher cortical functions are noted. A neurological consult examines the patient and suggests that he might have a subacute spongiform encephalopathy.
- ___8. The patient is a well-developed young man of 16 years of age who has been working on a combine crew this summer. He has a sudden onset of malaise and fever and complains of a global headache and a stiff neck and back. CSF exam reveals an elevated opening pressure, normal glucose, but elevated proteins. The predominating cell in the spinal fluid is a lymphocyte.

Answers to A

1. B Because the rhinoviruses are not here, you have to choose the coronaviruses.
2. F Herpes 1 is a major causative agent of encephalitis.
3. I Communal eating may lead to food contaminated by a food preparer with Norwalk virus. Characteristically, infants are less likely to be involved not having eaten the food, even though they may have attended the event.
4. C This is hand, foot, and mouth disease caused by Coxsackie A virus.
5. A Adenovirus
6. H Late sequela to measles (rubeola)
7. E This is prion disease. The only prion listed is Gerstmann-Straussler.
8. D The most common causative agent of aseptic meningitis is echovirus.

B. Match viral feature with agent.

- A. Adenovirus
- B. Rubella
- C. Influenza A
- D. Herpes 1
- E. Human immunodeficiency virus
- F. Human papilloma virus
- G. Hepatitis B
- H. Variola

- ___ 1. Genetic shift important to epidemiology
- ___ 2. Highest transmissibility rate by needle stick
- ___ 3. Oncogenic virus
- ___ 4. Self-coded envelope
- ___ 5. Teratogenic

Answers to B

- 1. **C** Shift is in influenza A, while drift also is notable in HIV.
- 2. **G** Hepatitis B is acknowledged to be more infectious than HIV.
- 3. **F** HPV has the strongest evidence for being carcinogenic. HIV is not considered directly oncogenic.
- 4. **H** Smallpox (variola) is the only one that makes its own envelope.
- 5. **B** Rubella is most noted for teratogenicity.

C. These are killers. If you can do these, you are in excellent shape! Write what you know about the virus, and then look on the list, rather than trying to do it in whatever order they have given you the characteristics.

Match features with viral types.

- A. Icosahedral, enveloped, double-stranded DNA
- B. Icosahedral, naked, double-stranded DNA
- C. Icosahedral, naked, single-stranded DNA
- D. Icosahedral, naked, positive single-stranded RNA
- E. Icosahedral, enveloped, positive single-stranded RNA
- F. Icosahedral, double-stranded, segmented DNA
- G. Helical, naked, negative single-stranded RNA
- H. Helical, enveloped, negative single-stranded RNA
- I. Helical, enveloped, negative, single-stranded, segmented RNA

- ___1. The baby was born at the 31st week of gestation. It was small for gestational age and showed jaundice. Petechiae were noted and the spleen and liver were enlarged. The pathology department reported that placental examination revealed chorioamnionitis, villitis, and owl's eye intranuclear inclusion bodies.
- ___2. The Peace Corps worker was bitten by a camp dog while helping in the construction of a dam near the village of Urundan in Central Africa. The animal was caged and 4 days after the attack, died. The head was sent to a reference laboratory and Negri bodies were found in Ammon's horn.
- ___3. A 4-year-old child of Mexican immigrant parents was brought to the hospital with a temperature of 101°F, a cough coryza, conjunctivitis, and a characteristic exanthem on the buccal mucosa opposite the first and second upper molars. The spots resemble tiny grains of white sand surrounded by inflammatory areolae.
- ___4. A 16-year-old high school dropout develops urticaria and arthralgia. He reports that his urine is dark in color and his sclera are yellow. He has not felt well for the past week or so and has quit smoking because the cigarettes started to taste funny.

Answers to C

- 1. A Owl's eye's inclusion bodies are found in cytomegalovirus, which is a Herpes virus. The viruses are DNA (double-stranded), enveloped, and icosahedral.
- 2. H Negri bodies are found in rabies, which is a negative single-stranded RNA virus, both helical and enveloped.
- 3. H Koplik spots are associated with measles, which is an enveloped, helical, negative, single-stranded RNA virus.
- 4. A Dane particles are the infectious particle of hepatitis B that are icosahedral, enveloped, double-stranded DNA viruses.

Microbial Genetics/ Drug Resistance

6

TERMINOLOGY

Polymerases

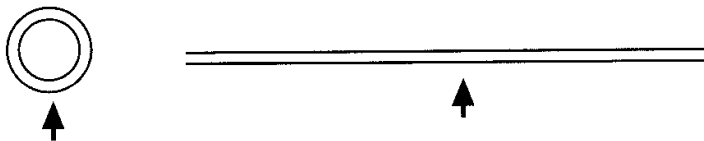
(major nucleic acid synthesizing polymerases)

_____ -dependent _____ polymerase
(template) (product)

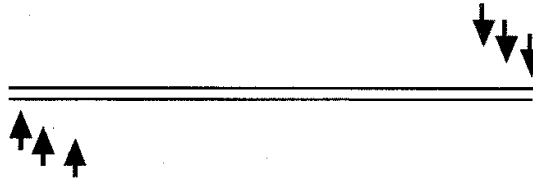
_____ -dependent _____ polymerase
(template) (product)

Nucleases

Endonucleases—cleave the nucleic acid backbone in the middle of the nucleic acid strand. Restriction endonucleases (a.k.a., site-specific endonucleases) recognize specific base sequences and make specific breaks. Bacterial cells methylate their own restriction nuclease sites so that their enzymes do not break down their own DNA. Endonucleases are not present in high concentration in the cells.



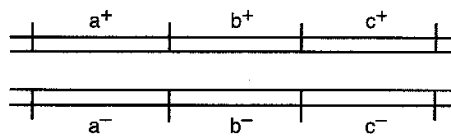
Exonucleases—remove end nucleotides sequentially from linear pieces of nucleic acid, ultimately totally breaking the piece of DNA down. Numerous exonucleases per cell.



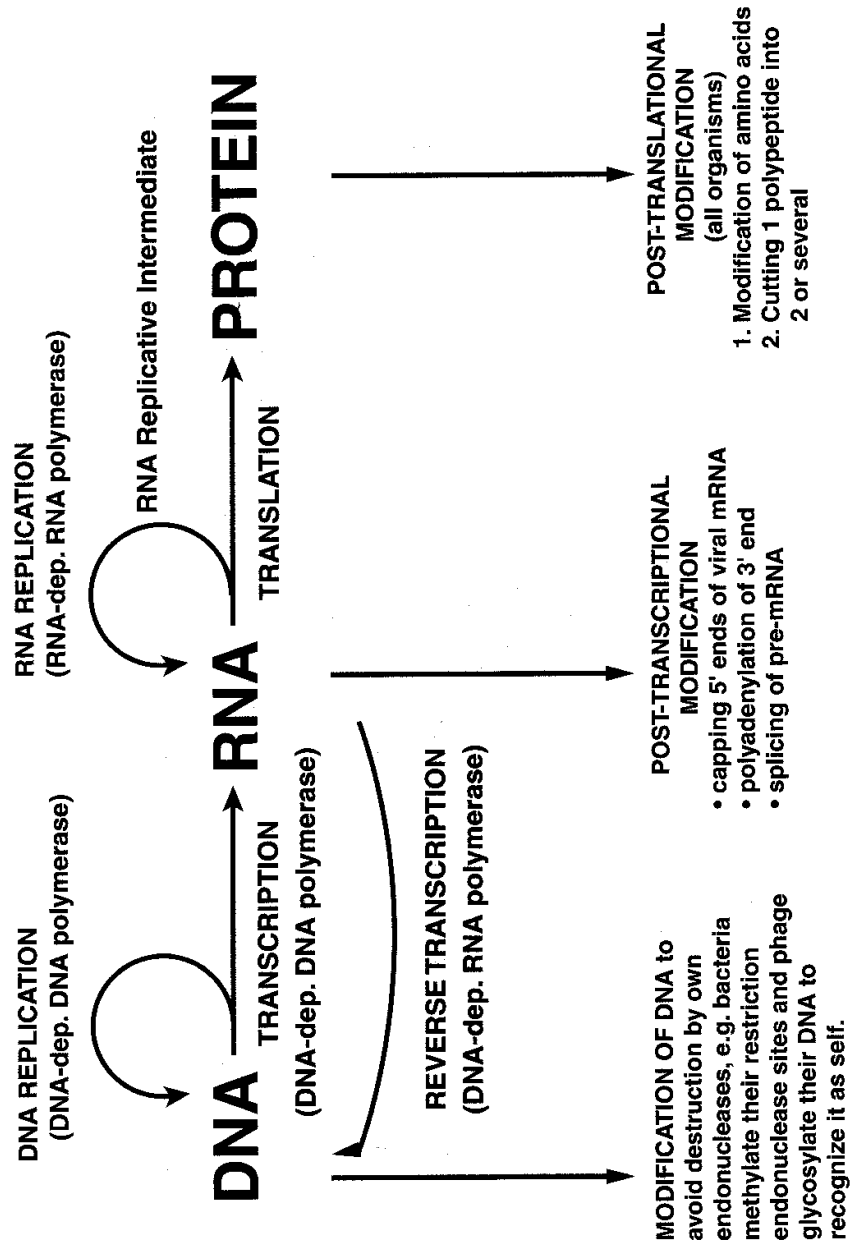
Alleles

Alternative forms of the same gene.

In the following two strands of DNA, *a*, *b*, and *c* are different genes but *a*⁺ and *a*⁻ are different alleles.



THE FLOW OF GENETIC INFORMATION

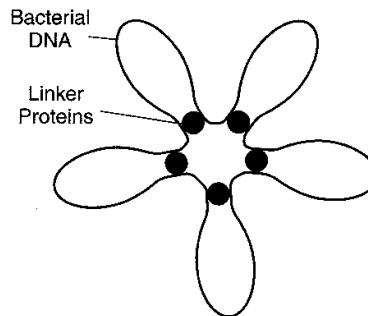


THE BACTERIAL GENETIC MATERIAL

Three different types of DNA may be found in a bacterial cell: bacterial DNA, plasmid DNA, or bacteriophage DNA.

Bacterial Chromosome (Genome)

- Most bacteria have only **one chromosome but often multiple copies** of it in the cell.
- Most bacterial chromosomes are a **large covalently closed, circular DNA molecule** (about 1,000 times the diameter of the cell).
- The chromosome is **organized into loops** around a proteinaceous center. A single-stranded topoisomerase (1 nick) will relax only the nicked loop, allowing DNA synthesis or transcription.
- Most have **around 2,000 genes**. (*E. coli* has about 4,500 kbases.)
- All **essential genes** are on the bacterial chromosome.



Plasmids

- Are **extrachromosomal genetic elements** found in bacteria (and eukaryotes)
- Are generally covalently closed, **circular DNA**
- Are **small** (1.5–400 kB)
- Can replicate autonomously in bacterial cells
- One subclass of plasmids, called **episomes**, may be integrated into the bacterial DNA. Episomes have insertion sequences matching those on the bacterial chromosome.
- Plasmids carry the genetic material for a variety of genes, e.g., the fertility genes directing conjugation (*tra* operon), many of the genes for antibiotic resistance, and most bacterial exotoxins.

Bacteriophage (= phage = bacterial virus) Genome

- Stable pieces of bacteriophage DNA may be present in the bacterial cell.
- These are generally repressed temperate phage DNA inserted into the bacterial chromosome (called a prophage).
- Besides the repressor protein, this prophage DNA may also direct synthesis of another protein. Most notable are gene products that make bacteria more pathogenic. This enhanced virulence is called **lysogenic conversion**.

INSERTION SEQUENCES/TRANSPOSONS

Transposons (Tn) and Insertion Sequences (IS)

- Are mobile genetic elements (DNA) that can move themselves or a copy from one molecule of DNA to another ("jumping genes")
- Are found in eukaryotic and bacterial cells and viruses
- Have sequences of indirect repeats of bases on each end
- Have at least one gene for a transposase (enzyme[s] involved in the "movement")
- Create additional mutations with their insertion in another totally unrelated gene
- May insert anywhere the transposase recognizes the specific sequence of nucleotides.
- Insertion creates direct repeats on each side of the transposon/IS.

Phage-Coded Pathogenic Factors or Lysogenic Conversion

O = O antigen of *Salmonella*

B = Botulinum toxin

E = Erythrotoxic exotoxins of *S. pyogenes* (SPE-A, -B, and -C)

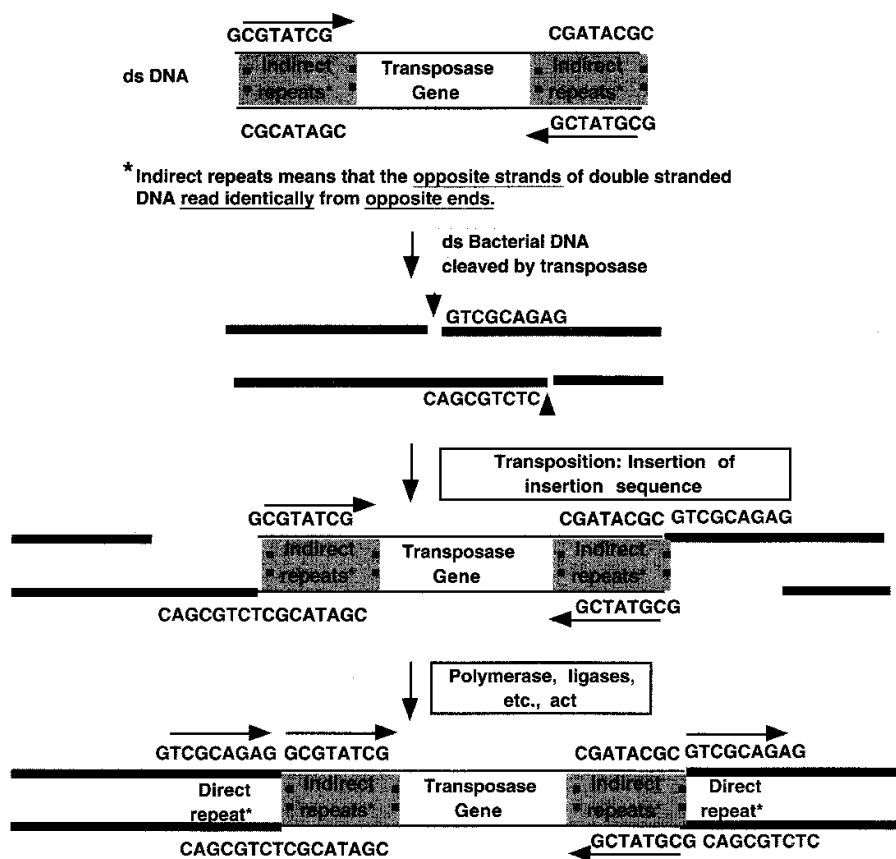
D = Diphtheria toxin
(OBED: a little bit pregnant with phage!)

Insertion Sequences (IS)

Minimal transposable elements (A, below)

- Have just the one gene for transposase.
- Have the terminal indirect repeats.
- Have promoters and transcription and translation-termination signals.

Insertion Sequence Showing Terminal Indirect Repeats* + Genes for Transposition



* Indirect repeats means that the opposite strands of double stranded DNA read identically from opposite ends.

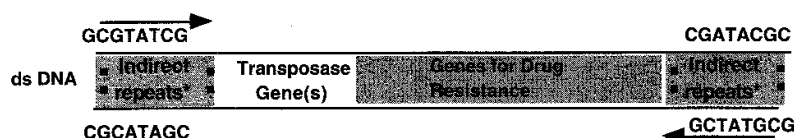
* Direct repeat means that the same strand read from the same direction is identical. It is created because a homologous strand was separated over a short space and then each strand used as a template.

Transposons (Tn)

Insertion sequences plus at least one other gene

- Include some genes for drug resistance
- Transposons play an important role in building multiple drug resistance plasmids. They also (but not as frequently) may move to the chromosome.

**Transposons have additional genes,
e.g., drug resistance genes**



REARRANGEMENT OF DNA WITHIN A BACTERIUM

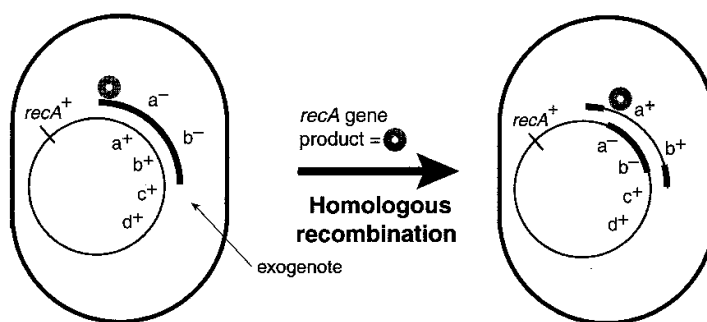
Homologous Recombination

Homologous recombination is a gene exchange process that may stabilize some genes introduced by transformation, conjugation, or transduction.

Imported bacterial DNA (transferred into a cell by transformation, conjugation or transduction) is on short linear pieces of DNA called exogenotes. Most linear DNA is not stable in cells because it is broken down by exonucleases.

Homologous recombination produces an "exchange" of pieces of DNA between the linear piece of DNA and it is near a homologous region on the stable (circular) bacterial chromosome. Homologous recombination requires:

- Several genes worth of homology or near homology between the DNA strands.
- A series of recombination enzymes/factors coded for by the recombination genes *recA*, *recB*, *recC*, and *recD* (with *recA* generally an absolute requirement).



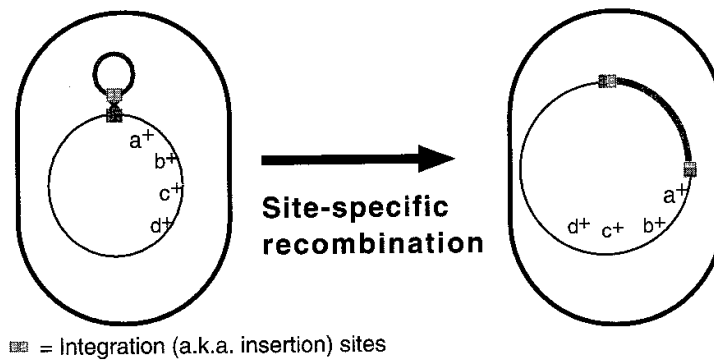
Genes ending up on the linear piece of DNA are lost.

Those on the circular molecule become part of the cell's **permanent** genetic make up.

Site-Specific Recombination

Site-specific recombination is the integration of a DNA molecule into a DNA with which it has no homology except for a small site on each DNA (called an attachment, integration or insertion site).

- Requires restriction endonucleases and restriction endonuclease sites on each DNA (often named an integration site or attachment site but basically insertion sequences) but DNA synthesis is not required.
- Since this process integrates rather than exchanging pieces of DNA, the end result is a molecule the sum of the two original molecules.



Three major roles of site-specific integration

- Integration of a fertility factor to make an Hfr cell
- Integration of temperate phage DNA into a bacterial chromosome to create a prophage
- Movement and insertion of transposons (transposition is the name of site-specific integration of transposons)

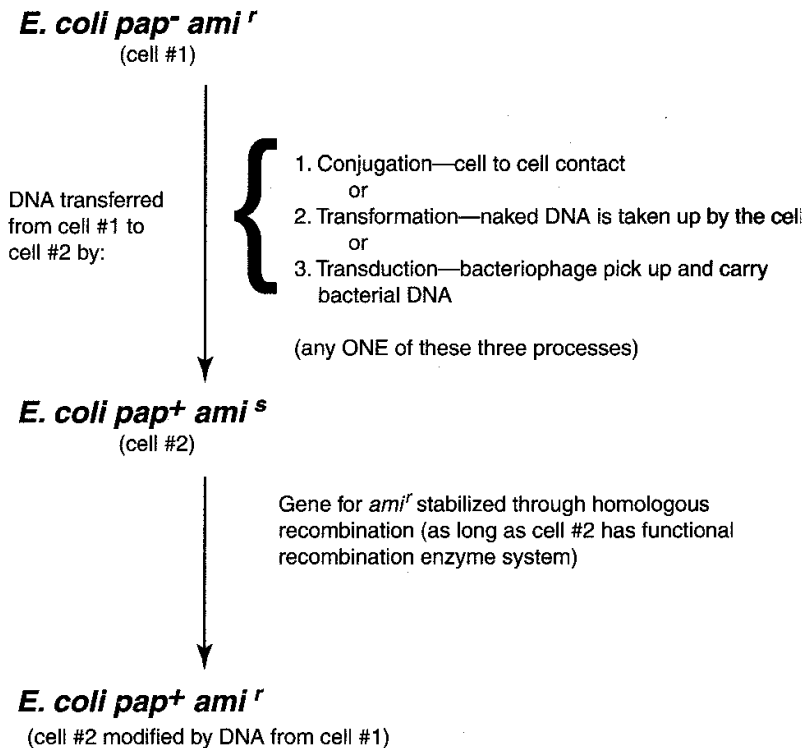
GENE TRANSFER

Overview

Bacterial reproduction is asexual so progeny are identical to parent cell with only rare mutations.

How do you get new genetic combinations in bacteria?

Answer: Gene transfer followed by stabilization of genes (recombination).

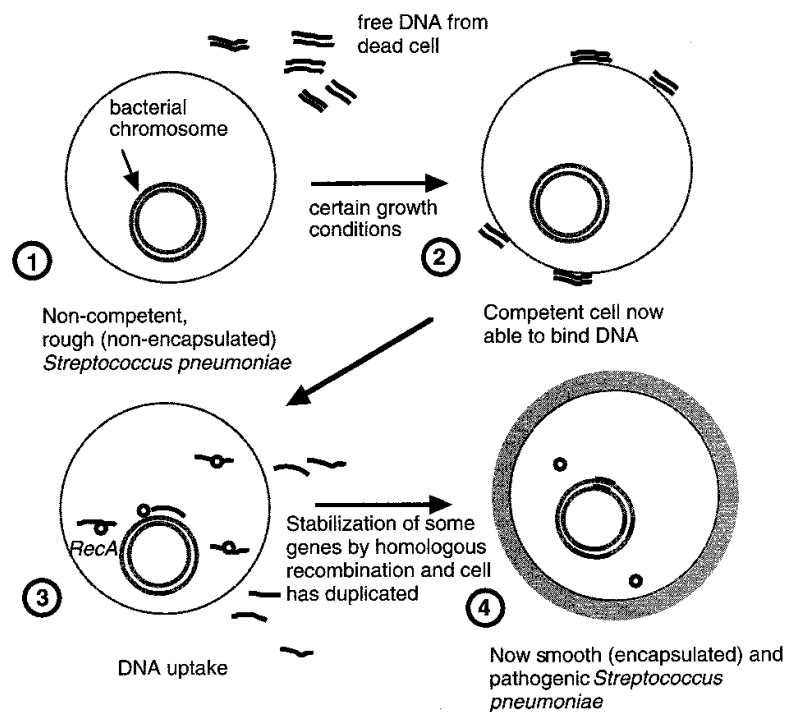


- Now *pap⁺* (initially linked to *ami^s*) is linked to *ami^r* instead, producing a new combination of genes and more significantly, a cell that can cause pyelonephritis and is amikacin resistant.
- (Could also have yielded *E. coli* that was *pap⁻ ami^s* or *pap⁻ ami^r* or the cell could have stayed *pap⁺ ami^s*.)

Transformation

Transformation is the uptake of naked DNA from the environment by competent cells.

- Cells become competent (able to bind short pieces of DNA to the envelope and import them into the cell) under certain environmental conditions (which you do not need to know).
- DNA (released from dead cells) is taken up.
- Newly introduced DNA is generally linear, homologous DNA from same type of cell but perhaps one that is genetically diverse.
- The steps of transformation of a non-encapsulated *Streptococcus pneumoniae* are shown below.



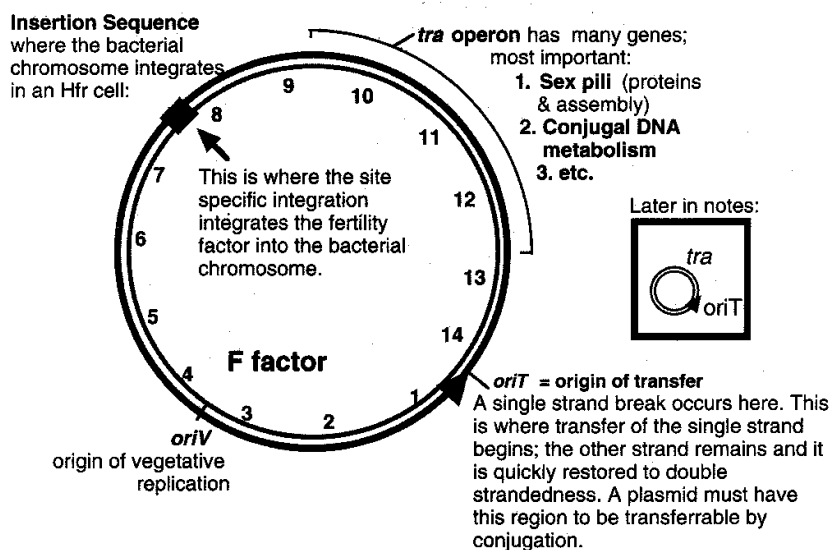
Conjugation

Conjugation is gene transfer from one bacterial cell to another involving direct cell-to-cell contact.

- **Fertility factors control** conjugation
- Sex pili (genes on F factor) play a role in establishing cell-to-cell contact.
- A single strand (or a portion thereof) of the double helix of DNA is transferred from the donor (or male) cell to the recipient or female cell.
- Bacterial genes transferred in by conjugation have to be stabilized by homologous recombination (i.e., in an Hfr $\times$ F⁻ cross). Plasmid genes transferred by conjugation circularize and are stable without recombination.
- Conjugation with recombination may produce new genetic combinations.

Donor (Male) Cells

- ALL have fertility plasmids known as F factors. F factors have a series of important plasmid "fertility" genes called the transfer or *tra* region which code for:
 - sex pili
 - genes whose products stabilize mating pairs
 - genes which direct conjugal DNA transfer, and other genes.
- Have a region called *oriT* (origin of transfer) where a single strand break in the DNA will be made and then *oriT* begins the transfer of one strand of the double helix.
- Many have insertion sequences where the plasmid can be inserted into the bacterial chromosome combining to make one larger molecule of DNA.
- A genetic map of an F factor is shown below.

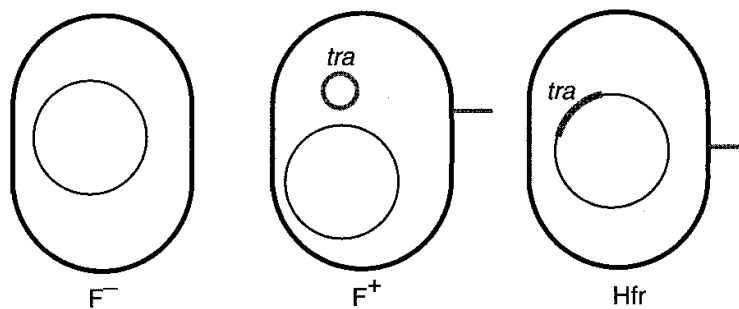


- Donor cells in which the **fertility plasmid** is in its **free state** are called **F⁺ cells**.
- Donor cells in which the **fertility factor** has **inserted** itself into the bacterial chromosome are called **Hfr cells**.

Recipient (Female) Cells: F^- Cells

- Recipient cells lack fertility factors and genes.
- In every cross, one cell must be an F^- cell.

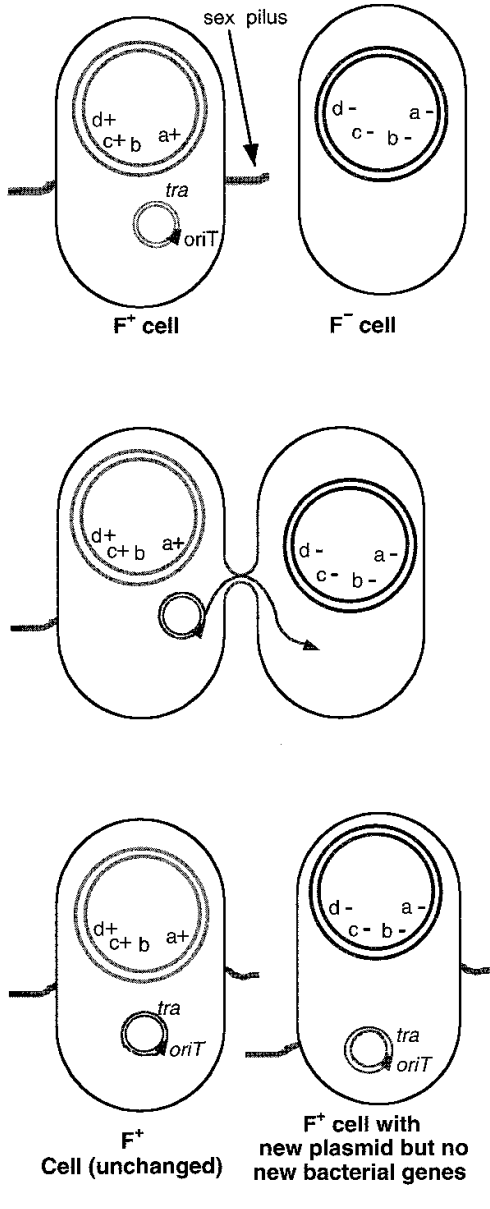
Mating Types



CONJUGAL CROSSES

There are two major types of crosses:

Conjugation: $F^+ \times F^-$ Mating



1 Important points: In the male or F^+ parent, the fertility factor is present but free from the bacterial chromosome. Transfer is uni-directional from male to female. $oriT$, as in every cross, will be transferred first and then the rest of the plasmid genes.

2 Note only a single strand of the plasmid DNA duplex is transferred. The area that is lost is reduplicated (shown as dotted lines) so that the donor always stays the same genotype. The last genes to be transferred are the tra region.

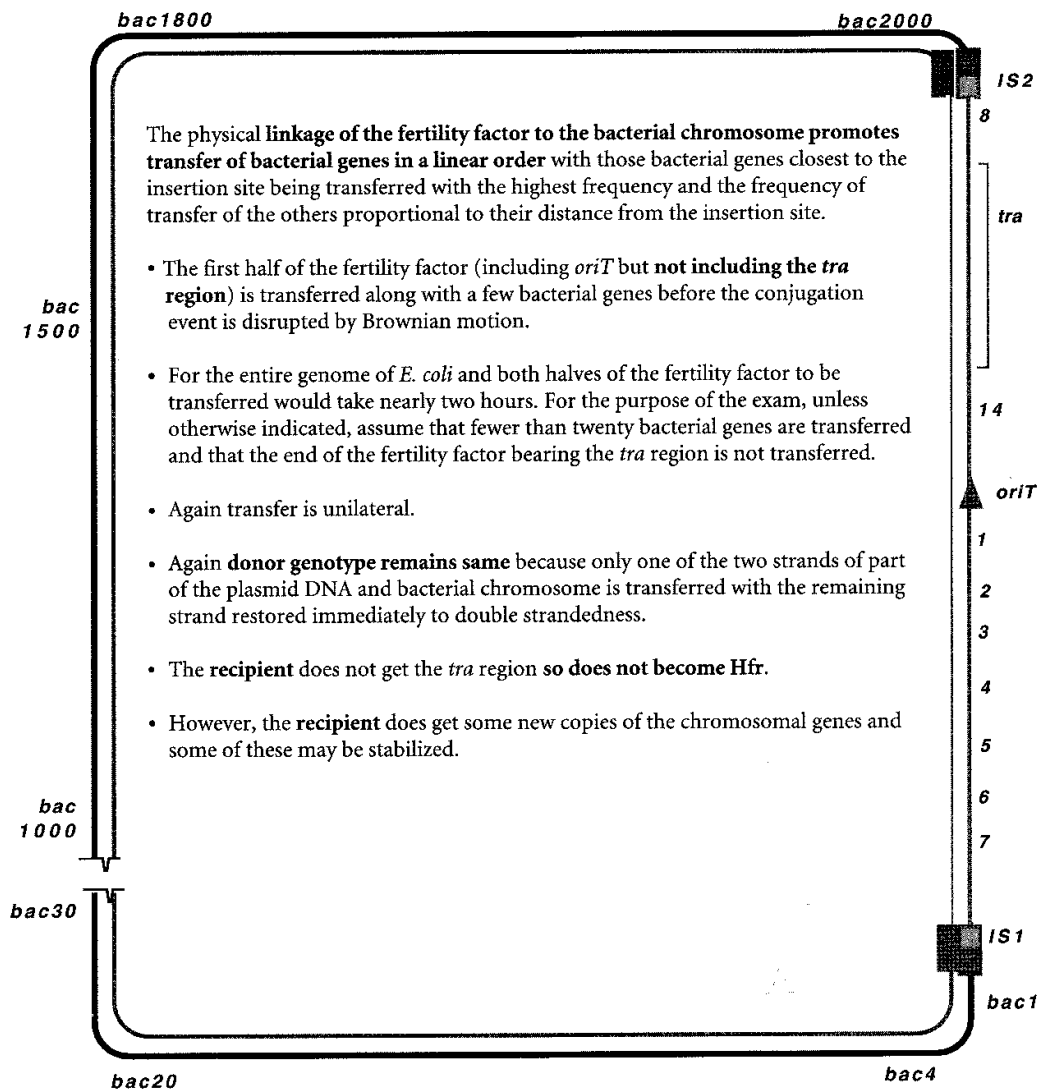
3 The transfer of the plasmid is fairly quick so assume that it is transferred in its entirety 100% of the time unless otherwise told. Note that the F^- cell undergoes a sex change becoming F^+ (male). These two F^+ cells can no longer mate. But no BACTERIAL genes are transferred.

Integrated Fertility Factor

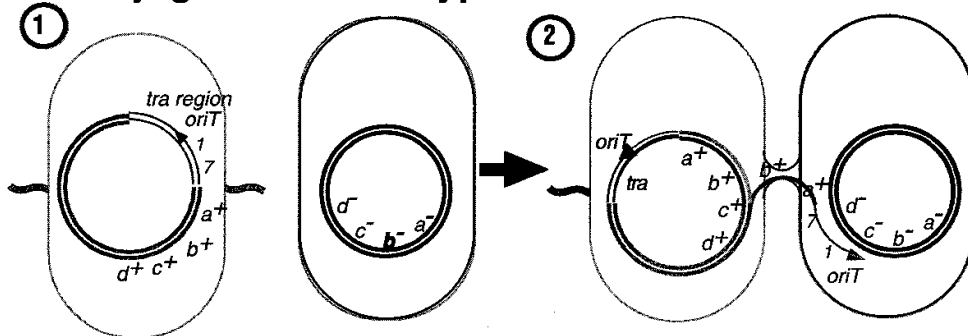
Hfr Chromosome (Bacterial chromosome with integrated F factor)



Bacterial genes are represented as **bac##** to remind you that there are generally several thousand bacterial genes and that this molecule of DNA is very large.



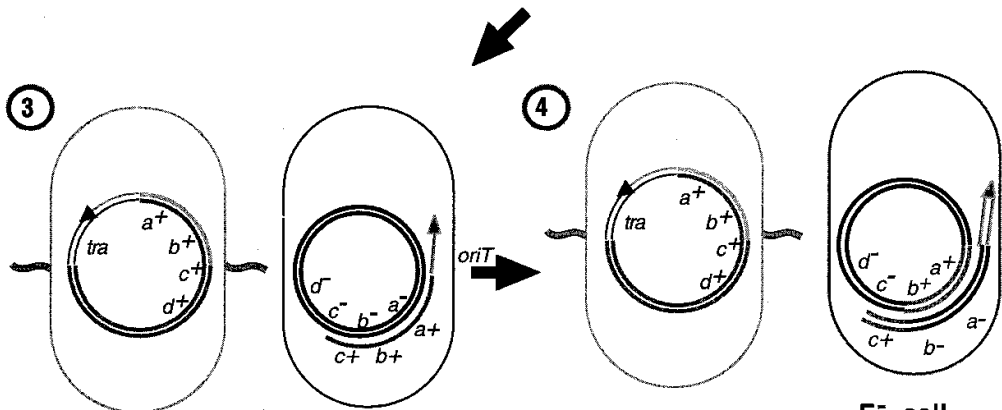
Conjugation: 2nd type of cross Hfr $\times$ F⁻



Hfr $a^+b^+c^+d^+$ $\times$ F⁻ $a^-b^-c^-d^-$

Important points: Fertility factor is integrated into the bacterial chromosome. In this cross *oriT* and the first half of the fertility factor (regions 1-7 on the F factor) will be transferred first (and in that order) and then the bacterial genes in the linear order away from the plasmid.

Note, that as with the F⁺ $\times$ F⁻ cross, only a single strand of the DNA duplex is transferred. The area that is lost is reduplicated so that the donor always stays the same genotype. The last genes to be transferred would be the *tra* region.



It takes approximately two hours for a complete transfer to occur. Because the cytoplasmic bridge and DNA strand is so fine, mating is normally interrupted before the transfer is complete. For the purpose of exam, assume that mating is interrupted and the recipient gets some new genes but does not become Hfr.

Hfr cell
(unchanged)

F⁻ cell
with new
bacterial genes:
 a^+ and b^+
(No sex change)

Transduction

Transduction is the transfer of bacterial DNA by a phage vector.

The phage picks up the bacterial DNA through an error in phage production.

There are two types of transduction: generalized and specialized.

A generalized transducing phage is produced by the phage putting a piece of bacterial DNA into its head. All genes have an equal chance of being transduced.

- Specialized transduction is dependent on integration of phage DNA into the bacterial chromosome at a specific site and then an error being made in its excision as the phage begins to replicate lytically.

To understand transduction, you need first to understand how a phage replicates normally so that you can understand how the errors are made.

Phage

= bacteriophage = bacterial virus

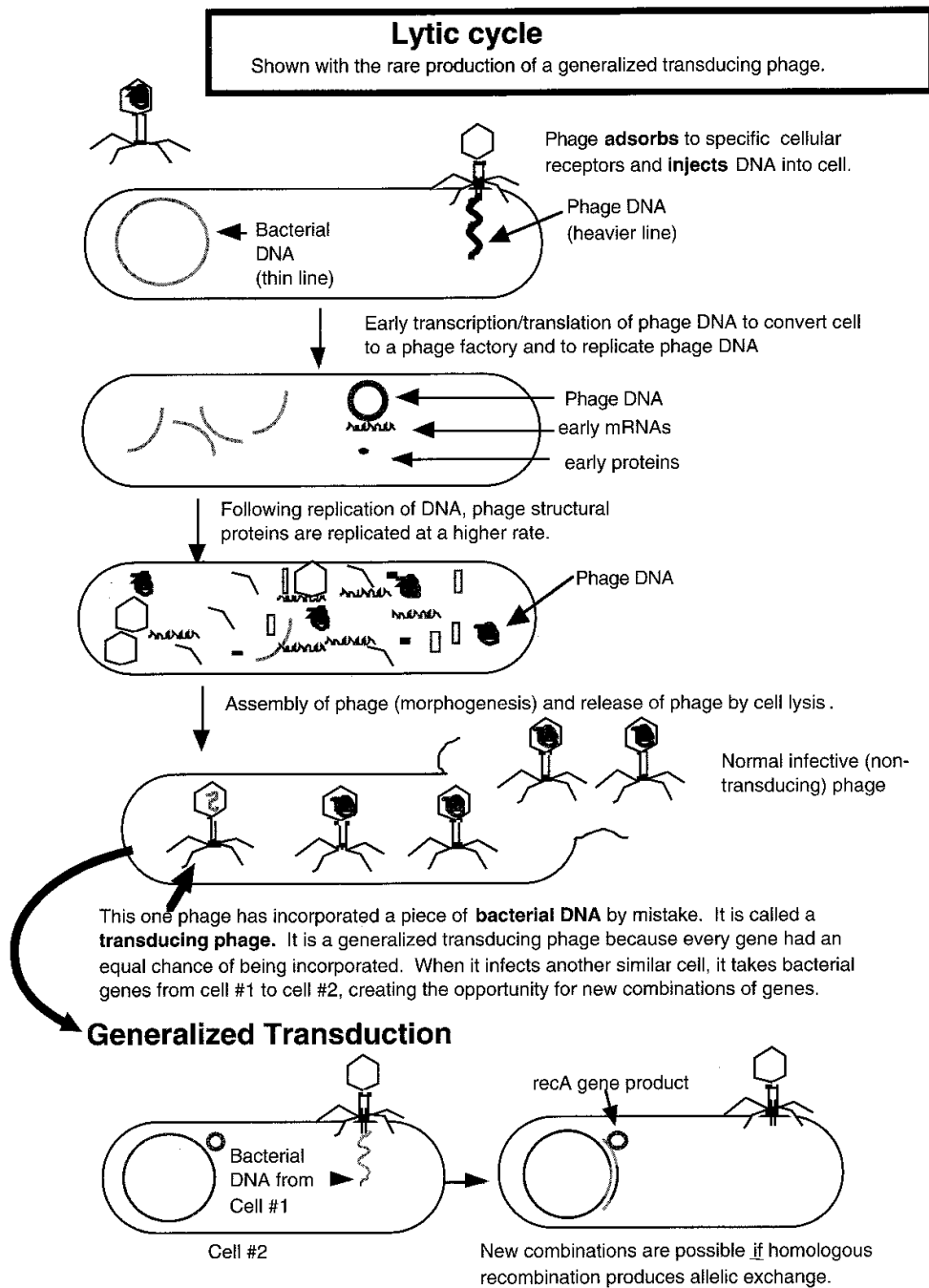
Come in two major types:

- **Virulent phage** infect bacterial cells, always making more virus and lysing the cells (lytic replication).
- **Temperate phage** often infect without lysing the cells because they have the ability to repress active phage replication and to stably integrate their DNA into the bacterial chromosome. In the absence of functional repressor protein, they also may replicate lytically.

Lytic Infection

Lytic infection, by phage or viruses, leads to production of viruses and their release by cell lysis.

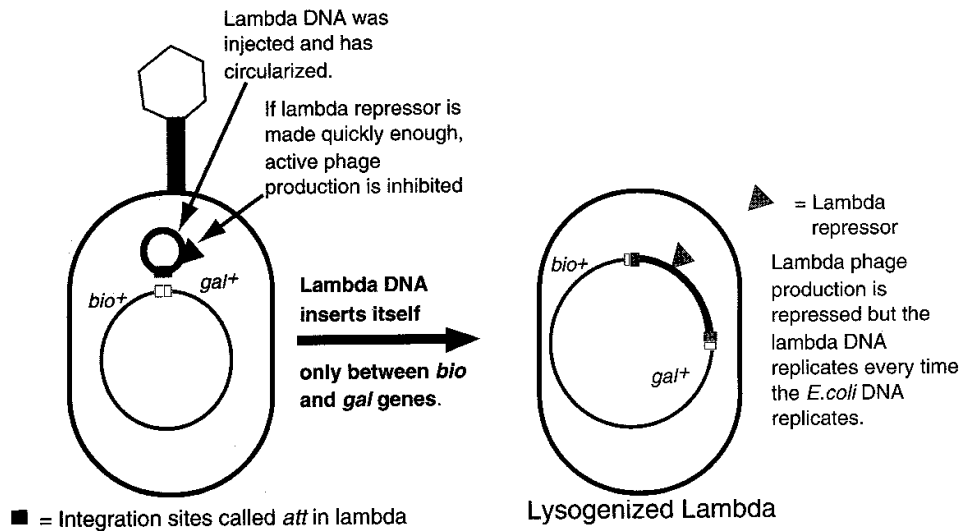
- **Virulent viruses can only go into lytic life and can only carry out generalized transduction.**
- The lytic (or productive) life cycle of virulent phage is shown below. It is entirely normal except for a mistaken incorporation of bacterial DNA into one phage head, creating a transducing virus, shown at the bottom of the next page. Transduction of another bacterial cell is shown following that.



Lysogeny and Specialized Transduction

Temperate phage may become prophage (DNA stably integrated) or replicate lytically.

- When repressor is made, temperate phage insert their DNA into the bacterial chromosome where it stably stays as a prophage.
- If the repressor gene gets mutated or the repressor protein gets damaged then the prophage gets excised from the bacterial DNA and is induced into the lytic production of virus. On rare occasions these temperate phage can produce either specialized or generalized transducing viruses. Lambda phage of *E. coli* is the best studied. Most temperate phages have only a single insertion site.
- Lambda inserts ONLY between *E. coli* genes *gal* and *bio* as shown below.



Lysogeny

Lysogeny is the state of a bacterial cell with a **stable phage DNA** (generally integrated into the bacterial DNA), **not undergoing lytic replication** either because it is **repressed** or **defective**. When the cell DNA replicates, the phage DNA also replicates and, as long as the repressor protein is not damaged, the lysogenic state continues ad infinitum. Defective phage (or defective viruses in the human equivalent) cannot go into an active replication unless a helper virus is present.

Phages that have both options (lytic replication or lysogeny) are called temperate phages. When a temperate phage first infects a cell there is a regulatory race that determines whether the repressor is made fast enough to prevent synthesis of phage components.

The lysogenized cell will replicate to produce two identical cells each with a prophage as long as the repressor gene product is present.

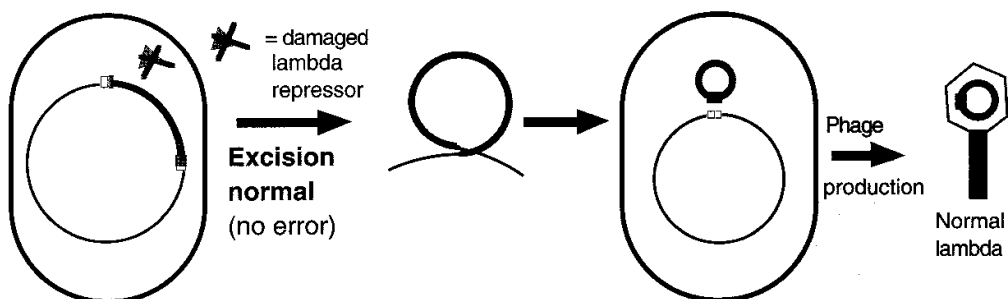
Significance of lysogeny:

- Can confer new properties to a genus such as toxin production or antigens:
 - O: Presence of specific prophage in *Salmonella* can affect O antigens.
 - B: Phage CE β or DE β cause *Clostridium botulinum* to produce Botulinum toxin.
 - E: Exotoxins A–C (erythrogenic or pyogenic) of *Strep. pyogenes*
 - D: Prophage beta causes *Corynebacterium diphtheriae* to make Diphtheria toxin.
- (Mnemonic for phage-mediated pathogenic factors = OBED)
- Model for retrovirus provirus
- Allows specialized transduction

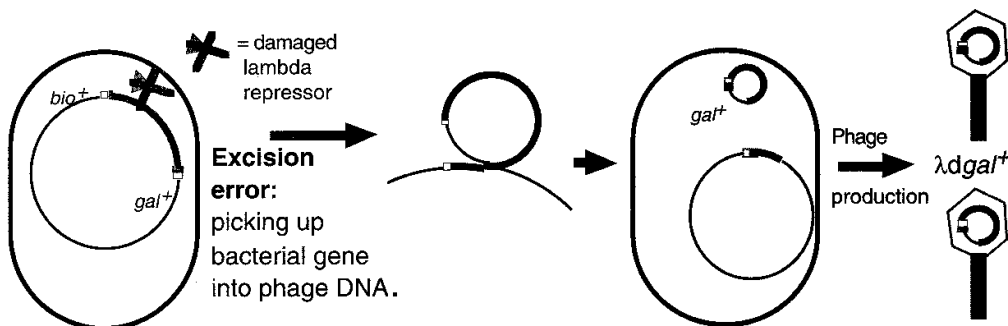
Induction

If the repressor is damaged (by UV, cold, or alkylating agents), then the prophage is excised and the cell goes into lytic replication of the phage. This process is called **induction**.

Most of the time this process is carried out **perfectly** as below and the cell produces **perfect** (non-transducing) **normal phage**.



Rarely, in the excision process, an **excisional error** is made and **one of the bacterial genes next to the insertion site is removed attached to the lambda DNA** and a little bit of lambda DNA is left behind.



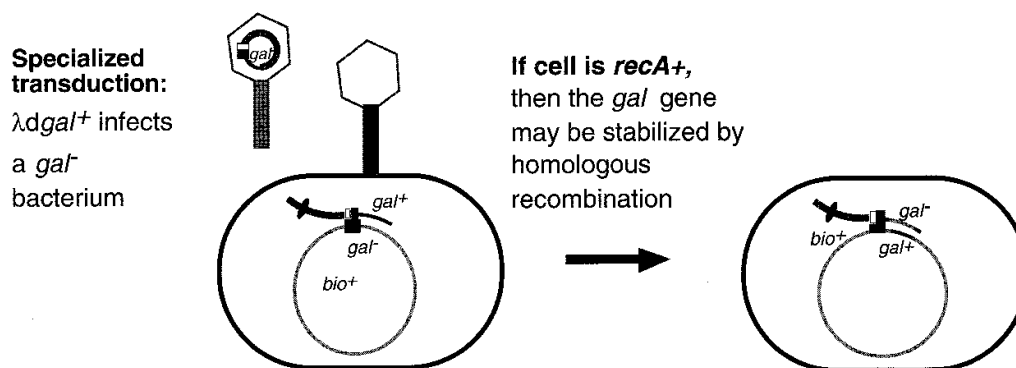
Because lambda has only one insertion site (between *gal* or *bio*), only *gal* or *bio* can be incorporated by excisional error.

Because all of the phage genes are still in the cell, phage are still made with the circular defective phage genome copied and put in each phage head. These are **specialized transducing phage** (only able to transduce *bio* or *gal*).

Induction with an excisional error is shown above.

Specialized Transduction

Bacterial genes picked up by error in the excision process are transferred to another generally closely related but often genetically distinct cell. If any genes on the exogenote are stabilized by recombinational exchange, then new genetic combinations occur.



Only those genes next to the phage insertion site can be transduced by specialized transduction.

Table I-6-1. Comparison of Transformation, Conjugation, and Transduction

Requirement	Transformation	Conjugation	Transduction
Is cell-to-cell contact required?	No	Yes	No
Does it require an antecedent phage infection?	No	No	Yes
Is competency required?	Yes	No	No
Is naked (free) DNA involved?	Yes	No	No
Is recombination required to stabilize new genes?	Yes	No for F ⁺ × F ⁻ Yes for Hfr × F ⁻	Yes

Table I-6-2. Comparison of Generalized and Specialized Transduction

	Generalized	Specialized
Mechanism	Error in assembly	<u>Error of excision</u> Requires stable insertion of prophage DNA (lysogeny)
What genes may be transferred?	Any	Only genes next to the insertion site

DRUG RESISTANCE

Overall problem

- Drug resistance is becoming such a significant problem that there are bacteria for which most antibiotics no longer work. Experts have begun to discuss the “post-antibiotic era.”
- Drug resistance can be transferred from one genus of bacteria to another, e.g., from your normal flora to a pathogen.
- Three general types of antibiotic resistance: intrinsic, chromosomal-mediated, and plasmid-mediated.

Intrinsic Drug Resistance

For example, a cell which has no mycolic acid will not be inhibited by isoniazid.

Chromosome-Mediated Antibiotic Resistance

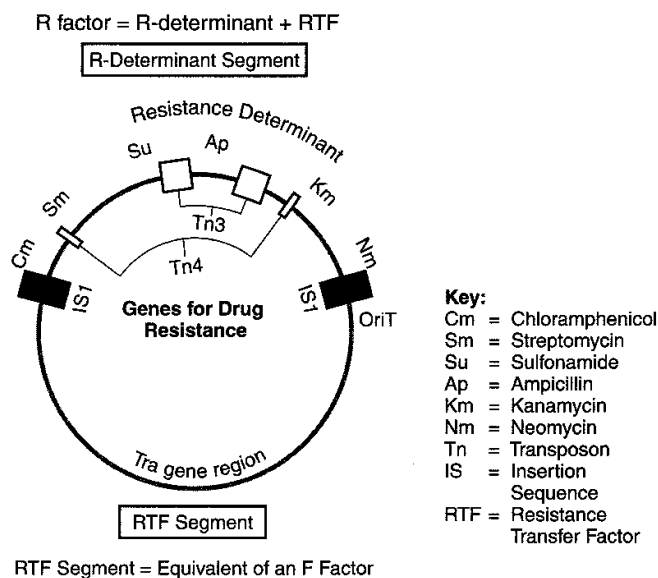
The genes that determine this resistance are located on the bacterial chromosome.

- Most commonly these genes modify the receptor for a drug so that the drug can no longer bind (e.g., a mutation in a gene for a penicillin binding protein [normal cell wall synthetic enzyme]).
- In general, causes low level drug resistance rather than high (exception: methicillin resistance in *Staphylococcus aureus* where a major PBP was mutated).
- But low level resistance may be clinically significant, e.g., in *Streptococcus pneumoniae* meningitis.

Plasmid-Mediated Drug Resistance

The genes that determine this resistance are located on plasmid.

- R factors are conjugative plasmids carrying genes for drug resistance.
 - One section of the DNA (containing *oriT* and the *tra* gene region) mediates conjugation.
 - The other section (R determinant) carries genes for drug resistance. Multiple genes seem to have been inserted through transpositional insertion into a “hot spot.”
 - A typical genetic map of an R factor (a conjugative drug-resistant plasmid) is shown on the next page:
- Plasmid-mediated resistance is created by a variety of mechanisms but often genes code for enzymes that modify the drug.



- **Nonconjugative plasmids**
 - Have lost their *tra* operon (genes) so have lost the ability to **DIRECT** conjugation.
 - But as long as they retain their *oriT*, nonconjugative plasmids may actually be transferred by conjugation as long as there is another fertility factor in the same cell with a functional *transfer* region.
 - The process may be referred to as **mobilization** and is able to occur because the genes in the *tra* region are soluble gene products that are **trans acting**. The region *oriT*, by contrast, is **cis acting**.

Table I-6-3. Plasmid-Mediated Mechanisms

Antimicrobial Agent	Mechanism
Penicillins and cephalosporins	Production of β -lactamase; cleavage of β -lactam rings
Aminoglycosides	Production of acetyltransferase, adenosyltransferase, or phosphotransferase; inactivation of drug by acetylation, adenosylation, or phosphorylation
Chloramphenicol	Production of acetyltransferase; inactivation of drug by acetylation
Tetracyclines	Increased efflux out of cell
Sulfonamides	Active export out of cell and lowered affinity of enzyme

Transfer of Drug Resistance

Neisseria gonorrhoeae

Nonconjugative plasmids: The segment with the *tra* operon has separated leaving a plasmid with the genes for drug resistance still linked to *oriT*.

Transferred by conjugation (mobilization): As long as the cell still has the *tra* region in the cell, it can direct conjugation, make the nick at *oriT* and mobilize the transfer of the nonconjugative plasmid.

Staphylococcus aureus (Methicillin Resistant = MRSA)

Resistance to methicillin is chromosomal, transferred by transduction. Most of the other antibiotic resistance is transferred by plasmids.

Gram-Negative Bacilli

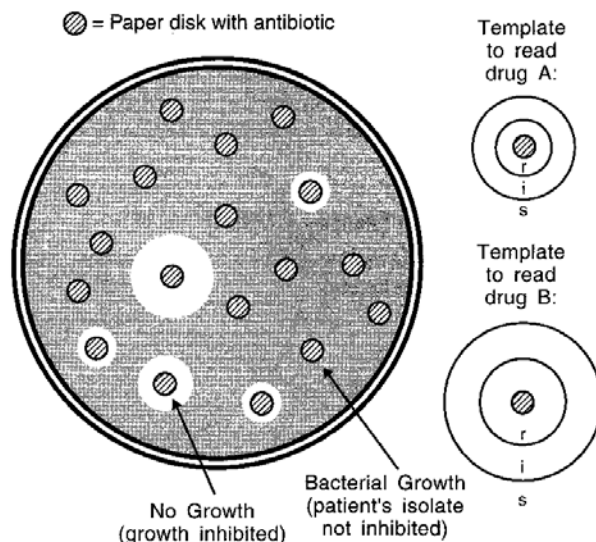
Plasmid mediated, transferred by conjugation.

ANTIBIOTIC SUSCEPTIBILITY TESTING

Kirby Bauer Agar Disk Diffusion Test

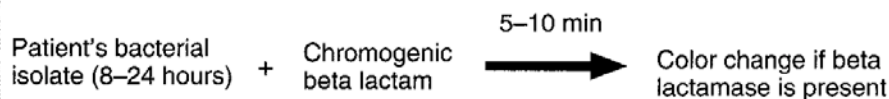
- Solid medium with patient's isolate swabbed on the entire plate surface
- Multiple paper disks, each with a single dried drug placed on plate
- Hydration and diffusion of drug set up a concentration gradient during incubation and growth of the bacteria.
- The diameter of the zones of inhibition must be measured to determine significance.
- Only qualitative (reported back as susceptible, intermediate, or resistant)
- Advantages: relatively cheap, easy, can test numerous antibiotics on one plate, wealth of information based on clinical correlation
- Disadvantage: qualitative

Antibiotic Susceptibility Testing Kirby Bauer Agar Diffusion Plate



"Rapid" Methods

Testing for specific enzymes and a very few probes for genes determining drug resistance are currently available but still require a culture of the patient's pathogen. One current example is β -lactamase testing, shown below.



Minimal Inhibitory Concentration (MIC)

MIC measures antibiotic inhibition.

- This is a dilution technique where **each container** (well of microtiter plate, test tube, or automated system bottle) **has one concentration of an antibiotic** with the patient's isolate. Always one control container has just the patient's isolate and growth medium with no antibiotic to make sure the inoculum is viable.
- **Lowest concentration showing no visible growth is the MIC.**
- In the example below, MIC = 2 $\mu\text{g/ml}$.
- This indicates levels needed to inhibit; it does not necessarily indicate killing levels, which is done with the MBC (see below).

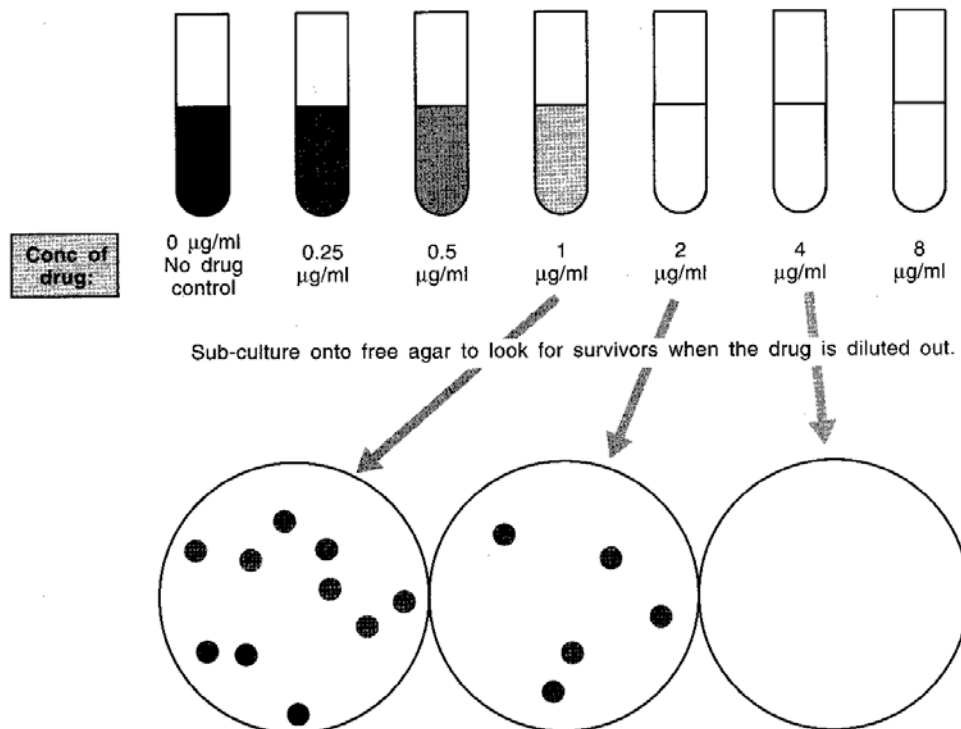
Minimal Bactericidal Concentration (MBC)

This measures the antibiotic killing (bactericidal activity).

- A dilution technique starting with the MIC containers and sub-plating onto solid medium. Because a small inoculum is used on the plate with a large volume of medium, this dilutes the drug way below the MIC and allows determination of viability of cells.
- Important to determine for treating immunocompromised patients whose immune system cannot kill the bacteria while they are inhibited.
- **The MBC is the lowest antibiotic concentration showing no growth on subculture to media without the antibiotic.** In the example below, the MBC would be 4 $\mu\text{g/ml}$.
- Formulas are increasingly replacing this test.

Minimal Inhibitory Concentration = MICs

1. Each container has one concentration of a drug.
2. Each container has identical inoculum of the patient's bacterial isolate
3. Must run a no drug control.
4. Lowest concentration showing no visible growth = MIC
(in example, MIC = 2 $\mu\text{g/ml}$)



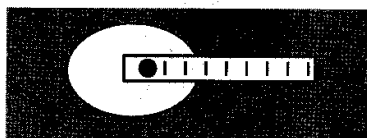
Minimal Bactericidal Concentration (MBC)

(Not routinely done in many hospitals but ordered when necessary)

The lowest drug concentration showing no growth on sub-culture to media without drugs = MBC
MBC in example would be 4 $\mu\text{g/ml}$.

E-Test (Agar Diffusion)

E-test uses a strip of plastic marked with a gradient unique to each antibiotic that has the dot of dried antibiotic on the underside. This is placed on an agar plate already swabbed with the patient's isolate and read after incubation. It produces a $\mu\text{g/ml}$ value that correlates fairly well with the Minimal Inhibitory Concentration.



Sterilization, Disinfection, Pasteurization

Sterilization: complete removal or killing of all viable organisms.

Disinfection: the removal or killing of disease-causing organisms. Compounds for use on skin: antiseptics.

Pasteurization: the rapid heating and cooling of milk designed to kill milk-borne pathogens such as *Mycobacterium bovis*, *Brucella*, and *Listeria*.

Physical Methods of Control

Heat = saturated steam

- Autoclaving (steam under pressure): 15 lbs pressure $\rightarrow$ 121°C 15–20 min (sterilizing)
- Dry heat—2 hr 180°C

Radiation

- UV: formation of thymine–thymine pairs on adjacent DNA bases

Filtration

- HEPA (High Efficiency Particulate Air) filters for air
- Nitrocellulose or other known pore-size filters
 - 0.45 μm filters out most bacteria except Mycoplasmas and other cell wall-less forms.
 - 0.22 μm will filter out all bacteria and spores.

Chemical Methods of Control

Agents damaging membrane

- **Detergents:** (surface active compounds) most notable the quaternary ammonium compounds like benzalkonium chloride—interact with membrane through hydrophobic end disrupting membrane.
- **Alcohols:** disrupt membrane and denature protein.
- **Phenols** and derivatives: damage membrane and denature proteins.

Agents modifying proteins

- **Chlorine:** oxidizing agent inactivating sulfhydryl-containing enzymes
- **Iodine and iodophors** (which have reduced toxicity): also oxidation of sulfhydryl-containing enzymes
- **Hydrogen peroxide:** oxidizing agent (sulfhydryl groups); catalase inactivates
- **Heavy metals:** (silver and mercury) bind to sulfhydryl groups inhibiting enzyme activity
- **Ethylene oxide:** alkylating agent (sterilizing agent)
- **Formaldehyde** and **glutaraldehyde:** denatures protein and nucleic acids and alkylates amino and hydroxyl groups on both

Modification of nucleic acids

- **Dyes:** like crystal violet and malachite green whose positively charged molecule binds to the negatively charged phosphate groups on the nucleic acids

Chapter Summary

The definitions of polymerases, nucleases, and alleles and the flow of genetic information are briefly reviewed.

Three types of DNA may be found in a bacterial cell: chromosomal, plasmid, and bacteriophage DNA.

The chromosomal DNA contains all the essential bacterial genes. Most bacteria have one chromosome but may have multiple copies of it. The chromosomal DNA exists as a large covalently closed circular strand, looped around a proteinaceous center, and contains about 2,000 genes. Each loop can be transcribed independently.

Plasmids are small, covalently closed circular DNAs that replicate autonomously and may be transferred from one bacterium to another. Episomes are a type of plasmid integrated into the bacterial chromosome. Plasmids may code for fertility factors, antibiotic resistance, and exotoxins.

Bacteriophage (viral) DNA may be inserted into the bacterial chromosome as a prophage by a temperate virus. Such an inserted viral gene sometimes directs the synthesis of a virulence factor, making the bacterium more pathogenic.

Transposons are elements of DNA that independently move from one site to another on a plasmid or chromosome. Insertion may cause mutations in unrelated genes and may cause repeats on both sides of the transposon. Transposons may carry genes for drug resistance and play an important role in the development of multiple drug resistance.

Homologous recombination is a process in which linear DNA introduced by transformation, conjugation, or transduction is exchanged for near-homologous DNA on the chromosome. This stabilizes the newly introduced DNA, which as a consequence has now become an integrated part of the bacterial genome.

Site-specific recombination is the integration of a nonhomologous circular DNA molecule into the bacterial genetic material.

Excision is the reversal of site-specific recombination in which an integrated prophage or plasmid may be removed from the bacterial genome.

(Continued)

Chapter Summary (continued)

New genetic information is introduced into bacterial cells by gene transfer followed by recombination. The modes of gene transfer are transformation, conjugation, and transduction.

Transformation is the incorporation of naked DNA from the environment. This process is probably of minor biologic import but was used to transform rough *Streptococcal* cells into smooth ones in the now classic experiments that confirmed the genetic role of DNA.

Conjugation is the direct transfer of DNA from one bacterial cell to another.

Conjugation occurs by direct contact mediated by a fertility (F) plasmid. Donor (male) cells are F^+ , and receptor (female) cells are F^- . A strand of the plasmid from the F^+ cell is transferred across the conjugal bridge formed by a sex pilus. The net result is that both cells end up with a complete copy of the plasmid.

A second type of conjugation occurs when a high-frequency recombination (Hfr) donor cell, which has its F factor incorporated into its chromosome, contacts an F^- recipient cell. In this case, the attempt to transfer the whole genome is aborted before completion, and only a portion of the genetic material is transferred and incorporated into the recipient's chromosome.

Transduction is the transfer of bacterial DNA by a bacterial virus (phage). Phages are either virulent (lytic) or temperate (lysogenic). Virulent phages always replicate and lyse their hosts. Temperate phages can repress their lytic behavior and stably integrate their DNA into the bacterial chromosome. If anything interferes with the repressive mechanisms, they revert to a lytic form. Generalized transduction occurs when a lytic phage lyses a bacterium and some bacterial DNA accidentally recombines with phage DNA and is subsequently transferred into another bacterial cell. Specialized transduction occurs when the excision of integrated phage DNA (a prophage) includes some chromosomal DNA. This sometimes occurs when a temperate virus is induced to become lytic by damage to the repressor system or infection with a "helper" virus.

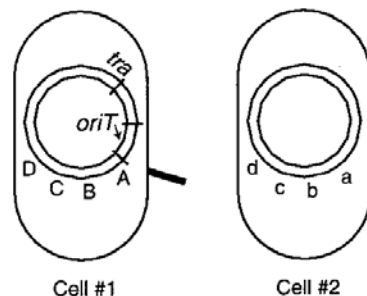
Drug-resistant bacteria have become a major medical problem. Drug resistance arises by three different mechanisms: intrinsic, chromosome-mediated, or plasmid-mediated. Intrinsic resistance is that which is normal for a species. Chromosome-mediated resistance is brought about by mutation, and plasmid-mediated resistance is transferred by conjugation. Transfer of plasmid DNA is the most important way of dispersing drug resistance among bacteria.

Antibiotic susceptibility may be tested by the Kirby-Bauer agar diffusion test, by testing for specific resistance causing enzymes, by the minimal inhibitory concentration test, by the minimal bactericidal test, or by the E (agar diffusion) test.

Pathogenic contamination may be controlled by sterilization or disinfection. Methods of control include heating (e.g., autoclaving, dry heat, or pasteurization); radiation, filtration, or chemicals that destroy the membrane (e.g., detergents, alcohols, or phenols); denaturing proteins (e.g., chlorine, iodine and iodophors, H_2O_2 , heavy metals, ethylene oxide, or formaldehyde); or modifying nucleic acids (dyes).

Review Questions

- What type of genetic material is created by repeated transpositional recombination events?
 - Chromosomal drug resistance genes
 - Genetic operon
 - Hfr chromosome
 - Insertion sequences
 - Multiple drug resistance plasmids
- Which genetic material is found in pathogenic *Corynebacterium diphtheriae* but not in nonpathogenic *C. diphtheriae*?
 - A diphthamide on EF-2
 - An episome
 - An F factor
 - An integrated temperate phage
 - Highly repetitive bacterial DNA
- How is a prophage created?
 - Through activation of the *recA* gene product of an exogenote
 - Through infection of a bacterial cell with a virulent bacteriophage
 - Through site-specific recombination of a temperate phage and bacterial DNA
 - Through infection of a bacterial cell with lambda phage, lacking the lambda repressor
 - Through excision of bacterial DNA and active lytic replication of a bacteriophage
- If one cell of type one (figure below) is mixed into a culture of 100 cells of type two (below), and culture conditions are optimized for conjugation BUT NOT for cell division, the cellular genotype that would predominate after overnight incubation would be that of
 - Cell #1
 - Cell #1 with new a, b, c, and d alleles
 - Cell #2 with new A, B, C, and D alleles
 - Cell #1 with a new a allele
 - Cell #2 with a new A allele
 - Cell #1 with new a and b alleles
 - Cell #1 with new A and B alleles



5. Assume the following cells have no other plasmids other than those mentioned. Which cell type would contain two molecules of DNA?
 - A. F^+
 - B. F^-
 - C. Hfr
6. Assume the cells whose genotype is listed below have no other plasmids than those indicated by the indicated genotype. Which bacterial cell is most likely to transfer chromosomal genes in linear order?
 - A. F^+
 - B. F^-
 - C. Hfr
7. What bacterial gene transfer process is most sensitive to extracellular nucleases?
 - A. Conjugation
 - B. Generalized transduction
 - C. Homologous recombination
 - D. Site-specific recombination
 - E. Specialized transduction
 - F. Transformation
8. Following specialized transduction, if any of the bacterial genes transferred in are to be stabilized, what process must occur?
 - A. Conjugation
 - B. Generalized transduction
 - C. Homologous recombination
 - D. Site-specific recombination
 - E. Specialized transduction
 - F. Transformation
9. The ability of a cell to bind DNA to its surface and import it is required for which genetic process?
 - A. Conjugation
 - B. Generalized transduction
 - C. Homologous recombination
 - D. Site-specific recombination
 - E. Specialized transduction
 - F. Transformation

10. Process by which bacterial or plasmid DNA may be mistakenly incorporated (during assembly) into one phage being produced by the lytic life cycle and then that DNA-transferred to another bacterial cell which may acquire some new genetic traits is called
 - A. Conjugation
 - B. Generalized transduction
 - C. Homologous recombination
 - D. Site-specific recombination
 - E. Specialized transduction
 - F. Transformation
11. Recombination is required for stabilization of genetic material newly transferred by all of the following processes EXCEPT
 - A. Movement of a transposon
 - B. Integration of a temperate bacteriophage
 - C. Transduction of a chromosomal gene
 - D. Conjugal transfer of an R factor
 - E. Transformation of a chromosomal gene
12. Lysogenic conversion
 - A. is a change in pathogenicity due to the presence of a prophage.
 - B. is the induction of a prophage to its virulent state.
 - C. is the conversion of a virulent phage into a temperate phage.
 - D. refers to the incorporation of a prophage into the chromosome.
 - E. is the immunity that a prophage confers on a bacterium.
13. Which of the following events is most likely due to bacterial transformation?
 - A. A formerly non-toxicogenic strain of *C. diphtheriae* becomes toxigenic.
 - B. A non-encapsulated strain of *Streptococcus pneumoniae* acquires a gene for capsule formation from an extract of an encapsulated strain.
 - C. A strain of *Neisseria gonorrhoeae* starts producing a β -lactamase encoded by a plasmid similar to a plasmid of another Gram-negative strain.
 - D. A gene for gentamicin resistance from an *E. coli* chromosome appears in the genome of a virulent bacteriophage that has infected it.
14. Which of the following mechanisms is most likely to be involved in multiple drug resistance transfer from one cell to another?
 - A. Specialized transduction of a chromosomal gene for drug resistance
 - B. Transformation of chromosomal genes
 - C. Transposition
 - D. Conjugation with one parent with a free plasmid carrying drug resistance
 - E. Conjugation with one parent with chromosomal drug resistance

15. Which of the following agents, if introduced into a growing culture of bacteria, would halt growth but, if then removed, would allow growth to resume?
 - A. Antiseptic
 - B. Bactericide
 - C. Bacteriostat
 - D. Disinfectant
 - E. Sterilizing Agent

Answers

1. **Answer: E.** Transposition or transpositional recombination is a form of site-specific recombination and is largely responsible for the creation of multiple drug resistant plasmids. Chromosomal drug resistance may arise by movement of a plasmid gene to the chromosome, but it is usually just a solitary gene and not a repetitive event. The Hfr chromosome arises through a single site-specific integration of a fertility factor with the bacterial chromosome.
2. **Answer: D.** This question is asking what carries the genetic code (or, more simply, **codes**) for diphtheria toxin, which must be some kind of DNA, which in turn means that the protein EF-2 can be immediately eliminated. The diphthamide on EF-2 is actually the substrate for the ADP-ribosylation done by the diphtheria toxin. Genes expressing the diphtheria toxin originally enter *C. diphtheriae* as part of the DNA of the temperate corynebacteriophage. Integration of this temperate phage results in a stable prophage, which directs the production of the diphtheria toxin.
3. **Answer: C.** Site-specific recombination of phage DNA into bacterial cell DNA by the process of lysogeny creates a prophage. The RecA gene product is necessary for homologous recombination with an exogenote but does not create a prophage. A virulent bacteriophage causes lysis of the host cell and not the production of prophage. The lambda phage is a temperate phage, which can cause lysogeny of infected cells, but the lambda repressor is necessary in such cases to prevent the lytic life cycle. **Choice E** might be the pathway a prophage may choose to reinitiate its bacteriophage lytic lifestyle, but it would not be a means to create a prophage.
4. **Answer: E.** This hypothetical condition describes the mixing of one Hfr cell with 100 F⁻ recipients. Over time, with no cell division occurring, the one Hfr cell would repeatedly conjugate with the F⁻ cells and transfer one strand of its chromosomal DNA in sequence, beginning with oriT and theoretically ending with the tra genes. The most frequently transferred bacterial genes also have the greatest likelihood of successful recombination; they are those closest to oriT; in this example, the A allele. The entire chromosome is so large that it is virtually never transferred in its entirety and thus, the tra genes would not be transferred. (Even if tra genes were transferred, oriT and tra genes have no homologous regions in the recipient cell chromosome and so would not successfully recombine within.) Thus, the recipient cell acquires only new chromosomal alleles and NOT the whole fertility factor and never changes phenotype to become an Hfr cell. Therefore, any of the answers with cell one (the Hfr parent) as the dominant type would be wrong.

The genes are transferred in linear order, so choice A will always be transferred more frequently than any of the later genes.

Therefore, given sufficient time for conjugation, the cell type that would be most numerous is that of the recipient genotype with a newly acquired allele close to oriT. This means

that the best answer is choice F: cell two with a new A gene. The farther from $oriT$ that the allele is, the less likely that it will be successfully transferred. The distractor, choice C, with all four alleles transferred in, is less likely.

5. **Answer: A.** The F^+ cell would contain both the bacterial chromosome and the fertility factor. The other two would just each have the bacterial chromosome (F^-) or the single DNA molecule of the chromosome with the integrated fertility factor.
6. **Answer: C.** Only F^+ and Hfr can donate genes to a recipient or F^- cell. The F^+ cell would transfer only plasmid genes. The Hfr would be the only one likely to transfer chromosomal genes.
7. **Answer: F.** The nucleic acid from the donor cell is not protected from the environment either by a cell or by a phage coat, but is instead naked and therefore subject to nucleases.
8. **Answer: C.** The DNA is transferred in as a linear piece and must be stabilized by homologous recombination.
9. **Answer: F.** The statement fits the definition of competency required for transformation.
10. **Answer: B.** This obviously is transduction, but what are your clues? First, it says "one phage" rather than all the phage in the cell (as for specialized). Then it also said plasmid DNA could be picked up. For specialized transduction, only episomal plasmid DNA (incorporated into the bacterial chromosome near an attachment site) could be picked up.
11. **Answer: D.** Transpositional movement actually involves a type of recombination called transposition that is a form of site-specific recombination. Site-specific recombination is also involved in integration of a temperate bacteriophage. Both transformation and transduction require homologous recombination as would transfer of Hfr DNA by conjugation. But either F factor or R factor DNA circularizes when it enters a new cell and thus is stable without recombination even as circular DNA is not subject to the cellular exonucleases.
12. **Answer: A.** D is a definition of lysogeny but lysogenic conversion is when lysogeny changes the characteristic of the lysogenized organism. In medicine this usually means an increased pathogenicity from the lysogeny.
13. **Answer: B.** A would require phage infection with a temperate coryneophage. B (the answer) is most likely to occur through transformation. C is most likely to take place through a conjugal transfer. D might occur by generalized transduction.
14. **Answer: D.** Multiple drug resistance is almost always plasmid-mediated, which rules out A, B, and E. Transposition is just generally within a cell moving a copy of the DNA to another molecule of DNA within the cell.
15. **Answer: C.** This is the classic description of a bacteriostatic agent.

Clinical Infectious Disease

7

These charts are designed for self-study after the organisms have all been reviewed in class. They represent the basics used in clinical scenarios on the USMLE.

Cover the last column on each chart and write the causative agent(s) on paper. Then think about how the organism causes disease. Is there a major virulence factor?

Abbreviations Used

→ means progressing on to	GU = genitourinary
~ means about or approximately	IC = immunocompromised
HIV+ = patient with known human immunodeficiency virus infection	Infl'd = inflamed
Can be used for anyone who is infected but often used for those who are HIV+ but do not have frank AIDS (in other words, CD4+ count >200)	Infl'n = inflammation
AIDS = acquired immunodeficiency syndrome (CD4+ count <200)	i.v. = intravenous
abd = abdominal	mo = month(s)
CF = cystic fibrosis	NF = normal flora
CMI = cell-mediated immunity	occ = occasional
CGD = chronic granulomatous disease	PMNs = polymorphonuclear cells
	pt = patient
	RBCs = red blood cells
	subQ = subcutaneous

If there are multiple causative agents, at the end of the description there may be a # with the abbreviation "CA." This means you should be able to list that number. If it specifically says "species," you should give species.

Table I-7-1. Diseases of Skin, Mucous Membranes, and Underlying Tissues

Type Infection	Case Vignette/Key Clues	Common Causative Agents
Furuncles, carbuncles	Neck, face, axillae, buttocks	<i>Staphylococcus aureus</i>
	Inflamed follicles from neck down	<i>Pseudomonas aeruginosa</i> (hot tub folliculitis)
Acne vulgaris	Inflammation of follicles and sebaceous glands	<i>Propionibacterium acnes</i>
Cutaneous lesions (may be from scratching mosquito bites, cats)	Initially vesicular; skin erosion; honey crusted lesions; catalase negative organism	<i>Streptococcus pyogenes</i>
	Initially vesicular but with longer lasting bullae; catalase positive organism	<i>Staphylococcus aureus</i>
Red raised butterfly facial rash	Dermal pain, edema, and rapid spread	<i>Streptococcus pyogenes</i> (Erysipelas)
Jaw area swelling with pain, sinus tract formation, yellow granules in exudate	Associated with carious teeth, dental extraction or trauma	<i>Actinomyces israelii</i> "lumpy jaw" (Actinomycosis)
Hot inflamed tissues	Deeper tissues from extension of skin lesions or wounds including surgical	Variety of bacteria: <i>S. aureus</i> , <i>S. pyogenes</i> , Gram - rods, <i>Clostridia</i> and anaerobes (cellulitis)
Vesicular lesions	Sometimes preceded by neurologic pain	Herpes
	Sometimes large	<i>Staphylococcus aureus</i>
SubQ granulomas/ulcers/cellulitis	Tropical fish enthusiasts; granulomatous lesion	<i>Mycobacterium marinum</i>
	Cellulitis following contact with saltwater or oysters	<i>Vibrio vulnificus</i>
	Solitary or lymphocutaneous lesions, rose gardeners or florists, sphagnum moss	<i>Sporothrix schenckii</i> (Rose Gardener's disease)
	Subcutaneous swelling (extremities, shoulders), sinus tract formation, granules; multiple CA	Bacteria: <i>Actinomyces</i> , <i>Nocardia</i> , Fungi: <i>Madurella</i> , <i>Pseudallescheria</i> (Mycetoma)
Malignant pustule	Pustule → dark red fluid-filled, tumor-like lesion → necrosis → black eschar surrounded by red margin	<i>Bacillus anthracis</i>
Enlargement from lymphatic blockage	Legs or genitalia with previous painless genital chancre	<i>Chlamydia trachomatis</i> L1-3
	Fever, headache, myalgia, inflammation and then lymphadenopathy and elephantiasis of limbs or genitalia	<i>Wuchereria</i> and <i>Brugia</i> ; mosquito spread
Burns, cellulitis	Blue-green pus, grape-like odor	<i>Pseudomonas aeruginosa</i>
Wounds	Surgical wounds (clean)	<i>Staphylococcus aureus</i>
	Surgical wounds (dirty)—list groups	<i>S. aureus</i> , <i>Enterobacteriaceae</i> , anaerobes
	Trauma - list groups	<i>Clostridium</i> , <i>Enterobacteriaceae</i> , <i>Pseudomonas</i>
	Animal bites	<i>Pasteurella multocida</i>
	Cat scratches resulting in lymphadenopathy with stellate granulomas	<i>Bartonella henselae</i>
	Shallow puncture wound through tennis shoe sole	<i>Pseudomonas aeruginosa</i>
Target lesion, generally with fever, headache	(Not necrotic) rashy border; bite site	<i>Borrelia burgdorferi</i>

Table I-7-2. Ear, Nose, Throat, Upper Respiratory System Infections

Type Infection	Case Vignette/Key Clues	Common Causative Agents
Acute otitis media	Red, bulging tympanic membrane, fever 102–103°; pain goes away if drum ruptures or if ear tubes are patent. -5CA	<i>Streptococcus pneumoniae</i> <i>H. influenzae</i> (often nontypeable, recurs) <i>Moraxella catarrhalis</i> RSV Rhinovirus
Otitis externa	Ear pain—list of organisms	Normal flora often involved Often mixed infections: <i>Staph aureus</i> (from NF)* <i>Candida albicans</i> (from NF) <i>Proteus</i> (water organism) <i>Pseudomonas</i> (water)
Malignant otitis externa	Severe ear pain in diabetic; life threatening	<i>Pseudomonas aeruginosa</i>
Sinusitis	Sinus pain; low-grade fever	As for acute otitis media
Oral cavity disease	Painful mouth—overgrowth of spirochetes and fusiform bacteria	<i>Fusobacterium</i> and treponemes (normal oral spirochetes)
	Sore mouth with thick white coating (painful red base under); increased risk: premature infants, AIDS, IC pts, pts on antibiotics, vitamin C deficiency	<i>Candida</i>
Sore throat	Inflamed tonsils/pharynx, which may be purulent and may develop abscesses; cervical lymphadenopathy, fever, $\pm$ stomach upset; $\pm$ sandpaper rash	<i>Streptococcus pyogenes</i> (Group A Strep) Rash indicates presence of erythrogenic exotoxin A
	White papules with red base on posterior palate and pharynx, fever	Coxsackie A
	Throat looking like Strep with severe fatigue, lymphadenopathy, fever $\pm$ rash	Epstein-Barr virus (Downey type II cells)
	Low-grade fever with a 1–2 day gradual onset of membranous nasopharyngitis and/or obstructive laryngotracheitis; bull neck from lymphadenopathy; elevated BUN; abnormal ECG; little change in WBC (toxin)	<i>Corynebacterium diphtheriae</i> (diphtheria)
Common cold	Rhinitis, sneezing, coughing; list CA with seasonal peaks	Rhinoviruses (Summer–Fall) Coronaviruses (Winter–Spring)

* NF = normal flora.

Table I-7-3. Eye Infections

Type Infection	Case Vignette/Key Clues	Common Causative Agents
Eyelid	Bilateral eye lid swelling, >10% eosinophilia, muscle pain; earlier GI Sx	<i>Trichinella</i>
	Stye; 2 CA	<i>Staphylococcus aureus</i> <i>Propionibacterium acnes</i>
	Unilateral inflammation at bite site often around eye or mouth; travel to Mexico, travel to Central or South America	<i>Trypanosoma cruzi</i>
Conjunctivitis neonate	Red itchy eye(s)/pus; onset 2–5 days	Bacterial pink eye <i>Neisseria gonorrhoeae</i>
	Red itchy eye(s)/pus; onset 5–10 days	<i>Chlamydia trachomatis</i> (serotype D–K U.S.) <i>Staphylococcus aureus</i>
Conjunctivitis	Neonate with “sticky eye”	<i>Staphylococcus aureus</i>
	Red itchy eye(s), thin exudate; pain, photophobia	Viral pink eye: adenovirus (more common than bacterial pink eye)
	Red eye, pus 3 CA	<i>S. aureus</i> Group A Strep, <i>Strep pneumoniae</i> (all Gram +) <i>Haemophilus influenzae</i> (<i>H. aegyptius</i>)
	Red eye, pus, presence of inclusion bodies in scrapings; CA with serotypes in U.S.	<i>Chlamydia trachomatis</i> serotypes D–K (inclusion conjunctivitis)
Chorioretinitis	Granulomas and intumed eye lashes, corneal scarring, blindness; CA with serotypes	<i>Chlamydia trachomatis</i> serotypes A, B, Ba, C (trachoma)
	Neonate or AIDS; 2 CA	<i>Toxoplasma</i> , CMV
Retinopathy with keratitis in baby	Mom i.v. drug abuser	<i>Treponema pallidum</i> (congenital syphilis)

Table I-7-4. Cardiac Symptoms

Chills, fever, arthralgia, myalgia, back pain, acutely ill, Janeway lesions; emboli	Developing a heart murmur; i.v. drug user	<i>Staphylococcus aureus</i>
	Not i.v. drug user	<i>Staphylococcus aureus</i>
Fever with vague symptoms with insidious onset, fatigue, weakness, weight loss, night sweats, anorexia, myalgias; murmur may have been long present; emboli	Poor oral hygiene or dental work	Viridians streptococci (55% of cases in native hearts)
	Biliary or urinary tract infection GU manipulation in elderly men	<i>Enterococcus faecalis</i>
Endocarditis in i.v. drug user		<i>Staphylococcus aureus</i> Viridians streptococci <i>Staph. epidermidis</i> <i>Aspergillus</i> (branching <45°) <i>Candida</i> (pseudohyphae) <i>Pseudomonas</i>

Table I-7-5. Middle and Lower Respiratory System Infections

Type Infection	Case Vignette/Key Clues	Most Common Causative Agents
Respiratory difficulty or obstruction	Inflamed epiglottitis; patient often 2–3 and unvaccinated	<i>Haemophilus influenzae</i> (epiglottitis)
	Infant with fever, sharp barking cough, inspiratory stridor, hoarse phonation	Parainfluenza virus (Croup)
Laryngotracheitis laryngotracheobronchitis		Viral etiology
Bronchitis	Wheezy; infant or child ≤ 5 years	RSV
	>5 years	<i>Haemophilus influenzae</i> , <i>Mycoplasma pneumoniae</i> , <i>Chlamydia pneumoniae</i>
	With cough >2 weeks, afebrile; >9	<i>Bordetella pertussis</i>
Pneumonia	Poorly nourished, unvaccinated baby/child; giant cell pneumonia with hemorrhagic rash	Measles: malnourishment $\uparrow$ risk of pneumonia and blindness
	Adults (including alcoholics) Lobar pneumonia or less commonly, bronchopneumonia	<i>Streptococcus pneumoniae</i> (rusty sputum)
	Neutropenic pts, burn patients, CGD, CF	<i>Pseudomonas</i>
	Pneumonia teens/young adults; bad hacking cough; initially non-productive cough	<i>Mycoplasma pneumoniae</i> (most common cause of pneumonia in school age children)
	Atypical with air conditioning exposure especially >50 yr, heavy smoker, drinker	<i>Legionella</i> spp.
	Atypical with bird exposure $\pm$ hepatitis	<i>Chlamydia psittaci</i>
	Foul smelling sputum, aspiration possible	anaerobes, mixed infection (<i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Peptococcus</i>)
	Alcoholic, abscess formation, aspiration, facultative anaerobic, Gram-negative bacterium with huge capsule	<i>Klebsiella pneumoniae</i> (currant jelly sputum)
	AIDS patients with staccato cough; “ground glass” x-ray; biopsy: honeycomb exudate with silver staining cysts	<i>Pneumocystis carinii</i>
Pneumonia with influenza	Primary infection	Influenza virus pneumonia
	Secondary	<i>Streptococcus pneumoniae</i>
Acute pneumonia or chronic cough with weight loss, night sweats	Over 55, HIV+, or immigrant from developing country	<i>Mycobacterium tuberculosis</i>
	Dusty environment with bird or bat fecal contamination (Missouri chicken farmers)	<i>Histoplasma capsulatum</i>
	Desert sand SW U.S.A.	<i>Coccidioides immitis</i>
	Rotting contaminated wood	<i>Blastomyces dermatitidis</i>

Table 1-7-6. Genitourinary Tract Infections

Type Infection	Case Vignette/Key Clues	Most Common Causative Agents
Urethritis	Gram-negative diplococci in PMNs in urethral exudate	<i>Neisseria gonorrhoeae</i>
	Culture negative, inclusion bodies	<i>Chlamydia trachomatis</i>
	Urease positive, no cell wall	<i>Ureaplasma urealyticum</i>
	Flagellated protozoan with corkscrew motility	<i>Trichomonas vaginalis</i>
	Frequent and painful urination, hematuria, and fever	(Cystitis) #1 <i>E. coli</i> , other Gram-negative enterics, <i>Pseudomonas</i> , <i>Proteus</i> <i>Staphylococcus saprophyticus</i>
Cervicitis	Young, newly sexually active individual; Gram-positive cocci	(pyelonephritis) <i>E. coli</i> , <i>Staphylococcus</i>
	As above with flank pain and prominent fever	
	Friable, inflamed cervix with mucopurulent discharge; probes or culture to distinguish	<i>Neisseria gonorrhoeae</i> (Gram-negative diplococci) <i>Chlamydia trachomatis</i> (non-staining obligate intracellular parasite) Herpes simplex (virus)
Vaginal itching, pain, discharge odor	Adherent yellowish discharge, pH >5, fishy amine odor in KOH, clue cells; Gram-negative cells dominate	(Bacterial vaginosis) overgrowth of <i>Gardnerella vaginalis</i> and anaerobes
	Vulvovaginitis, pruritis, erythema discharge: consistency of cottage cheese	<i>Candida</i> spp.
	Foamy, purulent discharge, many PMNs and motile trophozoites microscopically (corkscrew motility)	<i>Trichomonas vaginalis</i>
Genital lesions	Genital warts	Human papilloma virus (most common U.S. STD), <i>Treponema pallidum</i>
	Multiple painful vesicular, coalescing, recurring	Herpes
	Nontender ulcer healing spontaneously 2–10 weeks	<i>Treponema pallidum</i>
	Non-indurated, painful ulcer, suppurative with adenopathy, slow to heal	<i>Haemophilus ducreyi</i>
	Initial papule heals; lymph nodes enlarge and develop fistulas; genital elephantiasis may develop	<i>Chlamydia trachomatis</i> L1-3

Diarrhea

Dysentery

- Abdominal cramps, tenesmus, and pus and blood in the stool
- Usually associated with invasive bacterial disease in the colon

Diarrhea

- Refers to profuse watery feces
- Most commonly associated with increased secretion of fluid across the mucosal surfaces of the small intestine in response to a toxin or a viral infection
- No inflammatory cells

Table I-7-7. Diarrhea by Intoxication

Most Common Sources	Common Age Group Infected	Incubation Period	Pathogenesis	Symptoms	Duration of Symptoms	Organism
Ham, potato salad, cream pastries	All	1–6 hours	Heat stable enterotoxin is produced in food contaminated by food handler with skin lesion; food sits at room temperature	abd. cramps, vomiting, diarrhea; sweating and headache may occur, no fever	<24 hours	<i>Staphylococcus aureus</i>
Rice	All	<6 hours	Heat stable toxin causes vomiting		8–10 hours	<i>Bacillus cereus</i> : emetic form
Meat, vegetables	All	>6 hours	Heat labile toxin causes diarrhea (similar to <i>E. coli</i> LT)	Nausea, abd cramps, diarrhea	20–36 hrs	<i>Bacillus cereus</i> : diarrheal form

Table 1-7-8. Microbial Diarrhea: Organisms Causing Noninflammatory Diarrhea

Most Common Sources	Common Age Group Infected	Incubation Period	Pathogenesis	Symptoms	Duration of Symptoms	Organism
Day care, water, nosocomial, fecal-oral	Infants and toddlers, some older	1-3 days (fall, winter, spring)	Microvilli of small intestine blunted; mononuclear infiltrate in lamina propria; disaccharidase activity down; glucose coupled transport normal; lactose intolerance may cause build up and osmotic influx creating watery diarrhea	Noninflammatory watery diarrhea, vomiting, fever, and dehydration	5-7 days	Rotaviruses
Water, food, fecal-oral	Older kids and adults	18-48 hours	Jejunal biopsy shows blunting of microvilli; cytoplasmic vacuolization is seen along with mononuclear infiltrates of tissue; virus appears to decrease brush border enzymes causing malabsorption.	Diarrhea, nausea, and vomiting; fever in some	12-48 hours	Norwalk virus
Nosocomial	Young kids, IC	7-8 days	?	Diarrhea, fever, and vomiting	8-12 days	Adenovirus 40/41
Beef, poultry, gravies Mexican food	All	8-24 hours	Enterotoxin	abd cramps and watery diarrhea, rarely fever or vomiting	<24 hours	<i>Clostridium perfringens</i>
Water, food, fecal-oral	All ages	9-72 hours	Toxin stimulates adenylate cyclase and causes increase in cAMP in the small intestine without inflammation or invasion.	Profuse watery diarrhea with vomiting; fever may be present (rice water stools)	3-4 days	<i>Vibrio cholera</i>
Raw or undercooked shellfish prominent	Anyone eating raw shellfish	5-92 hours	Self-limited gastroenteritis mimicking cholera; there is a severe, rarer dysentery form, no clear enterotoxin; hemolysins, phospholipase and lysophospholipase; tests for invasiveness are negative.	Explosive watery diarrhea along with headache, abdominal cramps, nausea, vomiting, and fever.	Up to 10 days	<i>Vibrio parahaemolyticus</i>
Water, uncooked fruits and vegetables	All ages	12-72 hours	Heat labile toxin (LT) stimulates adenylate cyclase resulting in efflux of water and ions into the small intestine stable toxin stimulates guanylate cyclase	Watery diarrhea with some vomiting and sometimes fever	3-5 days	Enterotoxigenic <i>E. coli</i>
Food, water, fecal-oral	Infants in developing countries	2-6 days	Adherence to enterocytes through pili causes damage to adjoining microvilli.	Watery to profusely watery diarrhea	1-3 weeks	Enteropathogenic <i>E. coli</i>
Food, fecal-oral (hamburger)	50% <10 yrs., all	3-5 days	Verotoxin, which is a cytotoxin, causes bloody diarrhea with no invasion of the organism.	Abdominal cramps, watery diarrhea with blood (no fever)	7-10 days	Enterohemorrhagic <i>E. coli</i>
Water, day care, camping, beavers, dogs, etc.	All, children	5-25 days	Cysts ingested; excyst in the duodenum and jejunum; multiply and attach to intestinal villi by sucking disk.	Loose, pale, greasy diarrhea; mild to severe malabsorption syndrome	1-2 weeks to years	<i>Giardia lamblia</i>
Day care, fecal-oral, animals, homosexuals	Children, AIDS patients	2-4 weeks	Sporozoites attach to the epithelial surface of the intestine and replicate.	Mild diarrhea in immunocompetent; severe chronic diarrhea in AIDS	4 days to 3 weeks in AIDS; indefinite	<i>Cryptosporidium parvum</i>

Table 1-7-9. Microbial Diarrhea: Organisms Causing Inflammatory Diarrhea/Dysentery

Most Common Sources*	Common Age Group Infected	Incubation Period	Pathogenesis	Symptoms	Duration of Symptoms	Organism
Poultry, domestic animals, water, unpasteurized milk, day care, fecal-oral	All, especially <1 year and young adults	3-5 days	Multiply in the small intestine; invades epithelium resulting in inflammation and RBC and WBC in stools.	Diarrhea, abd pain, malaise enteritis with diarrhea, malaise, fever	1-2 days mild; <1 week normal self-limiting	<i>Campylobacter jejuni</i>
Poultry, domestic animals, water, day care, fecal-oral	All, especially infants and kids	8-48 hours	Adsoib to epithelial cells in terminal small intestine; penetrate to lamina propria of ileocecal region causing PMN response and PG response, which stimulates CAMP and watery diarrhea.	Diarrhea (occ bloody), abdominal cramps, abd tenderness, fever, and nausea w/occ vomiting osteomyelitis in sickle cell anemia	3-5 days; spontaneous resolution	<i>Salmonella</i> gastroenteritis
Water, day care, no animal reservoirs, fecal-oral	All, esp 6 mo to 10 yr.	1-7 days	<i>Shigella</i> colonize the small intestine producing at first an enterotoxin-induced watery diarrhea; ultimately the <i>Shigellae</i> penetrate the colon mucosa producing shallow mucosal ulcerations and dysentery; septicemia rare.	Watery diarrhea at first → lower abdominal cramps, tenesmus and abundant pus and blood in the stools (dysentery)	4-7 days; antibiotics can reduce spread	<i>Shigella</i>
Milk, wild domestic animals water, fecal-oral	All, esp older kids and young adults	2-7 days	The terminal ileum is infected with enlargement of the mesenteric lymph nodes; produces focal necrosis difficult to distinguish from appendicitis; organism is able to grow in cold; produces heat insensitive enterotoxin. Arthritis may occur.	Fever, diarrhea (frequently with leukocytes & blood in stools), abdominal pain; also a noninflammatory gastroenteritis	1 day - 3 weeks (avg. 9 days)	<i>Yersinia enterocolitica</i>
Associated with antibiotic use	Pt on antibiotics	NA	Intense inflammatory response creates the friable yellow plaque-like colonic lesions (pseudomembrane) associated with this disease	Mild diarrhea to severe colitis; abdominal cramps; spiking fever, systemic toxicity; blood, mucus, and pus in stools	Until antibiotic stopped; treat with metronidazole	<i>Clostridium difficile</i>
Food, water, fecal-oral	Adults	2-3 days	Similar to <i>Shigella</i> dysentery	Fever and cramps with blood and pus in the stools	1-2 weeks; fluid and electrolyte replacement	Enteroinvasive <i>E. coli</i>
Food, water, fecal-oral, tropical generally	All	2-4 weeks	Ingested cysts survive (trophozoites die) and multiply in the colon with invasion of the colon wall producing the characteristic flask-like lesions and extra-intestinal abscesses.	Gen. acute diarrhea with cramping; sometimes dysentery; ulceration of colon may produce peritonitis	Weeks to months Rx with metronidazole followed by iodoquinol	<i>Entamoeba histolytica</i>

Abbreviations: abd = abdominal; esp = especially; occ = occasional.

*Sources: water = those listed are the most common diarrhea diseases spread through water.

Day care = organisms listed are ones which have caused outbreaks in day care facilities, but note that any organism spread by the oral-fecal route may be a problem in this setting.

Milk = unpasteurized milk or dairy products.

Table 1-7-10. Other Gastrointestinal or Liver Infections

Signs and Symptoms	Case Vignette/Key Clues	Most Common Causative Agents
Jaundice, anorexia, nausea, right upper quadrant pain on palpation, cigarettes taste foul, elevated liver enzymes*	Food-borne (possibly contaminated raw oysters or clams); 14-45 days; without chronicity; sturdy naked RNA virus	Hepatitis A ("infectious" hepatitis) (picornavirus)
	i.v. drug abuse, needle stick; chronic carrier state, cirrhosis, primary hepatocellular carcinoma; DNA virus easily inactivated by alcohol	Hepatitis B ("serum" hepatitis) neonatal transmission (Hepadnavirus)
	Transfusion or i.v. drug abuse; acute illness is less severe than hepatitis B but chronicity is higher, with 60% of those infected having chronic active hepatitis; RNA, enveloped virus	Hepatitis C (Flavivirus)
Female with lower abdominal pain; onset often following menses	Enterically transmitted with high fatality in pregnant women, no chronic form	Hepatitis E (Calicivirus)
Acute abdominal pain	Adnexal tenderness, bleeding, deep dyspareunia, vaginal discharge, ± fever. Tenderness from cervical movement, possibly palpable inflammatory mass on bimanual exam	<i>Neisseria gonorrhoeae</i> or <i>Chlamydia trachomatis</i> or both or a variety of other organisms
	Intestinal blockage	<i>Ascaris lumbricoides</i> or potentially <i>Diphyllobothrium latum</i>
Bile duct blockage following surgery (anesthesia)		<i>Ascaris lumbricoides</i>
Peritonitis		Mixed flora often involving anaerobic normal flora: <i>Bacteroides fragilis</i> and facultative anaerobes such as <i>E. coli</i>
Cirrhosis	Travel history: Puerto Rico, Peace Corps, etc.; egg granulomas block triads → fibrosis	<i>Schistosoma mansoni</i>
Pancreatitis	i.v. drug use Generally with swelling of salivary glands	Hepatitis viruses Mumps virus

*Hepatitis may also occur with two other viruses: CMV and yellow fever virus or with toxoplasmosis or leptospirosis.

Table I-7-11. Changes in Blood Cells

Symptoms and Signs	Case Vignette/Key Clues	Most Common Causative Agents
Anemia	Megaloblastic	<i>Diphyllobothrium latum</i>
	Normocytic	Chronic infections
	Microcytic and hypochromic (iron deficiency anemia)	<i>Ancylostoma</i> , <i>Necator</i> , <i>Trichuris</i>
Patient with cyclic or irregular fever, decreased hemoglobin and hematocrit	Often foreign travel to tropics, schizonts in RBCs	<i>Plasmodium</i>
Reduced CD4 cell count		HIV
Increases in PMNs		Generally found in many extracellular bacterial infections
Increases in eosinophils		Allergy
		Helminths during migrations
Increases in mononuclear leukocytes (monocytes or lymphocytes)		Viral intracellular organisms <i>Listeria</i> , <i>Legionella</i> , <i>Leishmania</i> , <i>Toxoplasma</i> , <i>Pneumocystis</i>
Increases in lymphocytes (mononucleosis) Fever, fatigue, lymphadenopathy, myalgia, headache	Infectious mononucleosis Heterophile + Downey type II cells (reactive T cells) sore throat, lymphadenopathy, young adult	Epstein-Barr virus (EBV)
	Heterophile negative	CMV <i>Toxoplasma</i> <i>Listeria</i> (Listeriosis)
Lymphocytosis with hacking cough	Unvaccinated child hypoglycemic	<i>Bordetella pertussis</i>

Table 1-7-12. Central Nervous System Infections

Signs and Symptoms	Case Vignette/Key Clues	Most Common Causative Agents
Sepsis ± seizures, irritability, or lethargy; rarely bulging fontanelles or nuchal rigidity	Neonate to 2 months	<i>Streptococcus agalactiae</i> #1 (Gram-positive coccus) <i>E. coli</i> (Gram-negative rod) More rarely: <i>Listeria monocytogenes</i> (motile Gram-positive rod)
Headache, fever, confusion, lethargy, nuchal rigidity, vomiting	6 months to 2 years; no mention of <i>Haemophilus</i> vaccine; no indication that child is properly vaccinated	<i>Haemophilus influenzae</i> type B* (Gram-negative pleomorphic rod with polyribitol capsule)
	3 mo to young adult prodrome may be very rapid; child may be properly vaccinated; rash	<i>Neisseria meningitidis</i> (Gram-negative diplococcus with capsule; ferments maltose)
	<2 yrs Young adults to elderly	<i>Streptococcus pneumoniae</i> (Gram-positive coccus, catalase negative, alpha hemolytic, inhibited by optochin, lysed by bile)
	Renal transplant patient	<i>Listeria monocytogenes</i> (motile Gram-positive rod)
As above but less toxic and a more gradual onset (several days)		Viral: Enteroviruses (~70%): Coxsackie B, echovirus, poliovirus, Coxsackie A. Summer and fall but sporadically all year Mumps virus (now rare with vaccine) winter and spring most cases imported from Lymphocytic choriomeningitis (exposure to rodents, e.g., hamsters); most cases imported from S. America Herpes simplex type 2 or Varicella-Zoster
Several month prodrome (except in severely compromised) with signs of meningitis	Usually some underlying condition	Fungal, e.g., Cryptococcal, or if in Southwestern U.S.: <i>Coccidioides</i> If near U.S. great river beds with exposure to bird, bat feces: <i>Histoplasma capsulatum</i>
Prefrontal headache, high fever, disturbance of smell	Swimming and often diving in very warm polluted waters	<i>Naegleria</i>
Bell's palsy	Systemic disease	<i>Borrelia burgdorferi</i>
Guillain-Barré	With GI tract problems	<i>Campylobacter jejuni</i>
	With respiratory problems	Influenza

*By 1990, with day care centers and the dramatic increase in *Haemophilus meningitis*, *Haemophilus meningitis* became overall the most common. Since late 1990, when the conjugated vaccine went into use, there has been a dramatic decrease in *Haemophilus meningitis* in vaccinated kids.

(Continued)

Table 1-7-12. Central Nervous System Infections (*continued*)

Signs and Symptoms	Case Vignette/Key Clues	Most Common Causative Agents
Headache, and fever $\pm$ drowsiness, coma, hemiplegia, cranial nerve palsy, hallucinations, behavioral disturbances, and other focal neurological findings	Summer-fall, mosquito-borne from bird reservoirs (except for California encephalitis, which is a rodent reservoir)	Encephalitis with arboviruses: Western equine encephalitis (midwest and west U.S.) St. Louis encephalitis elderly most severe infections California encephalitis entire U.S. Eastern equine encephalitis all age groups but most common in young and old, highest morbidity of viral CNS infections; with mental retardation, seizures, personality changes in survivors
Headache, behavioral changes, lethargy, somnolence $\rightarrow$ focal deficit (especially frontal temporal lobe involvement)	Focal uptake of radionuclide, RBCs in CSF	Herpes simplex encephalitis (treatable)
Nerve palsies in a patient with tuberculosis and a low CSF glucose	Patient with low CMI	Tuberculous meningitis
Meningoencephalitis in immunocompromised patients		<i>Acanthamoeba</i> or <i>Toxoplasma</i>
Mass lesion (symptoms dependent on location of mass) and elevated intracranial pressure along with headache, mental changes, nausea, vomiting, fever with chills, and seizure	Generally following: sinus, ear, or dental infection, infection at distant site, head trauma, etc.	Don't do lumbar puncture; CT generally shows ring enhancing lesion; 45% mixed infections; Streptococci and <i>Bacteroides</i> are the two most commonly identified groups of bacteria
Child following a viral illness with pernicious vomiting, lethargy and irritability, which may lead to brain swelling	Perhaps indication of aspirin usage, though the linkage is not definitive	Influenza or varicella infection (Reye's syndrome)

Table I-7-13. Cerebrospinal Fluid Finding in Meningitis

Pressure	CSF Appearance	Cell Count (cells/mm ³)	Dominant Cell Type	Glucose mg/dL	Protein mg/dL	Condition
<100 mm H ₂ O	Clear	0-5	Lymphocytes	40-70	<40	Normal
Normal or +	Clear	0-500	Early: PMNs Late: lymphocytes	Normal or -	Normal or +	Viral infection
++	Opaque	1-60,000	PMNs	-	++	Bacterial infection
+	Clear	10-500	Early: PMNs Late: lymphocytes	-	+ to ++	Fungal infection

- Below normal range, + above normal range

Table I-7-14. Selected Rashes

Type Rash	Progression	Other Symptoms	Disease	Causative Agent/Toxin
Erythematous maculopapular rash (sandpaper-like rash)	Trunk and neck → extremities	Sore throat, fever, nausea	Scarlet fever	<i>Strep. pyogenes</i> Exotoxin A-C
Diffuse erythematous, macular sunburn-like rash	Trunk and neck → extremities with desquamation on palms and soles	Acute onset, fever >102°F, myalgia, pharyngitis, vomiting, diarrhea, hypotension leading to multi-organ failure	Toxic shock syndrome	<i>Staph. aureus</i> TSST-1
Perioral erythema, bullae, vesicles, desquamation	Trunk and neck → extremities, except tongue and palate; large bullae and vesicles precede desquamation	Abscess or some site of infection	Staphylococcal skin disease: scalded skin disease & scarletina	<i>Staph. aureus</i> Exfoliatin
Petechiae → purpura	Trunk → extremities; spares palms, soles, and face	Fever, rash, headache, myalgias, and respiratory symptoms	Epidemic typhus	<i>Rickettsia prowazekii</i> ? endotoxin
Petechiae → purpura	Ankles and wrists → generalized with palms and soles	Fever, rash, headache, myalgias, and respiratory symptoms	Rocky Mountain spotted fever (most common on East Coast)	<i>Rickettsia rickettsii</i> ? endotoxin
Petechiae → purpura	Generalized (all over)	Abrupt onset, fever, chills, malaise, prostration, exanthem → shock	Early meningococemia	<i>N. meningitidis</i> endotoxin
Skin: maculopapular; mucous membrane: condyloma	Generalized involving the palms and soles	Fever, lymphadenopathy, malaise, sore throat, splenomegaly, headache, arthralgias	Secondary syphilis	<i>Trep. pallidum</i> endotoxin
Confluent erythematous maculopapular rash	Head → entire body	Cough coryza, conjunctivitis, and fever (prodrome), oral lesions, exanthem, broncho-pneumonia and ear infections	Measles	Rubeola virus Rash from T cell destruction of virus-infected cells in capillaries

Table I-7-15. Osteomyelitis

Type Infection	Case Vignette/Key Clues	Most Common Causative Agents
Fever, bone pain with erythema and swelling, some patients (diabetic particularly, may have associated cellulitis)	Adults, children, and infants without major trauma or special conditions	<i>Staphylococcus aureus</i>
	Neonates (<1 mo)	<i>Staphylococcus aureus</i> Group B <i>Streptococcus</i> Gram-negative rods (<i>E. coli</i> , <i>Klebsiella</i> , <i>Proteus</i> , <i>Pseudomonas</i>)
	Sickle Cell Anemia*	<i>Salmonella</i>
	Trauma	<i>Pseudomonas</i>

* Sickle cell anemia patients are functionally asplenic and may have defective opsonic and alternate complement pathway activities. The most common bacterial infections include

- Encapsulated organisms
Streptococcus pneumoniae
Haemophilus influenza type b
Neisseria meningitidis
Salmonella enteritidis
- Osteomyelitis due to *Salmonella* sp
- Pneumonia, bacteremia, and meningitis are all a problem.

Table I-7-16. Arthritis Related to Infections

Type Infection	Case Vignette/Key Clues	Most Common Causative Agents
Pain, redness, low-grade fever, tenderness, swelling, reduced joint mobility	#1 overall except in the 15–40 age group where gonococcal is more prevalent	<i>Staphylococcus aureus</i>
	Multiple joints	From septicemia, e.g., staphylococci, gonococcal
	15–40 years; mono- or polyarticular	Gonococcal arthritis
	Prosthetic joint	Coagulase negative staphylococci
	Viral	Rubella and hepatitis B parvovirus
	Chronic onset, monoarticular	<i>M. tuberculosis</i> or fungal
	Large joint resembling Reiter's following tick bite or erythema migrans	<i>Borrelia burgdorferi</i>
Postinfectious (Reiter's)	Following gastrointestinal infection	<i>Salmonella</i> , <i>Shigella</i> , <i>Campylobacter</i> , or <i>Yersinia enterocolitica</i>

Chapter Summary

This is a self-study chapter consisting of 16 tables organized by organ system. Each table describes the major infectious disease associated with that organ. The information presented has particular relevance to the USMLE. Following these tables are 23 (A–Z) USMLE-style clinical vignettes with open-ended questions and answers. (Please note: This section is not reviewed on video.)

CASE HISTORIES

Case A: A 28-year-old known alcoholic man presents with fever and productive cough. He was basically well until 3 days ago when he noticed perspiration, cough, shaking, chills, and headache. His cough has been associated with the production of a yellowish-green sputum, which occasionally was tinged with brownish streaks, but was not foul smelling. A Gram stain shows Gram-positive cocci in pairs and short chains.

- A. What laboratory tests could you use to identify the genus?

Answer: Catalase test (negative) to genus.

- B. When plated on blood agar, what other bacterium might you isolate and confuse the causative agent with, and why? What test(s) could distinguish the two?

Answer: Viridans Strep; optochin and bile.

- C. What procedure would you perform to type the isolate?

Answer: Quellung reaction with known antibodies to capsule (not antibodies to cell-wall antigens).

Case B: The patient in Case A developed meningitis and died.

- A. What would be the expected CSF cell count?

Answer: High.

- B. What would be the expected CSF protein and sugar values?

Answer: Protein high, sugar low.

Case C: A 60-year-old male presents with ulceration of the fingers and sensory loss.

- A. What is the status of his CMI?

Answer: Leonine facial appearance and the multitude of lesions is strongly suggestive of lepromatous leprosy, so CMI is depressed and lepromin test is negative.

- B. Does the organism grow intracellularly or extracellularly?

Answer: Intracellularly-obligate pathogen.

Case D: A patient presents with multiple, crusted and oozing, honey-colored lesions.

A. What is the skin infection?

Answer: Impetigo.

B. What two bacteria would you expect to isolate on culture?

Answer: Strep pyogenes (often honey-colored crusted) and/or Staph aureus (often longer-lasting vesicular or with bullae).

C. How would you separate the two?

Answer: Catalase test positive for Staphylococcus, negative for Streptococcus

Case E: A patient had intermittent bouts of general malaise, fever with weight loss, and progressive anemia. She presents also with a heart murmur.

A. What additional physical sign might occur and what causes it?

Answer: Splinter hemorrhages as seen in subacute bacterial endocarditis (also seen in trichinosis and sometimes without either of these diseases).

B. What is her underlying condition and the most commonly involved bacteria?

Answer: Damaged heart valve; Viridans streptococci (associated with bad oral hygiene or dental work) or Enterococcus faecalis or E. faecium if she has had bowel surgery.

C. How would you distinguish the colony on blood agar?

Answer: Alpha hemolytic not inhibited by optochin; a viridans streptococcus.

Case F: A mother brings her 2-year-old son to the emergency room because of fever and a stiff neck. Examination reveals an acutely ill child with a temperature of 104°F. CSF is Gram stained, examined in a rapid test, and also cultured. A Gram stain shows pleomorphic, gram-negative rods.

A. In an unvaccinated child, what laboratory test could confirm the identity of the isolate?

Answer: Meningitis screen, a series of immunologic rapid identification tests (usually EIAs using known antibodies), followed by growth of CSF sediment or filtrate on special media and drug susceptibilities.

B. What growth factors are required to grow the isolate on blood agar?

Answer: X = hemin and V = NAD.

Chocolate agar provides both X and V.

C. What is the drug of choice?

Answer: Cefotaxime or Ceftriaxone.

D. What part of his routine health care is he likely missing?

Answer: His Haemophilus vaccination and perhaps others.

Case G: A 23-year-old woman presents with lower back pain, fever, and dysuria of 3 days' duration. Urinalysis reveals many white blood cells (WBC) and WBC casts. Gram stain of the uncentrifuged urine reveals numerous Gram-negative bacilli per oil immersion field. On culture, extremely motile bacteria form waves of confluent growth.

A. What is the most important biochemical characteristic of this organism? Why?

Answer: Urease producing Enterobacteriaceae; kidney stones induced.

Case H: A child developed a unilateral mucopurulent conjunctivitis 10 days after birth. A conjunctival specimen was sent to the laboratory and inoculated into tissue culture cells. Iodine-staining inclusion bodies were produced.

A. What is unusual about the chemical makeup of the organism?

Answer: ATP defective mutant, also muramic acid missing from peptidoglycan.

B. What are the two forms of the organism?

Answer: Elementary bodies (extracellular) and reticulate bodies = replicating forms.

C. What would you see on Gram stain?

Answer: Nothing in the cells—poorly Gram staining.

D. What serologic type caused the child's problem?

Answer: If U.S. kid, serotype D-K.

Case I: A 27-year-old attorney is hospitalized. He was in excellent health until two days earlier when he noted malaise, fatigability, and profound anorexia. He remembers approximately 6–8 weeks ago receiving a tattoo while vacationing in the Caribbean.

A. How would you confirm your clinical diagnosis?

Answer: HBsAg and IgM to HbcAg.

B. What is meant by the "window"?

Answer: A time period between the end of the detectable presence of HBsAg and the beginning of the production of antibody to HbsAg.

Hbc antibody and HBeAg are present.

C. What antigen's presence in the past 6 months is indicative that the patient is entering a carrier state?

Answer: HBsAg past 6 months.

D. What antigen correlates with viral production?

Answer: HbeAg.

E. Does the virus carry a virion associated polymerase? If so, what kind?

Answer: Yes, RNA-dependent DNA polymerase (Hepatitis B replicates through an RNA intermediate.)

Case J: A young man became ill with a sore throat and swollen tonsils, marked fatigue, cervical adenopathy, a palpable spleen, and a pruritic erythematous rash that started after self-administration of ampicillin.

- A. What is the most likely disease? What are the most common laboratory diagnostic tests? What does the antibody test measure?

Answer: Infectious mononucleosis; monospot test (measures heterophile antibody which is not specific to EBV antigen) plus CBC.

- B. What type of cells are the Downey type II cells?

Answer: T lymphocytes. (Reactive cells not infected.)

- C. What cells does the virus infect? Through what receptor does the lymphocytic infection begin?

Answer: EBV infects epithelia cells and B lymphocytes, whose receptor is CD21 = CR2.

Case K: A 35-year-old woman presents with a unilateral vesicular rash.

- A. The most likely diagnosis is

Answer: Shingles.

- B. Describe the virion's nucleic acid.

Answer: Linear dsDNA.

- C. Patient had a previous history of what other disease?

Answer: Chickenpox.

Case L: A 27-year-old man presents to the hospital emergency room with a cough, chest pain, and fever. Two days before admission he developed a nonproductive cough. Rales are heard. Gram stain of sputum was negative. Sputa cultures on blood agar were also negative. Culture on a special medium containing cholesterol, purines, and pyrimidines produced colonies in 10 days. Serology 3 weeks later (when he returned because of persistent cough but feeling better) showed cold agglutinins.

- A. What is the probable causative agent?

Answer: Mycoplasma pneumoniae.

- B. Why did the organism not show up on the Gram stain?

Answer: Organism does not have a cell wall and does not stain with either the primary or counterstain in the Gram stain.

- C. What antibiotics do you NOT use?

Answer: Penicillin/cephalosporin.

Case M: A 27-year-old woman presents to the hospital with a fever of 104°F, dyspnea, and cough. Approximately three days ago she noted a papular rash on the face and trunk. At the time of admission there were many skin lesions: some papular, some vesicular, and some umbilicate. She also has similar lesions in her mouth, and her exanthem is shown.

A. The most likely agent causing this condition is:

Answer: Varicella-Zoster.

B. What two viral coded enzymes made both by VZV and HSV make acyclovir a useful drug in treating severe active VZV infections?

Answer: Thymidine kinase and herpes coded DNA polymerase.

Case N: A markedly sick 3-year-old boy presents to the family physician because of fever, dyspnea, cough, and photophobia. In addition, he has a maculopapular eruption on the face and trunk. On examination, it is observed that he has pinpoint gray-white areas with a red base on his buccal mucosae.

A. What is the most probable disease and causative agent?

Answer: Measles; rubeola.

B. What type of vaccine is available and could have prevented these symptoms?

Answer: Attenuated, single serotype of rubeola.

C. To what viral family does the agent belong?

Answer: Paramyxoviruses.

D. Does the virion carry a polymerase? Why?

Answer: Yes, because it is a negative RNA virus.

Case O: A 35-year-old worker at a plant nursery seeks his physician for a suppurative lesion on one of his fingers. A smear is taken of the drainage and stained. Cigar-shaped yeasts are detected.

A. What is the causative agent?

Answer: Sporothrix schenckii.

B. Is the fungus dimorphic or monomorphic?

Answer: This is DIMORPHIC FUNGUS consistent with Sporothrix. You can tell from cigar-shaped yeast (in tissues generally tough to visualize) and hyphae with sleeve and rosettes arrangement of conidia in culture.

C. Treatment

Answer: Itraconazole but oral KI in milk given will also clear up.

Case P: A 24-year-old female returned to the United States after spending six months in Mexico. Ten days ago she started to have attacks of diarrhea and developed abdominal distention. After lunch on the 10th day she noticed marked abdominal discomfort. While the pain was initially mid-abdominal, by 10:00 p.m. it became located predominantly in the right-lower quadrant. She went to the local emergency room where on physical examination it was noted that she had rebound tenderness and a fever of 37.9°C. She became nauseous and vomited.

Laboratory studies revealed a white blood cell count of 20,000/mm³ with a pronounced eosinophilia. The emergency room physician diagnosed the problem as acute appendicitis. An appendectomy was performed. A smear was made of the appendiceal exudate and knobby oval to barrel-shaped structures measuring 35 × 55 μm were observed.

A. What is the most likely cause of her appendicitis?

Answer: Ascaris lumbricoides migrating into the appendiceal orifice.

B. What major group does the organism belong to?

Answer: Nematodes

Case Q: A 50-year-old Missouri farmer was referred to the hospital because of malaise, weakness, weight loss, fever, and a palpable spleen. Examination of the mouth reveals a painless ulcerated lesion. A punch biopsy of the lesion is obtained and submitted for laboratory study. Histologic study revealed oval structures measuring 2–5 μm, packing the macrophages.

A. What is the most likely causative agent, and what are the distinctive forms?

Answer: Histoplasma capsulatum with the intracellular oval yeasts and the tuberculate macroconidia (and microconidia) in the hyphal state.

B. Where in nature will you find the fungus in large numbers?

Answer: Histoplasma capsulatum: Great central riverbed plains. Chicken coops in Missouri 100% infected. Indianapolis has had an ongoing outbreak and has major problems with it disseminating in their AIDS patients; NY City also high.

Case R: A 64-year-old male is hospitalized because of dementia. One month prior to admission he complained of headaches. On physical examination it is noted that he has nuchal rigidity, disorientation to time and place, and marked confusion. Lumbar puncture reveals 100 WBCs, which are predominantly lymphocytes, protein 85 mg/dl and sugar 45 mg/dl (concomitant blood sugar is 90 mg/dL).

Despite attempts to treat, three weeks after admission he died. An autopsy was performed. Brain sections were stained with H&E. Encapsulated, oval yeasts were revealed in the infected tissue.

A. What is your diagnosis, and why?

Answer: Cryptococcal meningitis; capsule present on yeast in tissues.

B. Who usually acquires the infection?

Answer: Immunosuppressed for meningitis; pulmonary generally only in pigeon breeders or people exposed to extremely high doses.

C. What tests on CSF should have been run?

Answer: The India ink test is very insensitive and cannot rule out Cryptococcal meningitis. Latex particle agglutination is much more sensitive. India ink is still frequently on the exam.

Case S: A 34-year-old accountant presents to the emergency room because of headache and fever of 3 days' duration. The day before admission his wife noted mild confusion and irritability. Lumbar puncture revealed an opening pressure of 300 mm, 200 red blood cells, 90% of the WBCs which are lymphocytes, sugar of 85 mg/dL (concomitant blood sugar of 110 mg/dL), and protein of 65 mg/dL. Bacteriologic smears (and ultimately also the bacterial cultures) were negative, as were India ink preparations. All latex particle agglutination tests for fungal and bacterial capsules done on the patient's CSF were also negative. The patient's condition did not improve despite appropriate therapy, and he died 10 days after hospitalization.

A. What is the most likely diagnosis?

Answer: Herpes simplex encephalitis

B. Does the virus have an envelope?

Answer: Icosahedral with envelope.

C. Where within the cell does the virus replicate?

Answer: Nucleus for both DNA synthesis and assembly.

D. What other members belong to the same family?

Answers: EBV, Varicella-Zoster, Cytomegalovirus.

Case T: A markedly dehydrated patient presents with diarrhea. His stool culture grew organisms only when grown on an alkaline medium. The isolate was oxidase-positive.

A. What is the mechanism of the produced enterotoxin?

Answer: This is Vibrio cholerae: ADP ribosylation → activating adenylate cyclase → increased cAMP (leaves Gs locked in the active state).

B. How would you describe his stool specimen?

Answer: Rice water.

Case U: A patient presents with anogenital warts.

A. What is the virus that probably caused the tumors?

Answer: Human papilloma virus.

B. What serotypes are most commonly associated with this clinical presentation?

Answer: 6 and 11

C. Are they premalignant?

Answer: Rarely.

D. What serotypes are most commonly associated with cervical intraepithelial neoplasia?

Answer: 16, 18, and 31. These are sexually transmitted.

Case V: A girl received a bone marrow transplant for the treatment of leukemia. Nine weeks after the transplant her temperature rose, she became dyspneic, and died. Impression smears were taken from the cut surface of the lower lobe of the left lung. The smears were stained with H&E. Intranuclear inclusions with perinuclear clearing were found.

A. Why did the patient develop the pneumonia?

Answer: Immunocompromised—No T cells.

B. How would you describe what you would see (using only two words)?

Answer: Owl's eyes: cells with prominent basophilic intranuclear inclusion bodies.

C. What is the virion's nucleic acid type? To what viral family does it belong?

Answer: dsDNA; Herpes viruses.

Case W: A young woman developed a feverish illness with painful swelling of her knee, elbow, and wrist joints. She has a sparse rash on the distal parts of her limbs, consisting of small hemorrhagic pustules with an erythematous base. A smear was obtained from the exudate of the exanthem and Gram stained. The stain showed intracellular gram-negative diplococci.

A. What disease does she have?

Answer: Disseminated gonococcal infection.

B. Do pili play a role in the pathogenesis?

Answer: Yes, for attachment to epithelial surfaces—colonizing factor along with outer membrane proteins and antigenic variation.

Comparative Microbiology

8

MORPHOLOGY/TAXONOMY

Spore-Forming Bacteria (Have Calcium Dipicolinate)

Bacillus

Clostridium

Non-motile Gram-Positive Rods

Corynebacterium diphtheriae

Nocardia

Clostridium perfringens (rest of the pathogenic *Clostridia* are motile)

Bacillus anthracis (most other *Bacillus* species are motile)

Acid Fast Organisms

Mycobacterium

Nocardia (partially acid fast)

Cryptosporidium oocysts

Legionella micdadei

Isospora oocysts

Bacteria and Fungi That Characteristically Have Capsules

The “biggies” can be remembered by the mnemonic: Some Killers Have Pretty Nice Capsules!

Streptococcus pneumoniae

Klebsiella pneumoniae

Haemophilus influenzae

Neisseria meningitidis

Cryptococcus neoformans (only encapsulated fungal pathogen)

Pseudomonas aeruginosa—slime producer especially in cystic fibrosis patients’ lungs

Bordetella pertussis

Other Important Capsule Producers

E. coli meningeal strains have capsule, mostly K₁

Bacillus anthracis—poly D-glutamate capsule

Salmonella typhi—(virulence) capsular antigen

Streptococcus pyogenes when first isolated; non-immunogenic (but anti-phagocytic) hyaluronic acid capsule

Biofilm Producers

Staphylococcus epidermidis (catheter-related infections)

Streptococcus mutans (dental plaque)

Pigment Production

Pseudomonas aeruginosa—pyocyanin (blue-green, fluorescein)

Serratia—red pigment

Staphylococcus aureus—yellow pigment

Photochromagenic and scotochromagenic *Mycobacteria*—Carotenoid pigments (yellow and orange)

Corynebacterium diphtheriae—black to gray

Unique Morphology/Staining

Metachromatic staining—*Corynebacterium*

Lancet-shaped diplococci—*Pneumococci*

Kidney bean-shaped diplococci—*Neisseriae*

Bipolar staining—*Yersinia pestis*

Gulls wings—*Campylobacter*

Table I-8-1. Viral Cytopathogenesis

Inclusion Bodies	Virus
Intracytoplasmic (Negri bodies)	Rabies
Intracytoplasmic acidophilic (Guarnieri)	Poxviruses
Intracytoplasmic and intranuclear (Owl's eye)	Cytomegalovirus
Intranuclear (Cowdry's)	Herpes simplex virus Subacute sclerosing panencephalitis (measles) virus
Syncytia formation	Virus
Present	Herpes simplex virus Varicella-zoster Paramyxovirus Respiratory syncytial virus HIV

PHYSIOLOGY

Table I-8-2. Metabolism*

Aerobes	Anaerobes	Microaerophilic
<i>Mycobacterium</i>	<i>Actinomyces</i>	<i>Campylobacter</i>
<i>Pseudomonas</i>	<i>Bacteroides</i>	<i>Helicobacter</i>
<i>Bacillus</i>	<i>Clostridium</i>	
<i>Nocardia</i>	<i>Fusobacterium</i>	
<i>Corynebacterium diphtheria</i>	<i>Prevotella</i>	
	<i>Propionibacterium</i> (aerotolerant)	
	<i>Eubacterium</i>	
	<i>Lactobacillus</i> (aerotolerant)	

*Most others are considered facultative anaerobes.

Enzymes

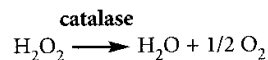
Oxidase

- All *Enterobacteria* are oxidase negative.
- All *Neisseria* are oxidase positive (as are most other Gram-negative bacteria).

Urease Positive

- *Helicobacter*
- All *Proteus* species produce urease; this leads to alkaline urine and may be associated with renal calculi.
- *Ureaplasma* (renal calculi)
- *Cryptococcus* (the fungus)
- *Nocardia*

Catalase



Staphylococci have catalase, *Streptococci* do not.

Most anaerobes lack catalase.

Catalase positive organisms are major problems in Chronic Granulomatous Disease (CGD):

- *Staphylococcus aureus* (all staphylococci)
- *Pseudomonas aeruginosa*
- *Candida*
- *Aspergillus*
- Enterobacteriaceae, especially *Klebsiella*

Coagulase Positive

- *Staph aureus*
- *Yersinia pestis*

DETERMINANTS OF PATHOGENICITY**Genetics****Genes Encoding Pathogenic Factors Reside on:**

- The bacterial **chromosome**
 - Choleragen
 - Endotoxin
 - Shiga toxin
 - Mnemonic (Chromosomal Encoded Somethings)
- A **plasmid**
 - Most toxins and multiple drug resistance
- A **bacteriophage chromosome** stably integrated into the host DNA as a prophage.
 - Virulence modified by the stable presence of phage DNA in bacterial cell = lysogenic conversion.

Examples:

- O** = *Salmonella* O antigen
B = Botulinum toxin (phage CE β and DE β)
E = Erythrogenic toxin of *Streptococcus pyogenes*
D = Diphtheria toxin (Corynephage β)
 Mnemonic: **OBED** (or a little pregnant with phage).

Antigenic Variation

- Neisseria gonorrhoeae* (pili)
Borrelia recurrentis
Trypanosoma brucei

Table I-8-3. Disease Due to Toxin Production

Bacterium	Disease	Activity of Toxin
<i>Corynebacterium diphtheriae</i>	Diphtheria	ADP ribosylation of EF-2 results in inhibition of protein synthesis
<i>Clostridium tetani</i>	Tetanus	Binds to ganglioside in synaptic membrane, blocks release of glycine
<i>Clostridium botulinum</i>	Botulism	Prevents release of acetylcholine
<i>Vibrio cholerae</i>	Cholera	Choleragen stimulates adenylate cyclase
<i>E. coli</i> (ETEC)	Travelers' diarrhea	LT stimulates adenylate cyclase
<i>Clostridium difficile</i>	Diarrhea	Toxin A and B inhibit protein synthesis and cause loss of intracellular K ⁺

EF-2 = eukaryotic elongation factor-2.

Heat Stable Toxins

60°C

- *Staphylococcus aureus* enterotoxin
- ST toxin of *E. coli*
- *Yersinia enterocolitica* toxin

100°C

- Endotoxin

Toxins with ADP-Ribosylating Activity

Table I-8-4. Toxins with A-B ADP-Ribosylating Transferase Activity

Toxin	ADP-Ribosylated Host Protein	Effect on Host Cell
<i>Pseudomonas</i> Exotoxin A Exotoxin S	eEF-2 unknown	Inhibits translocation during protein synthesis
Diphtheria toxin	eEF-2	Inhibits translocation during protein synthesis
<i>E. coli</i> heat-labile toxin (LT)	G-protein (G _s)	Increases cAMP in intestinal epithelium causing diarrhea
Cholera toxin	G-protein (G _s)	Increases cAMP in intestinal epithelium causing diarrhea
Pertussis toxin	G-protein (G _i)	Increases cAMP causing lymphocytosis and increased insulin secretion

A is the ADP-ribosyl transferase.

B binds to cell receptor and translocates the A subunit into the cell.

Table I-8-5. Invasive Factors

Invasive Factor	Function	Bacteria
All capsules	Antiphagocytic	See earlier list with morphology
Slime layer (capsule or glycocalyx)	Antiphagocytic	<i>Pseudomonas</i>
M protein	Antiphagocytic	Group A Streptococci
A protein	Antiphagocytic	<i>Staph. aureus</i>
Lipoteichoic acid	Attachment to host cells	All Gram-positive bacteria
<i>N. gonorrhoeae</i> pili	Antiphagocytic	<i>N. gonorrhoeae</i>

Table I-8-6. Extracellular Enzymes

Enzyme	Function	Bacteria
Hyaluronidase	Hydrolysis of ground substance	Group A Streptococci
Collagenase	Hydrolysis of collagen	<i>Clostridium perfringens</i> <i>Prevotella melaninogenica</i>
Kinases	Hydrolysis of fibrin	<i>Streptococcus</i> <i>Staphylococcus</i>
Lecithinase (alpha toxin)	Damage to membrane	<i>Clostridium perfringens</i>
Heparinase	May contribute to thrombophlebitis	<i>Bacteroides fragilis</i>
IgA Proteases	Colonizing factor	<i>Neisseria</i> <i>Haemophilus</i> <i>Strep. pneumoniae</i>

Ability to Survive and Grow in Host Cell

Obligate Intracellular Parasites

Cannot be cultured on inert media. Virulence is due to the ability to survive and grow intracellularly where the organism is protected from many B-cell host defenses.

- Bacteria
 - All Rickettsiae
 - *Chlamydia trachomatis*
 - *Chlamydia psittaci*
 - *Mycobacterium leprae*
- Viruses
 - All are obligate intracellular parasites.
- Protozoa
 - *Plasmodium*
 - *Toxoplasma gondii*
 - *Babesia*
 - *Leishmania*
 - *Trypanosoma cruzi* (amastigotes in cardiac muscle)
- Fungi
 - None

Obligate Parasites That Are Not Intracellular

(e.g., cannot be cultured on inert media but are found extracellularly in the body)

- *Treponema pallidum*
- *Pneumocystis carinii*

Facultative Intracellular Parasites of Humans

- Bacteria
 - Francisella tularensis*
 - Listeria monocytogenes*
 - Mycobacterium tuberculosis*
 - Brucella* species
 - Non-tuberculous *mycobacteria*
 - Salmonella typhi*
 - Legionella pneumophila*
 - Yersinia pestis*
 - Nocardia*
 - Borrelia burgdorferi*
- Fungi
 - Histoplasma capsulatum*
- Protozoa
 - Trypanosoma*

EPIDEMIOLOGY/TRANSMISSION

Bacteria That Have Humans as the Only Known Reservoir

Mycobacterium tuberculosis
M. leprae (armadillos in Texas)
Shigella species
Salmonella typhi
Rickettsia prowazekii (epidemic typhus)
 Group A β -hemolytic streptococcus
Neisseria meningitidis and *N. gonorrhoeae*
Corynebacterium diphtheriae
Streptococcus pneumoniae
Treponema pallidum
Chlamydia trachomatis

Zoonotic Organisms

(Diseases of animals transmissible to humans)
Bacillus anthracis
Salmonella species except *S. typhi*
Leptospira
Borrelia
Listeria monocytogenes
Brucella species
Francisella tularensis

Pasteurella multocida (cat bites)
Vibrio parahaemolyticus (from fish)
Vibrio vulnificus (oysters)
Yersinia pestis, *Y. enterocolitis*, *Y. pseudotuberculosis*
Campylobacter fetus, *C. jejuni*
Most Rickettsia
Chlamydia psittaci (birds)
Coxiella

Arthropod Vectors in Human Disease: Insects

- Lice
 - Epidemic or louse-borne typhus (*Pediculus h. humanus*)
 - Epidemic relapsing fever
 - Trench fever
- True bugs
 - Chagas' disease (American trypanosomiasis)—kissing bugs (*Reduviidae*)
- Mosquitoes
 - Malaria (*Anopheles* mosquito)
 - Dengue (*Aedes*)
 - Mosquito-borne encephalitides: WEE, EEE, VEE, SL
 - Yellow Fever (*Aedes*)
 - Filariasis
- Sandflies
 - Leishmanias
 - Sandfly fever (viral)
 - Bartonellosis
- Midge
 - Filariasis
- Blackflies
 - Onchocerciasis
- Deerflies and horse flies
 - Loaloasis
 - Tularemia
- Tsetse flies
 - African trypanosomiasis
- Fleas
 - Plague
 - Endemic typhus

Arthropod Vectors That Are Not Insects

- Ticks
 - Rocky Mountain spotted fever (*Dermacentor*, *Amblyomma*)
 - Colorado tick fever (*Dermacentor*)
 - Lyme disease (*Ixodes*)
 - Ehrlichia*
 - Babesiosis (*Ixodes*)
 - Tularemia
 - Recurrent fever or tick-borne relapsing fever (*Ornithodoros*, a soft tick)
- Mites
 - Scrub typhus (*Leptotrombium*) (transovarial transmission in vector)
 - Rickettsialpox

Parasitic Infections Transmitted by Eggs

Enterobius vermicularis (pinworm)
Ascaris lumbricoides (roundworm)
Toxocara canis
Echinococcus granulosus
Taenia solium

Bacterial and Fungal Infections That Are Not Considered Contagious

(i.e., no human-to-human transmission)
Nontuberculous mycobacterial infections, e.g., *Mycobacterium avium-intracellulare*
Non-spore forming anaerobes
Legionella pneumophila
All fungal infections except the dermatophytes

Infections That Cross the Placenta

Toxo
Other (Syphilis)
Rubella
CMV
Herpes and HIV
<5% perinatal hepatitis B could possibly have been acquired by crossing placenta.

- Viruses
 - Cytomegalovirus
 - Rubella
 - Herpes II (in primary infection)
 - Coxsackie B
 - Polio
 - HIV

- Parasites

Toxoplasma gondii

- Bacteria

Treponema pallidum

Listeria monocytogenes

Spread by Respiratory Droplet

Streptococcus pyogenes (Group A)

Streptococcus pneumoniae

Neisseria meningitidis

Mycobacterium tuberculosis

Bordetella pertussis

Haemophilus influenzae

Corynebacterium diphtheriae

Mycoplasma pneumoniae

Influenza

Rubella

Measles

Chickenpox

Pneumocystis carinii

Spread by Inhalation of Organisms from the Environment

Histoplasma

Coccidioides

Blastomyces

Nontuberculous mycobacteria, e.g., *M. avium-intracellulare* (MAC)

Legionella

Chlamydia psittaci

Pseudomonas (also spread by ingestion and contact)

Coxiella burnetii (the only Rickettsia that is stable in the environment)

Spread by Oral/Fecal Route

(Infections may be spread by oral sex.)

Salmonella

Shigella

Campylobacter

Vibrio

Yersinia enterocolitica

Yersinia pseudotuberculosis

Bacillus cereus

Clostridium

Staphylococcus (also other routes commonly)

Enteroviruses

Rotavirus

Norwalk agent

Hepatitis A

Polio virus

Toxoplasma—cat feces

Entamoeba

Giardia

Balantidium

All nematodes of interest

Echinococcus—dog feces

Contact: (Person-to-Person) Nonsexual

Impetigo (*Strep* and *Staph*)

Staphylococcus

Herpes I

Epstein-Barr (kissing)

Hepatitis B (all body fluids)

Contact: Sexual

Chlamydia HPV

Neisseria HIV

Treponema Herpes II

Trichomonas CMV

PATHOLOGY

Organisms That Produce Granulomas (Persistent Antigen)

Fran Likes My Pal Bruce And His Blasted Cockerspaniel (in) Blessed Salt Lake City.
(Mnemonic by M. Free.)

(ic) = intracellular organism

Francisella (ic)

Listeria (ic)

Mycobacterium (ic)

Treponema pallidum

Brucella (ic)

Actinomyces

Histoplasma (ic)

Blastomyces

Coccidioides

Berylliosis

Schistosoma species, sarcoid

Lymphogranuloma venereum (ic)

Cat scratch fever

Infections Causing Intracerebral Calcifications

Toxoplasma

CMV

Cysticercosis

LABORATORY DIAGNOSIS

Special Stains

- Silver stains
 - Dieterle—*Legionella*
 - Gomori methenamine—*Pneumocystis*, fungi
- Acid fast (Ziehl-Neelsen or Kinyoun)
 - Mycobacterium*, *Nocardia* (partially AF), *Legionella micdadei*, *Cryptosporidium*, and *Isospora*
- India ink—*Cryptococcus* (if negative not a reliable diagnostic method)
- Calcofluor white—fungi
- Giemsa
 - Blood protozoa (*Plasmodium*, *Babesia*, *Trypanosoma*, *Leishmania*)
 - Histoplasma capsulatum* in RES cells

Name Tests

<u>Tests</u>	<u>Disease</u>
PPD or Tuberculin (Mantoux)	TB
Lepromin	Leprosy
Fungal skin tests	Clinically valuable only to demonstrate exposure or anergy
CAMP test	<i>Streptagalactiae</i> carriers
Elek test	Toxin producing <i>C. diphtheriae</i> strains
Weil-Felix	Rickettsia (with <i>Proteus</i> strain OX antigens)

Unusual Growth Requirements

Haemophilus (most species require one or both)

- **X factor** = protoporphyrin IX, the precursor of **hemin**
- **V factor** = NAD (nicotinamide dinucleotide) or NADP

Mycoplasma

- Cholesterol

Salt

- *Staph aureus* will grow on high salt media.
- Group D enterococci will grow on 6.5% NaCl.
- *Vibrio parahaemolyticus* requires NaCl to grow and grows at 6.5%.

Cysteine requirement for growth

- Four Sisters Ella of the Cysteine Chapel (mnemonic by M. Free)
Francisella, *Legionella*, *Brucella*, and *Pasteurella*

Cultures that must be observed for a long time

- *Mycobacterium tuberculosis* and all non-tuberculous mycobacteria **except rapid growers**
- *Mycoplasma pneumoniae*
- *Brucella* sp
- Systemic fungal pathogens (*Blastomyces*, *Histoplasma*, and *Coccidioides* in U.S.)

TREATMENT/PREVENTION

Treat Prophylactically

- *Neisseria meningitidis* (household and day care contacts—**vaccination also used in outbreaks**)
- *Mycobacterium tuberculosis* with a recent skin test conversion or known household (i.e., significant) exposure; or persons under 35 with a positive skin test who have never been treated
- *Haemophilus influenzae* B (unvaccinated household contacts <6 years old)—also vaccinate
- *Neisseria gonorrhoeae* (sexual contacts)
- *Treponema pallidum* (sexual contacts)
- *Yersinia pestis*

Vaccines Available in the U.S.

Inactivated Vaccines

- Pertussis (killed whole cell in DTP)
- *Vibrio cholera*
- Influenza virus
- Salk polio (killed)—all primary vaccinations in U.S., including IC patients
- Rabies (HDC or RVA)
- Japanese encephalitis and several other encephalitis vaccines
- Hepatitis A

Live, Attenuated Vaccines

- *Francisella tularensis*
- Measles (rubeola)
- Rubella
- Mumps (killed vaccine available for IC patients)
- Sabin polio (oral)
- Smallpox
- Yellow fever
- Varicella-Zoster

Live, Pathogenic Virus (in enteric-coated capsules)

- Adenovirus

Toxoid: Chemically Modified Toxin—Vaccines

- Tetanus
- Diphtheria
- Pertussis toxoid (in DTaP)

Recombinant Vaccines or Subunit Vaccine

- Hepatitis B—HBsAg (produced in yeast)
- *Haemophilus*—purified capsular polysaccharide conjugated to protein
- *Neisseria meningitidis*—capsular polysaccharides
- Pneumococcal—capsular polysaccharide (23 serotypes)

Chapter Summary

The microorganisms, listed by their taxonomic classification, are divided into the following morphologic groups: spore-forming bacteria, nonmotile Gram-positive rods, acid-fast organisms, capsulated bacteria and fungi, biofilm producers, pigment producers, and bacteria with unique morphology or staining properties. Table I-8-1 lists viruses with inclusion bodies and those that form syncytia.

The microorganisms are listed according to the following properties related to their physiology: aerobes, anaerobes, and microaerophiles and oxidase, urease, catalase, and coagulase activity.

The microorganisms are listed according to the following properties related to virulence: genetic coding of pathogenic factors, ability to undergo antigenic variation, ability to cause toxin-induced diseases, production of heat-stable toxins, production of toxins that have ADP-ribosylation activity, production of invasive factors, production of extracellular enzymes, and ability to survive and grow in host cells.

The microorganisms are listed according to the following properties related to epidemiology: bacteria having only human reservoirs; microorganisms that normally cause disease in animals but also can cause human disease; organisms transmitted to humans by arthropod vectors; parasites transmitted to humans as eggs; parasites and bacteria that cause nontransmissible diseases; organisms that cross the placenta; and organisms spread by respiratory droplets, by inhalation, by the fecal/oral route, and by personal nonsexual or sexual contact.

The microorganisms also are listed according to the pathologic production of granulomas and intracerebral calcifications.

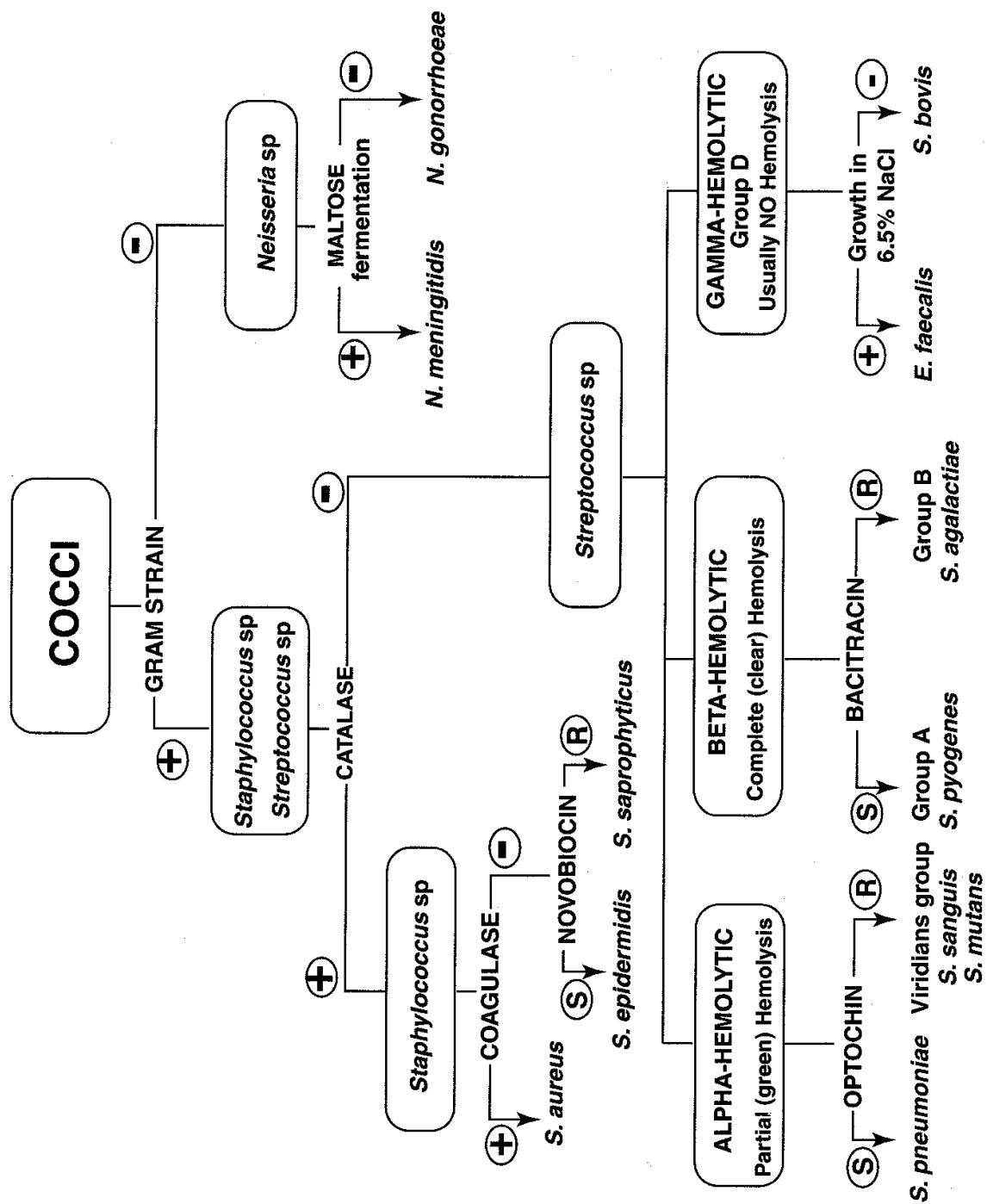
The organisms that can be identified by special stains or by named tests are cataloged.

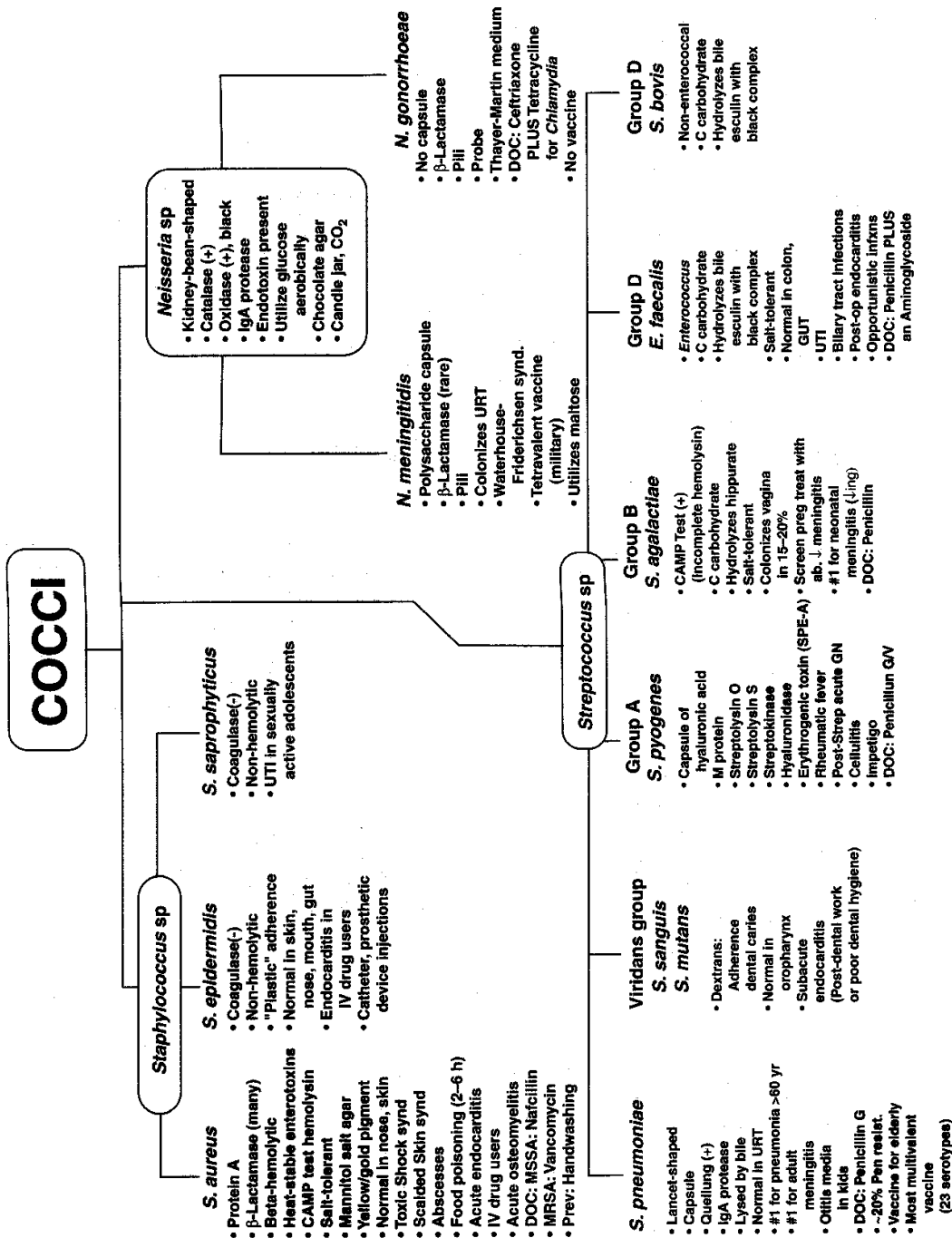
Fastidious organisms and their unusual growth requirements are listed.

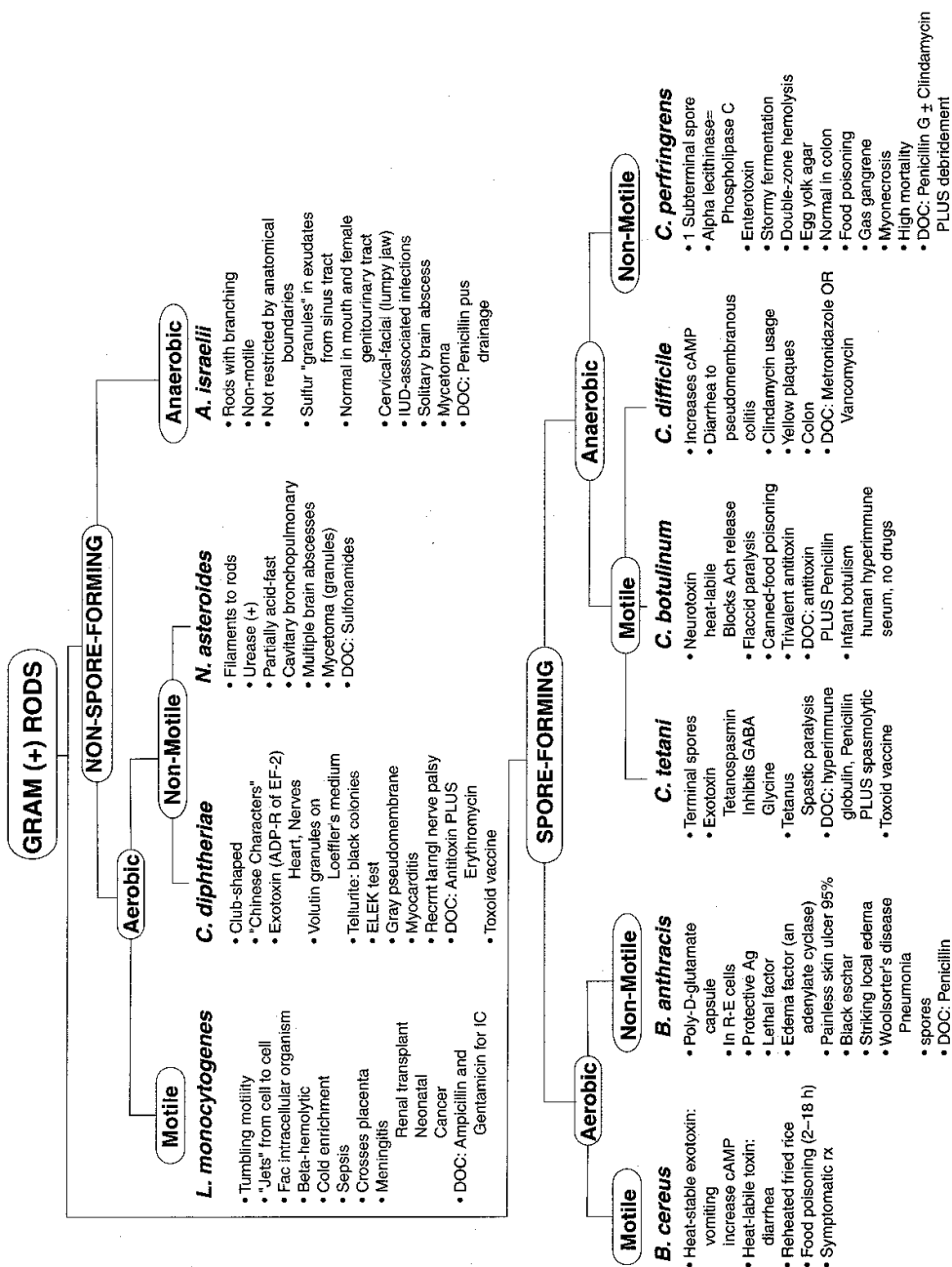
Organisms responding to prophylactic treatments (avoidance, etc.) and for which vaccines are available in the United States are listed. The list includes inactive, live attenuated, live pathogenic, toxoid, and recombinant or component (subunit) vaccines.

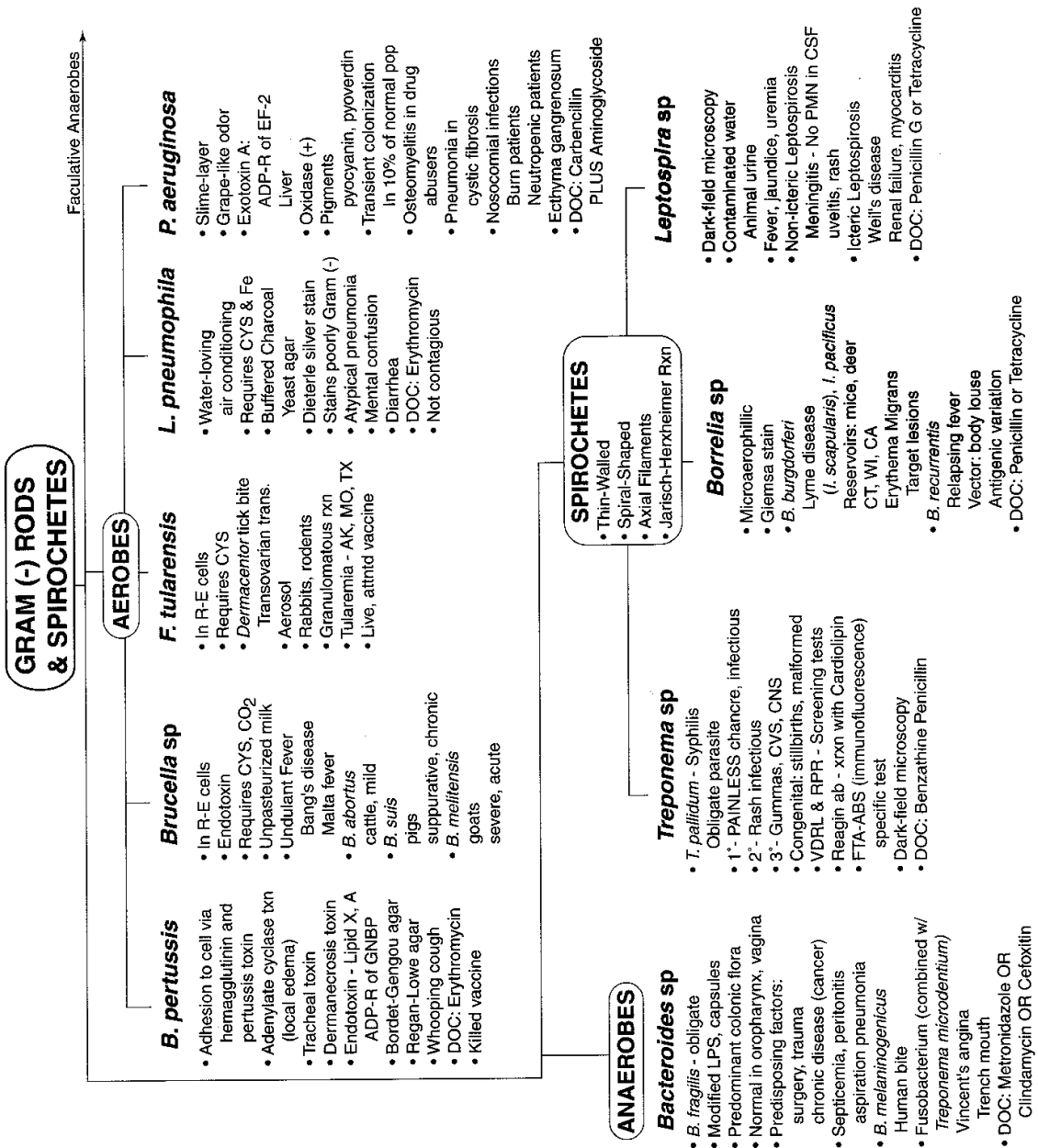
Flow Charts/Clue Sheets

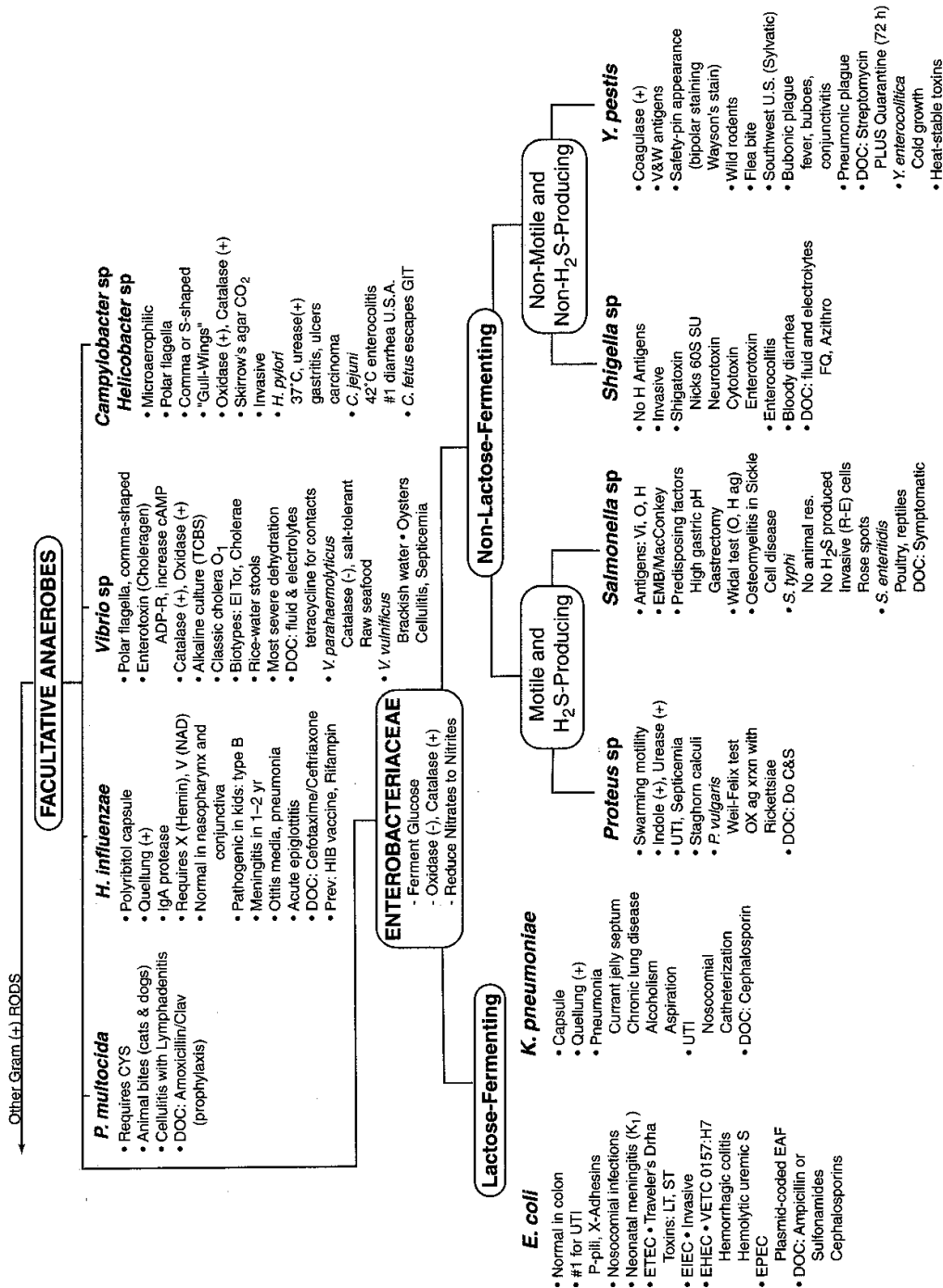


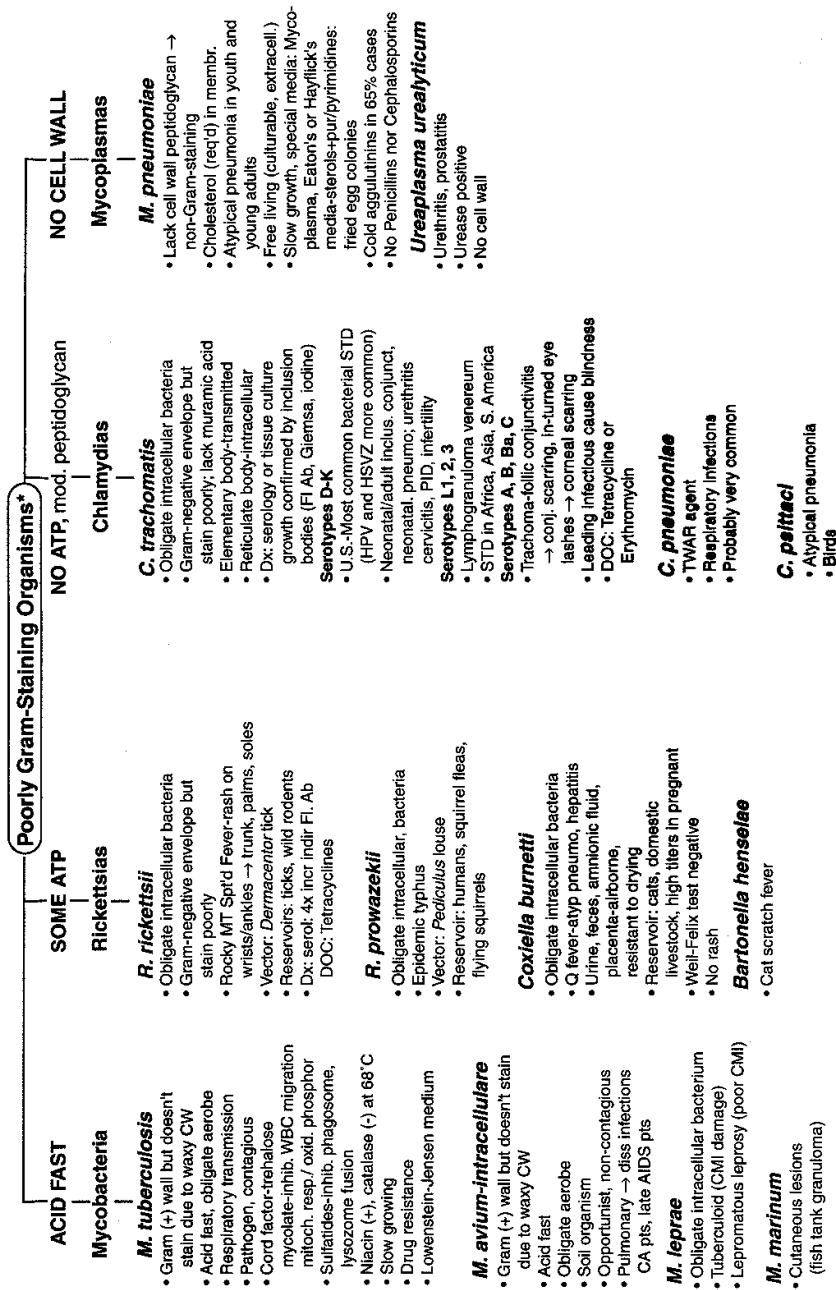




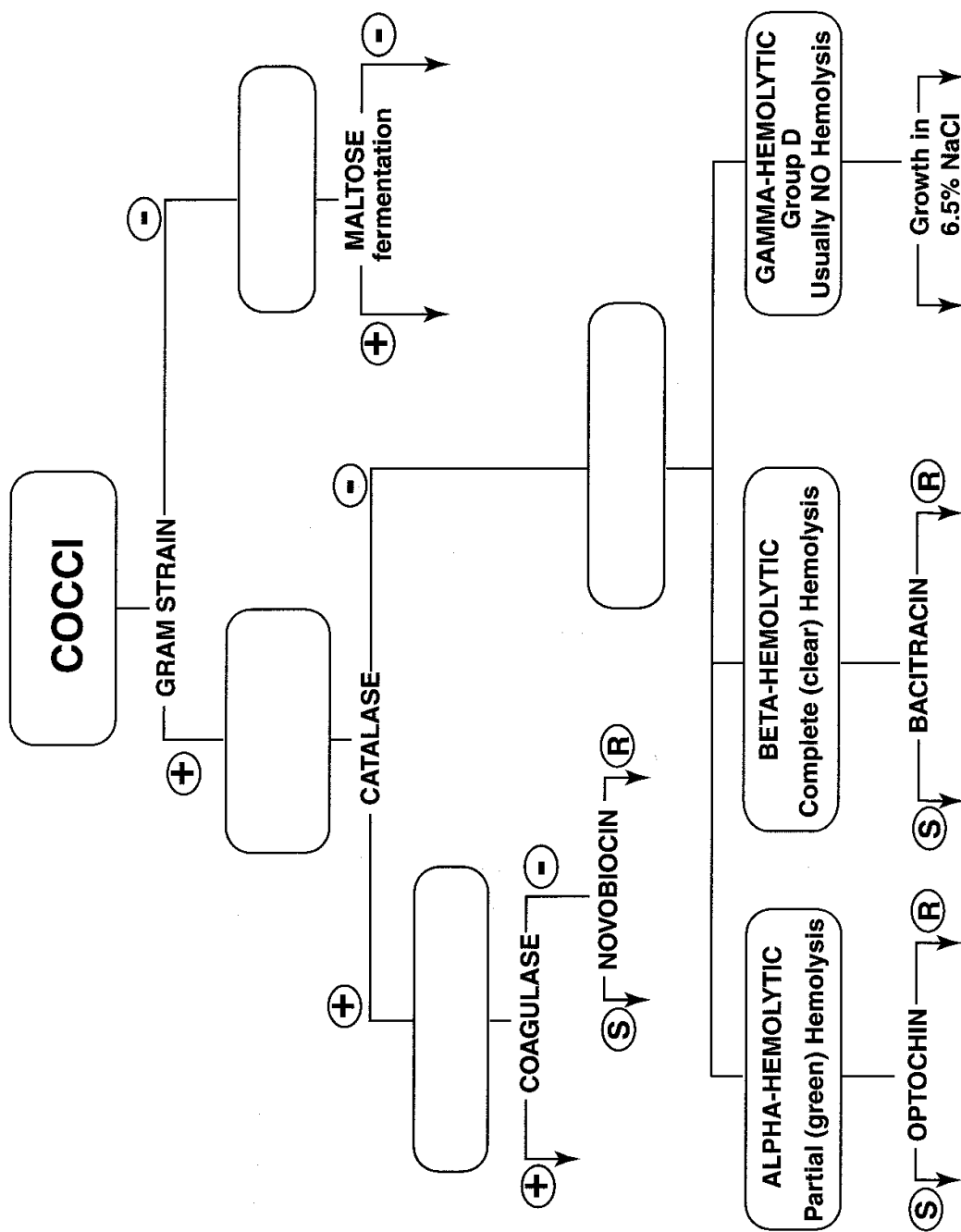


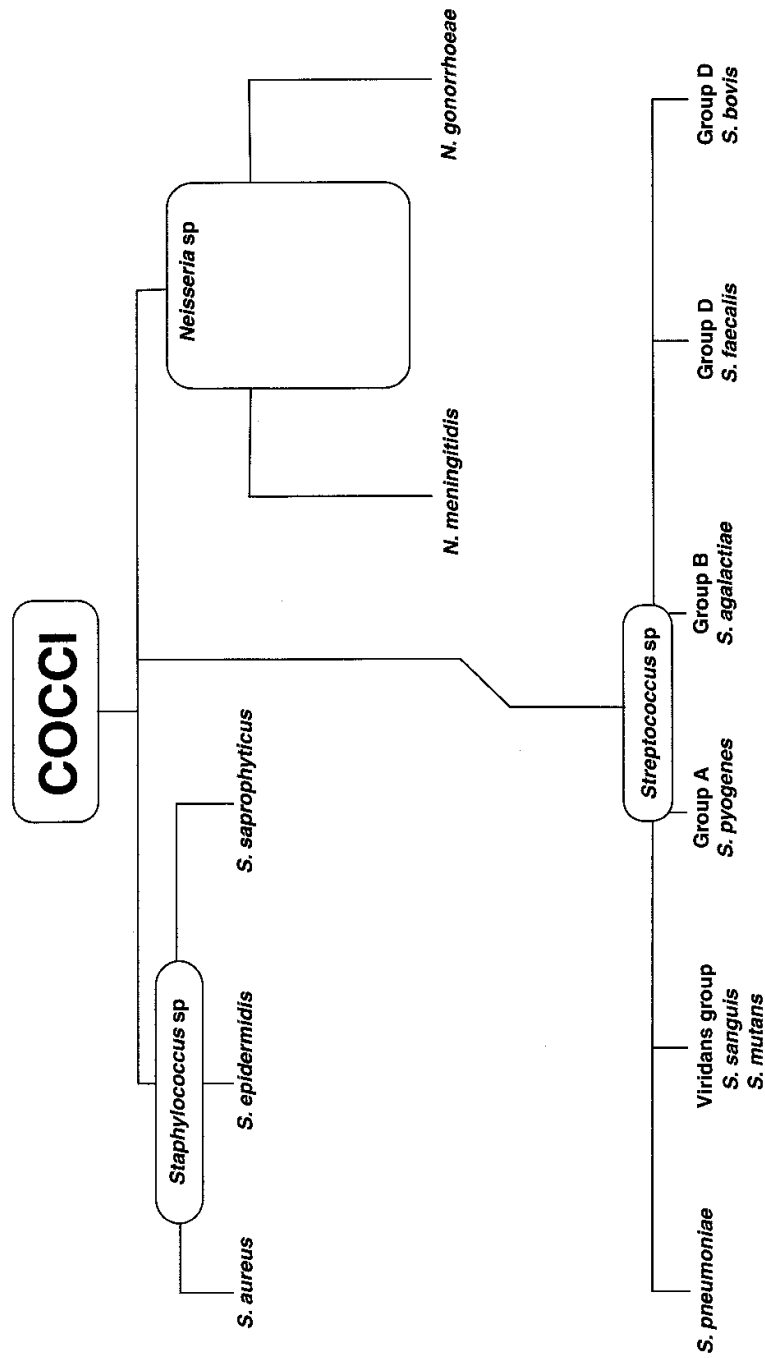


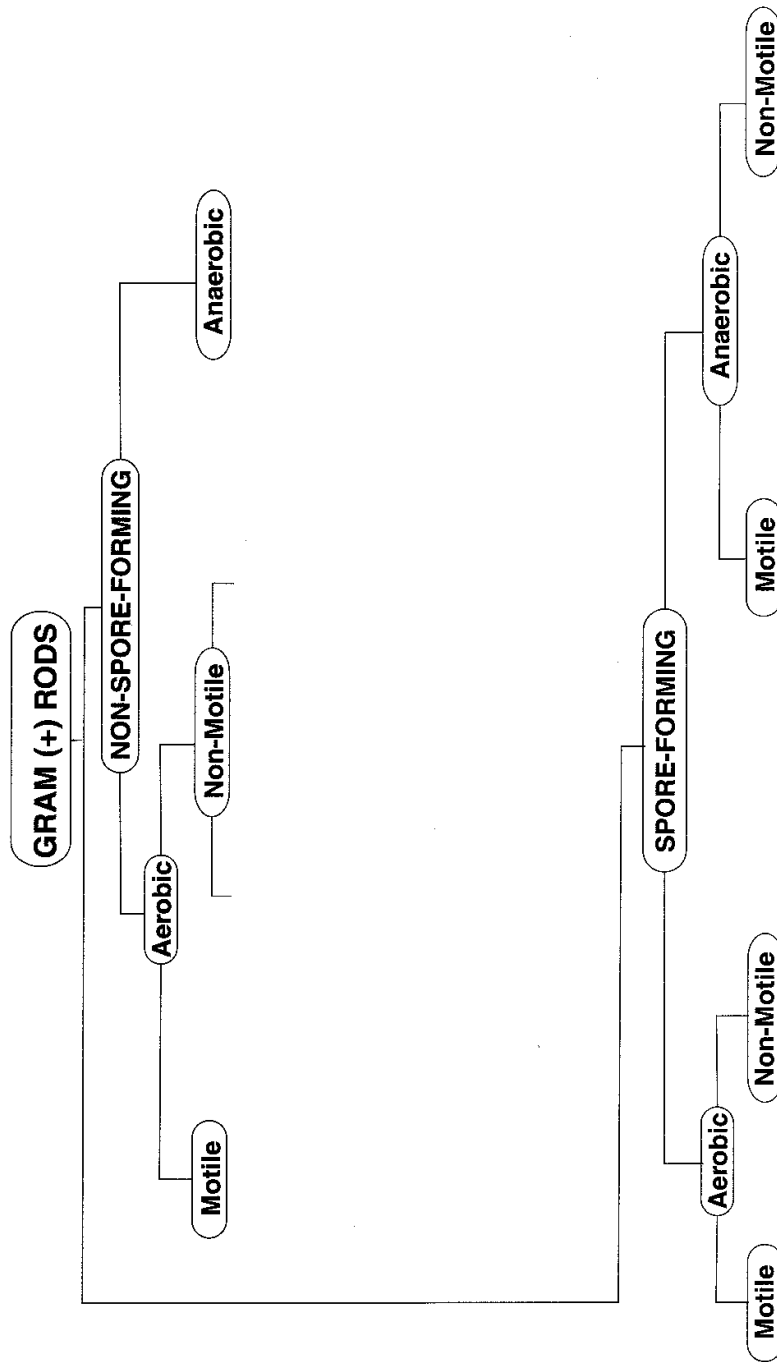


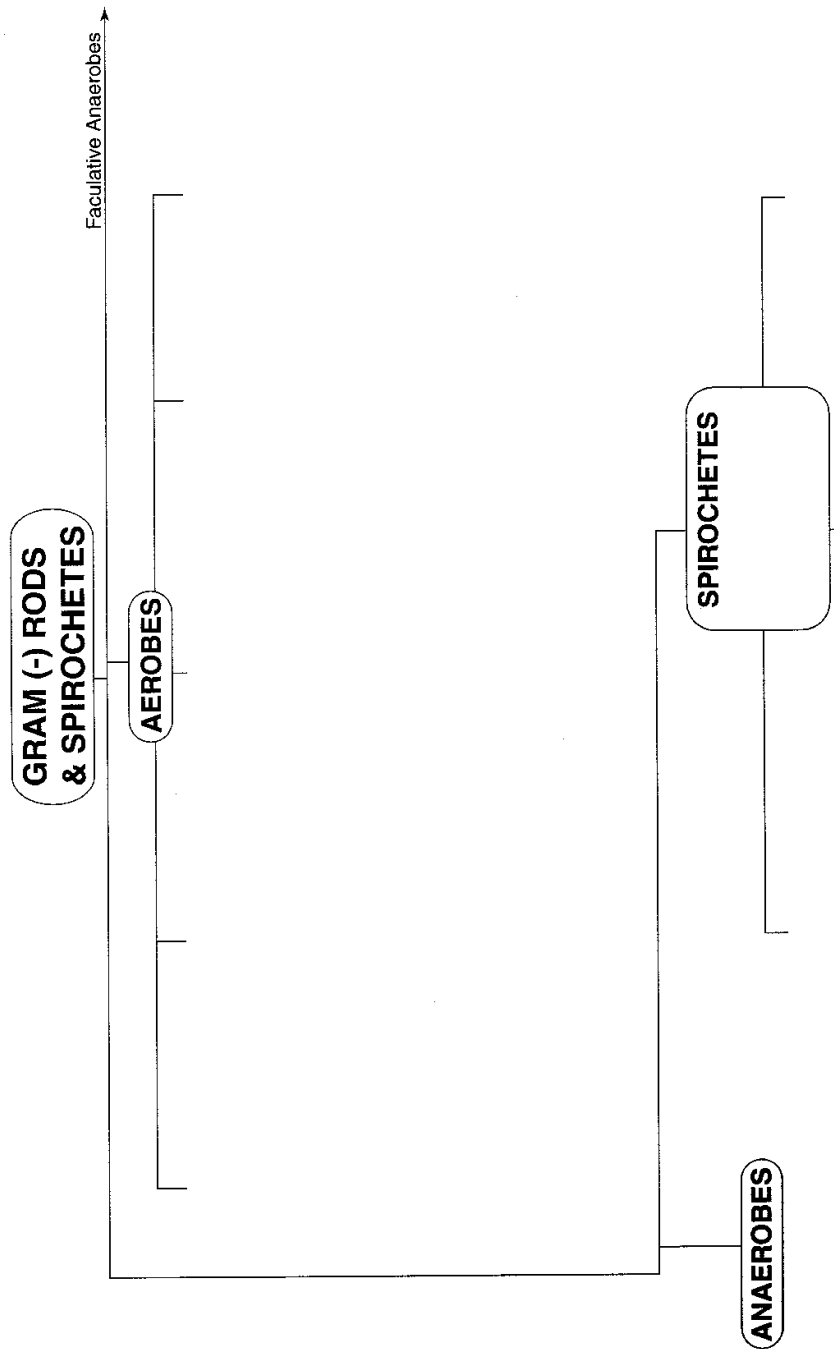


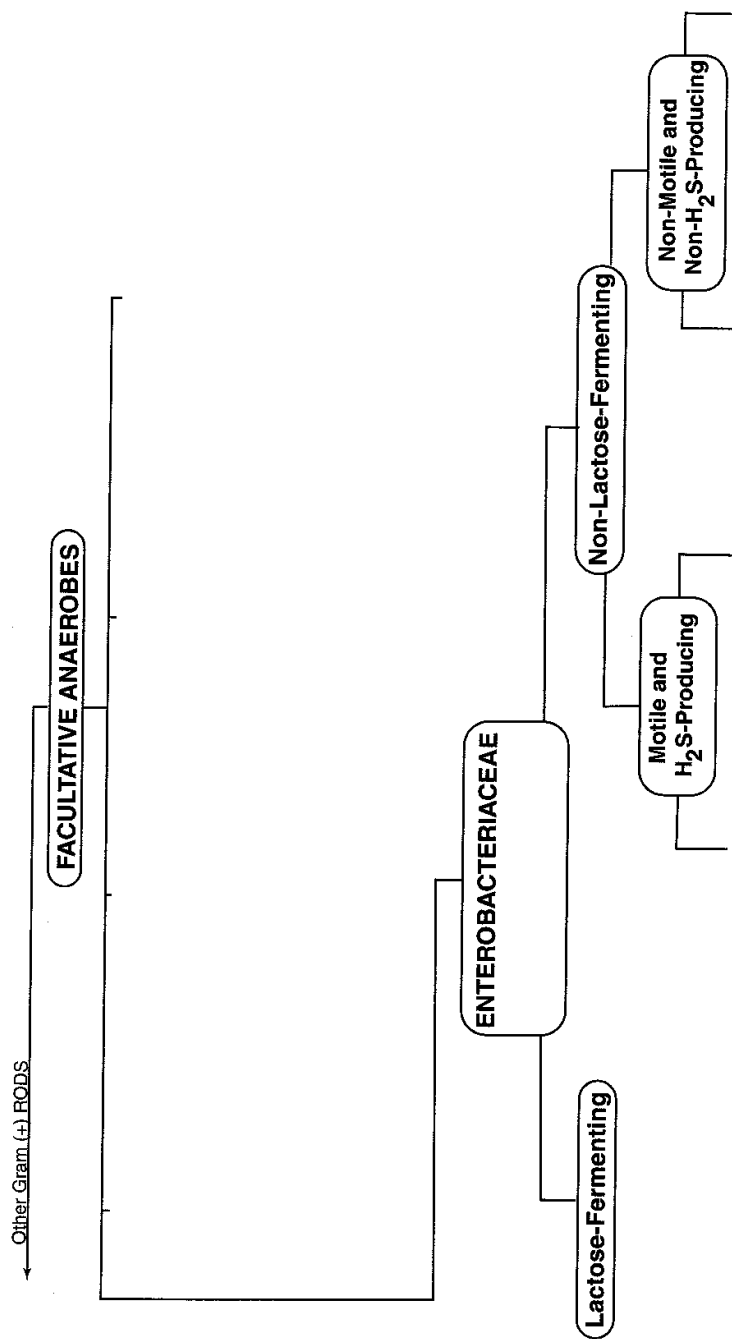
*Also note that *Legionella* and the spirochetes (*Treponema*, *Leptospira*, and *Borrelia*)—all Gram-negative—do not show up reliably with Gram stain.

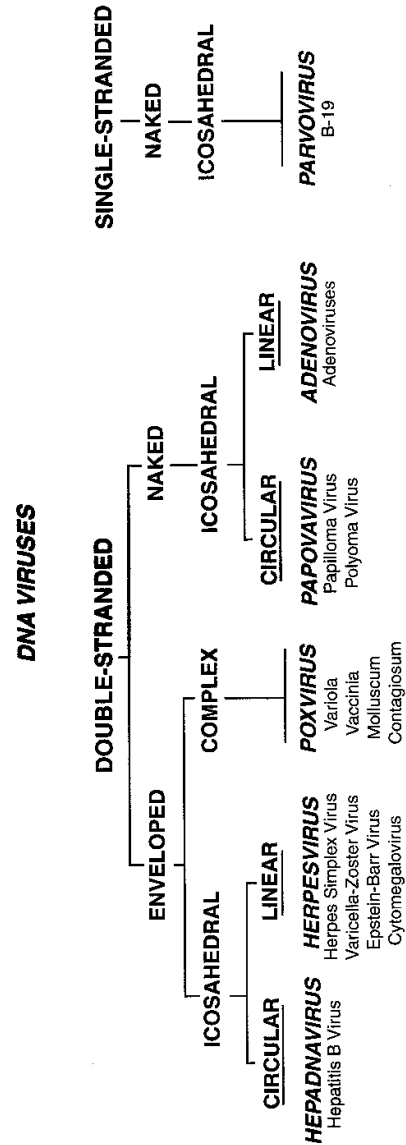
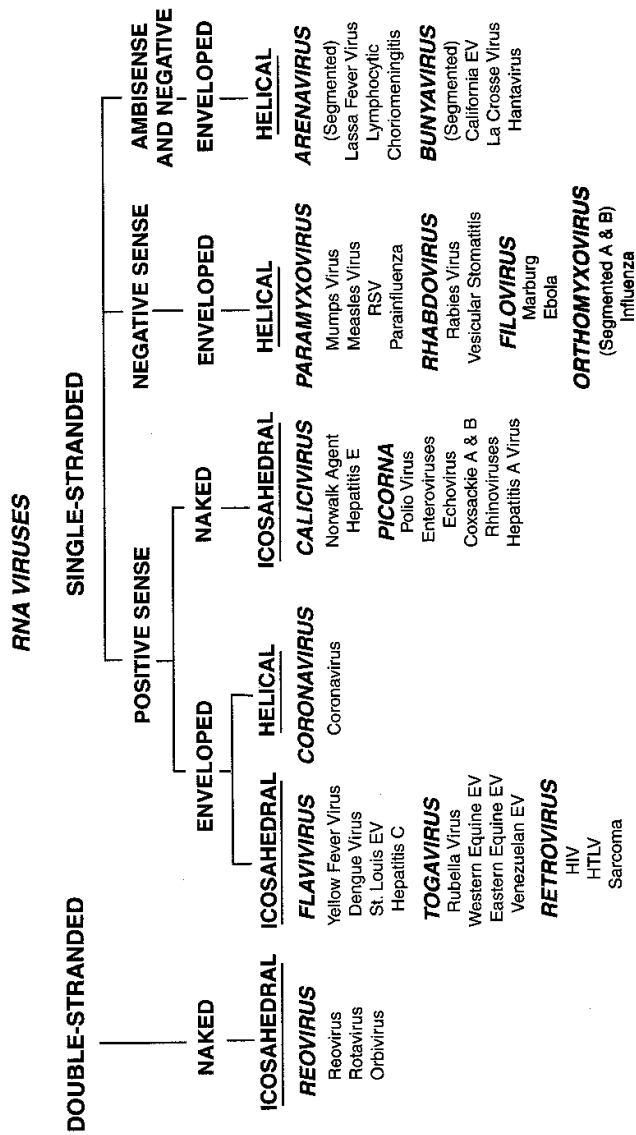


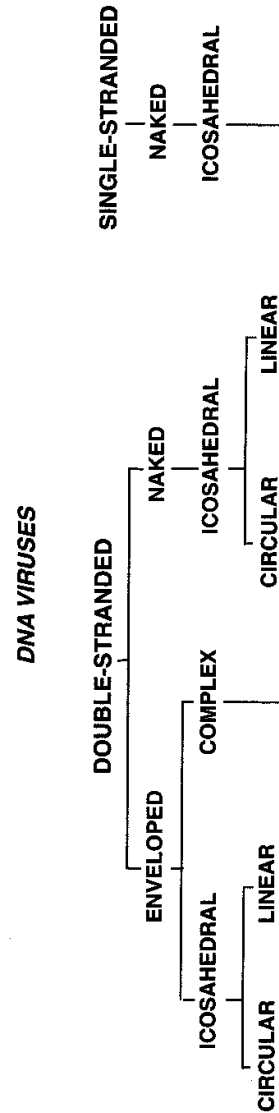
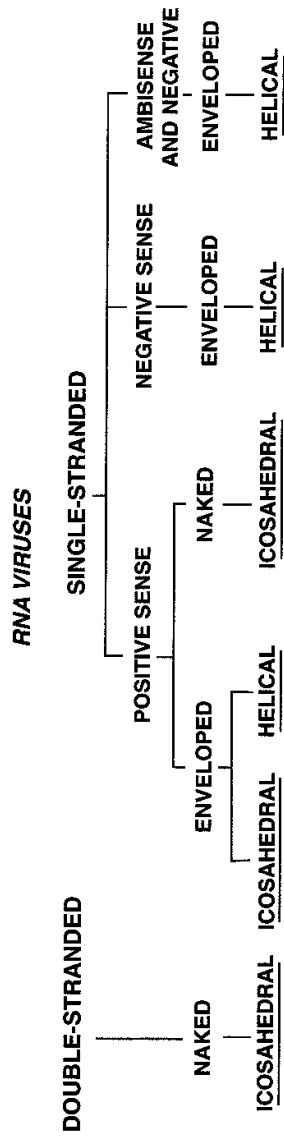












SECTION II

Immunology

Overview of the Immune System

1

What the USMLE Requires You To Know

- Components of the innate and adaptive immune responses
- Attributes of innate and adaptive immune responses
- Interactions between innate and adaptive immune responses

The **immune system** is designed to produce a coordinated response to the introduction of foreign substances or **antigens** into the body. It is organizationally divided into two complementary arms: the **innate** (or **native** or **natural**) immune system and the **adaptive** (or **acquired** or **specific**) immune system.

Innate immunity provides the body's early line of defense against microbial invaders. It comprises four types of defensive barriers:

- **Anatomic** or physical (skin, mucous membranes)
- **Physiologic** (temperature, pH, and chemicals such as lysozyme, complement, and some interferons)
- **Phagocytic** (monocytes, neutrophils, macrophages)
- **Inflammatory** events

Innate immune defenses have in common that they:

- Are **present intrinsically** with or without previous stimulation
- Have **limited specificity** for shared structures of microbes
- Are **not enhanced** in activity by repeated exposure
- Have **limited diversity** of expression

Once the barriers of the innate immune response have been breached, the adaptive immune response is activated in an antigen-specific fashion to provide for the elimination of antigen and lasting protection from future challenge. The components of the adaptive immune system are:

- **Lymphocytes** (T cells and B cells) and plasma cells (end cells of B-lymphocyte differentiation)
- **Antigen-presenting cells** (macrophages, B cells, and dendritic cells)

In A Nutshell

The immune system has two arms:

- Innate
- Adaptive

In A Nutshell

The Innate Arm (Anatomic, Physiologic, Phagocytic, Inflammatory)

- Present intrinsically
- Nonspecific
- No memory
- Limited diversity

In A Nutshell

The Adaptive Arm (Lymphocytes and Their Products)

- Inducible
- Specific
- Memory
- Extensive diversity
- Self versus non-self distinction
- Self-limiting

Adaptive immune defenses have in common that they are:

- **Specific** for particular antigens and are specialized to provide the best protection
- **Diverse** in their specificity
- Enhanced with each repeated exposure (express **immunologic memory**)
- Capable of **self/non-self** recognition
- **Self-limiting**

These features of adaptive immunity are designed to give the individual the best possible defense against disease. **Specificity** is required, along with **memory**, to protect against persistent or recurrent challenge. **Diversity** is required to protect against the maximum number of potential pathogens. **Specialization** of function is necessary so that the most effective defense can be mounted against diverse challenges. The ability to distinguish between invaders and one's own cells and tissues (**self versus non-self**) is vital in inhibiting a response to one's own cells (**autoimmunity**). **Self-limitation** allows the system to return to a basal resting state after a challenge to conserve energy and prepare for the challenge by new microbes.

Table II-1-1. Comparison of Innate and Adaptive Immunity

Characteristics	Innate	Adaptive
Specificity	For structures shared by groups of microbes	For specific antigens of microbial and nonmicrobial agents
Diversity	Limited	High
Memory	No	Yes
Self-reactivity	No	No
Components		
Anatomic and chemical barriers	Skin, mucosa, chemicals (lysozyme, interferons α and β), temperature, pH	Lymph nodes, spleen, mucosal-associated lymphoid tissues
Blood proteins	Complement	Antibodies
Cells	Phagocytes and natural killer (NK) cells	Lymphocytes (other than NK cells)

In A Nutshell

- Antibodies and complement enhance phagocytosis.
- Antibodies activate complement.
- Cytokines stimulate adaptive and innate responses.

The innate and adaptive arms of the immune response do not operate independently of one another.

- Phagocytic cells process and display antigen to facilitate stimulation of specific T lymphocytes.
- Macrophages secrete immunoregulatory molecules (**cytokines**), which help trigger the initiation of specific immune responses.
- T lymphocytes produce cytokines, which enhance the microbicidal activities of phagocytes.
- Antibodies produced by plasma cells bind to pathogens and activate the complement system to result in the destruction of the invaders.
- Antibodies produced by B lymphocytes bind to pathogens and assist with phagocytosis (**opsonization**).

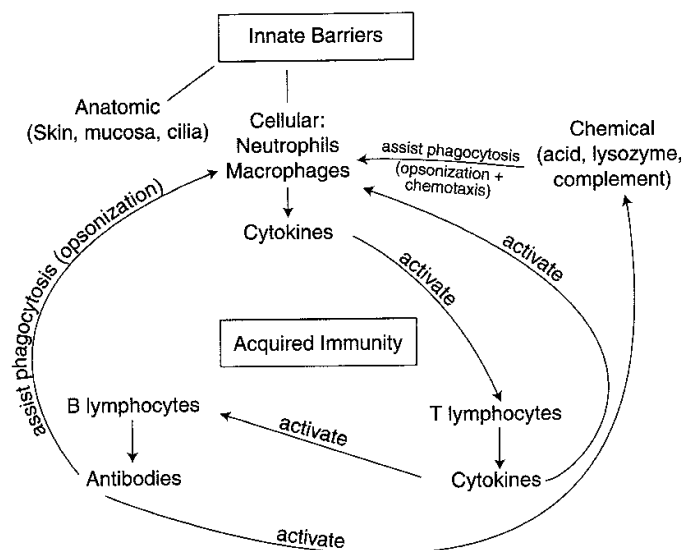


Figure II-1-1. Interaction Between Innate and Adaptive Immune Responses

Chapter Summary

- The immune system has two arms, innate and adaptive.
- The innate arm is a barrier system consisting of anatomic, physiologic, phagocytic, or inflammatory components.
- The innate arm is present intrinsically, has limited specificity and diversity, and is not enhanced by repeated exposure.
- The adaptive arm consists of T and B lymphocytes and antigen-presenting cells.
- Adaptive immune responses are specific, diverse, self-limiting, capable of self versus non-self recognition, and display memory.
- The innate and adaptive arms interact with and augment each other through soluble substances such as antibodies, complement, and cytokines.

Cells of the Immune System

2

What the USMLE Requires You To Know

- The cells of the immune system, their origin, tissue distribution, and function
- The structure and function of antigen-recognition molecules of B and T lymphocytes
- The make-up of the signal transduction complex of B and T lymphocytes
- The basic mechanism of gene-segment rearrangement to generate receptor diversity

ORIGIN

The cells of the immune system arise from a pluripotent stem cell in the bone marrow. Differentiation of this cell will occur along one of two pathways, giving rise to either a common lymphoid progenitor cell or a common myeloid progenitor cell. The common lymphoid progenitor cell gives rise to B lymphocytes, T lymphocytes, and natural killer (NK) cells. The myeloid progenitor gives rise to erythrocytes, platelets, basophils, mast cells, eosinophils, neutrophils, monocytes, macrophages, and dendritic cells.

In A Nutshell

- The lymphoid progenitor makes B cells, T cells, and NK cells.
- The myeloid progenitor makes red blood cells, platelets, basophils, mast cells, eosinophils, neutrophils, monocytes, macrophages, and dendritic cells.

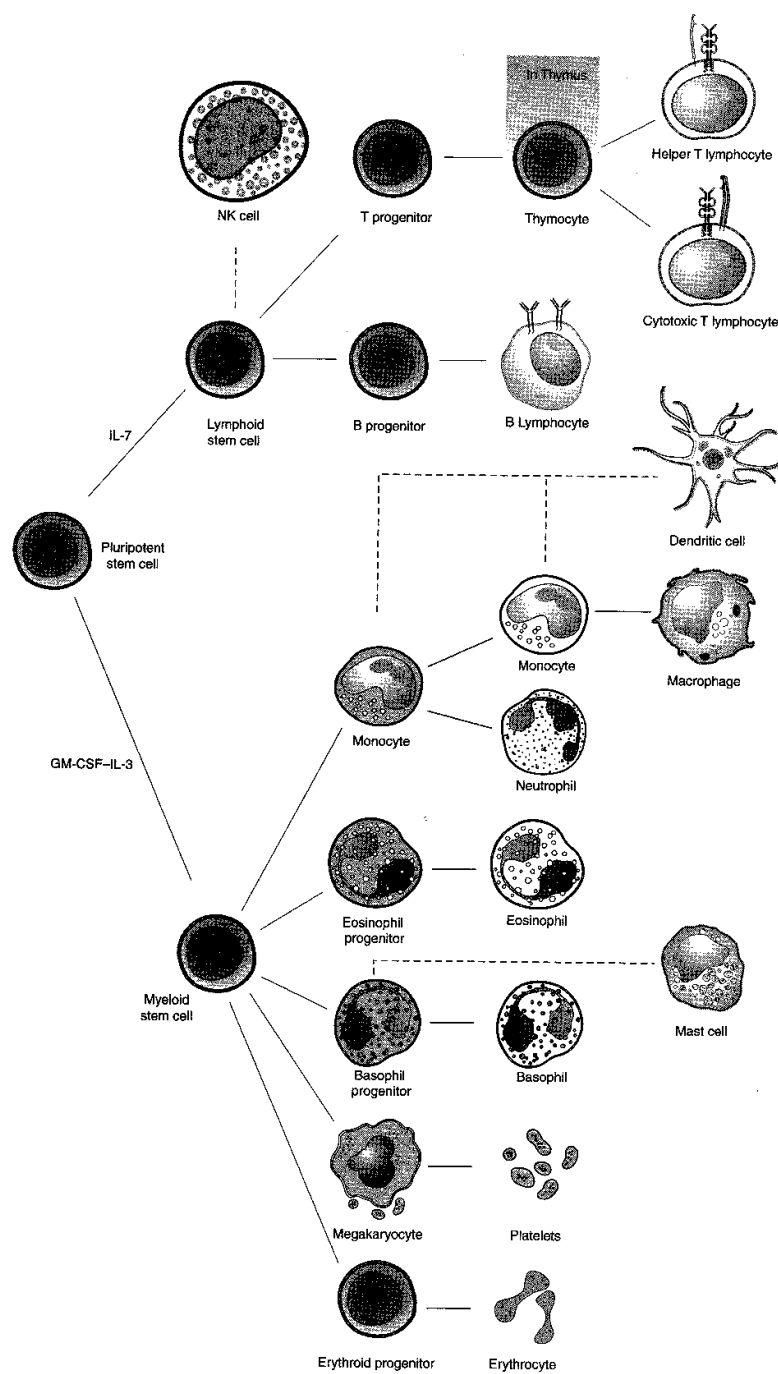


Figure II-2-1. The Ontogeny of Immune Cells

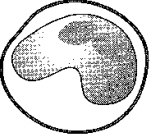
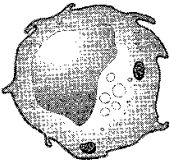
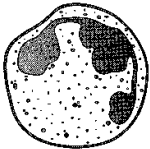
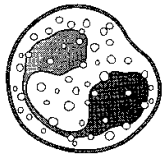
FUNCTION

The white blood cells of both myeloid and lymphoid stem cell origin have specialized functions in the body once their differentiation from the bone marrow is complete. Cells of myeloid heritage perform relatively stereotyped responses and are thus considered members of the innate branch of the immune system. Cells of the lymphoid lineage perform finely tuned, antigen-specific roles in immunity.

In a Nutshell

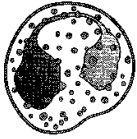
- Myeloid cells are in the innate branch.
- Lymphoid cells (except NK cells) are in the adaptive branch.

Table II-2-1. Myeloid Cells

Myeloid Cell	Tissue Location	Identification	Function
Monocyte 	Bloodstream, 0–900/ μ L	Horseshoe-shaped nucleus	Phagocytic, differentiate into tissue macrophages
Macrophage 	Tissues	Ruffled membrane, cytoplasm with vacuoles and vesicles	Phagocytosis, secretion of cytokines
Dendritic cell 	Epithelia, tissues	Long cytoplasmic arms	Antigen capture, transport, and presentation
Neutrophil 	Bloodstream, 1,800–7,800/ μ L	Multilobed nucleus; small pink granules	Phagocytosis and activation of bactericidal mechanisms
Eosinophil 	Bloodstream, 0–450/ μ L	Bilobed nucleus, large pink granules	Killing of antibody- coated parasites

(Continued)

Table II-2-1. Myeloid Cells (*continued*)

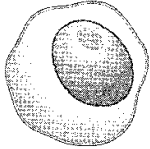
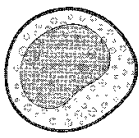
Myeloid Cell	Tissue Location	Identification	Function
	Bloodstream, 0-200/ μ L	Bilobed nucleus, large blue granules	Nonphagocytic, release pharmacologically active substances during allergic responses
	Tissues, mucosa, and epithelia	Small nucleus, cytoplasm packed with large blue granules	Release of granules containing histamine, etc., during allergic responses

In A Nutshell

- B lymphocytes are generated and mature in the bone marrow.
- T lymphocytes undergo maturation in the thymus.
- NK cells are large, granular lymphocytes.

Although lymphocytes in the bloodstream and tissues are nearly morphologically indistinguishable at the light microscopic level, we now know that there are several distinct but interdependent lineages of these cells: **B lymphocytes**, so called because they complete their development in the **bone marrow**, and **T lymphocytes**, so called because they pass from their origin in the bone marrow into the **thymus**, where they complete their development. Both have surface membrane receptors designed to bind specific antigens. The third type of lymphocyte, the **natural killer (NK) cell**, is a **large, granular lymphocyte** that recognizes certain tumor and virus-infected cells (See Chapter 8).

Table II-2-2. Lymphoid Cells

Lymphoid Cell	Location	Identification	Function
	Bloodstream, 1,000–4,000/ μ L; lymph nodes, spleen, submucosa, and epithelia	Large, dark nucleus, small rim of cytoplasm	B cells produce antibody T helper cells regulate immune responses Cytotoxic T cells (CTLs) kill altered or infected cells
	Bloodstream, $\leq 10\%$ of lymphocytes	Lymphocytes with large cytoplasmic granules	Kill tumor/virus cell targets or antibody-coated target cells
	Lymph nodes, spleen, mucosal-associated lymphoid tissues, and bone marrow	Small dark nucleus, intensely staining Golgi apparatus	End cell of B-cell differentiation, produce antibody

THE ANTIGEN RECOGNITION MOLECULES OF LYMPHOCYTES

Each of the cells of the lymphoid lineage is now clinically identified by the **characteristic surface molecules** that they possess, and much is known about these structures, at least for B and T cells. The B lymphocyte, in its mature ready-to-respond form (the naive B lymphocyte), wears molecules of two types of antibody or immunoglobulin called IgM and IgD embedded in its membrane. The naive T cell wears a single type of genetically related molecule, called the **T-cell receptor (TCR)**, on its surface. Both of these types of antigen receptors are encoded within the immunoglobulin superfamily of genes and are expressed in literally millions of variations in different lymphocytes as a result of complex and random rearrangements of the cells' DNA.

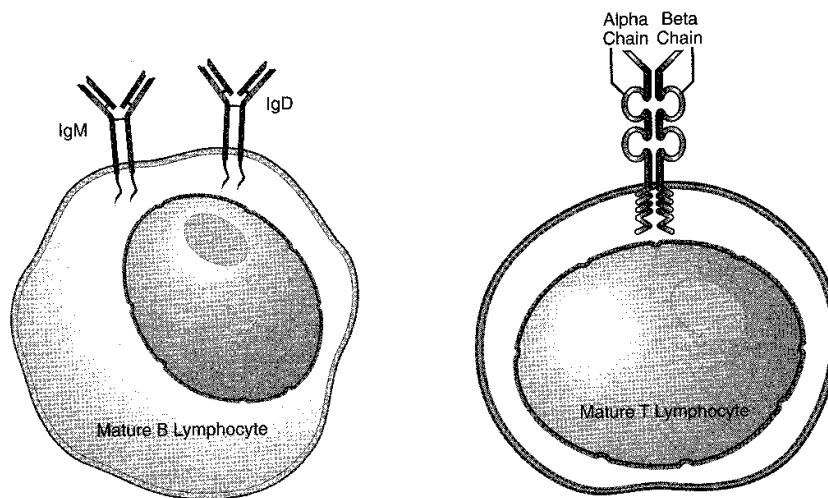


Figure II-2-2. Antigen Receptors of Mature Lymphocytes

The antigen receptor of the B lymphocyte, or **membrane-bound immunoglobulin**, is a four-chain glycoprotein molecule that serves as the basic monomeric unit for each of the distinct antibody molecules destined to circulate freely in the serum. This monomer has two identical halves, each composed of a long, or **heavy chain** (μ for immunoglobulin [Ig] M and δ for IgD), and a shorter, **light chain** (κ or λ). A cytoplasmic tail on the carboxy-terminus of each heavy chain extends through the plasma membrane and anchors the molecule to the cell surface. The two halves are held together by disulfide bonds into a shape resembling a "Y," and some flexibility of movement is permitted between the halves by disulfide bonds forming a **hinge region**. On the N-terminal end of the molecule where the heavy and light chains lie side by side, a "pocket" is formed whose three-dimensional shape will accommodate the noncovalent binding of one, or a very small number, of related antigens. The unique three-dimensional shape of this pocket is called the **idiotype** of the molecule, and although two classes (**isotypes**) of membrane immunoglobulin (IgM and IgD) are coexpressed (defined by amino acid sequences toward the carboxy terminus of the molecule), only one idiotype or **antigenic specificity** is expressed per cell (although in multiple copies). Each human individual is capable of producing hundreds of millions of unique idiotypes.

In a Nutshell

- The naive B-cell antigen receptors are IgM and IgD.
- The T-cell antigen receptor is made of α and β chains.

In A Nutshell

- Membrane-bound Ig has two heavy and two light chains.
- A "hinge" region joins the heavy chains.
- The idiotype of the molecule resides in the N-terminal pocket of heavy and light chains.
- The isotype of the molecule is determined by domains toward the C-terminus.

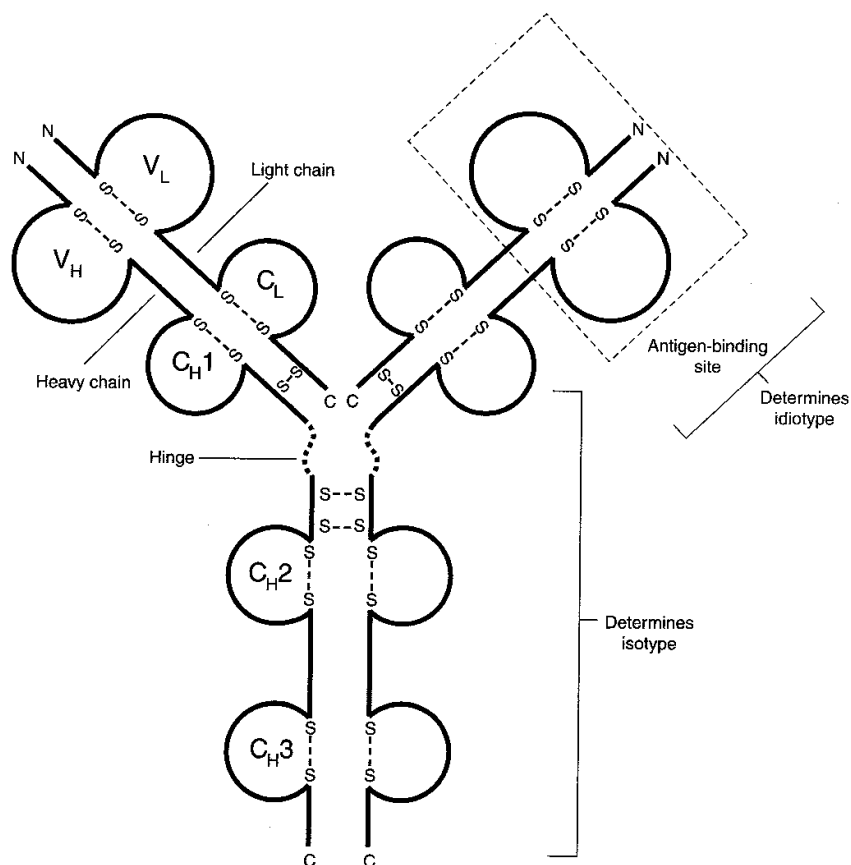


Figure II-2-3. The B-Lymphocyte Antigen Recognition Molecule (Membrane-Bound Immunoglobulin)

In A Nutshell

- The T-cell receptor has α/β chains.
- It binds peptides presented by antigen-presenting cells.
- The molecule is rigid.
- The molecule is always cell-bound.

In A Nutshell

- B cells recognize unprocessed antigens.
- T cells recognize cell-bound peptides.

In A Nutshell

- The B-cell signal transduction complex is Ig- α , Ig- β , CD19, and CD21.
- The T-cell signal transduction complex is CD3.

The antigen receptor of the T lymphocyte is composed of two glycoprotein chains that are similar in length and are thus designated α and β chains. A small proportion of T cells expresses an alternative TCR composed of γ and δ chains, but the function of these cells is poorly understood. On the carboxy-terminus of α/β chains, a cytoplasmic tail extends through the membrane for anchorage. On the N-terminal end of the molecule, a groove is formed between the two chains, whose three-dimensional shape will accommodate the binding of a small antigenic **peptide** presented on the surface of an antigen-presenting cell (macrophage, dendritic cell, or B lymphocyte). This groove forms the idio- type of the TCR. Notice that there is no hinge region present in this molecule, and thus its conformation is quite rigid.

The membrane receptors of B lymphocytes are designed to bind **unprocessed antigens** of almost any chemical composition, whereas the TCR is designed to bind only **cell-bound peptides**. Also, although the B-cell receptor is ultimately modified to circulate freely in the plasma as secreted **antibody**, the TCR is never released from its membrane-bound location.

In association with these unique antigen-recognition molecules on the surface of B and T cells, accessory molecules are found whose function is in signal transduction. Thus, when a lymphocyte

binds to an antigen complementary to its idiotype, a cascade of messages transferred through its **signal transduction complex** will culminate in intracytoplasmic phosphorylation events, which will activate the cell. In the B cell, this signal transduction complex is composed of two single-chain immunoglobulin relatives known as Ig- α and Ig- β and two other molecules designated CD (cluster of differentiation) 19 and 21. In the T cell, the signal transduction complex is a multichain structure called CD3.

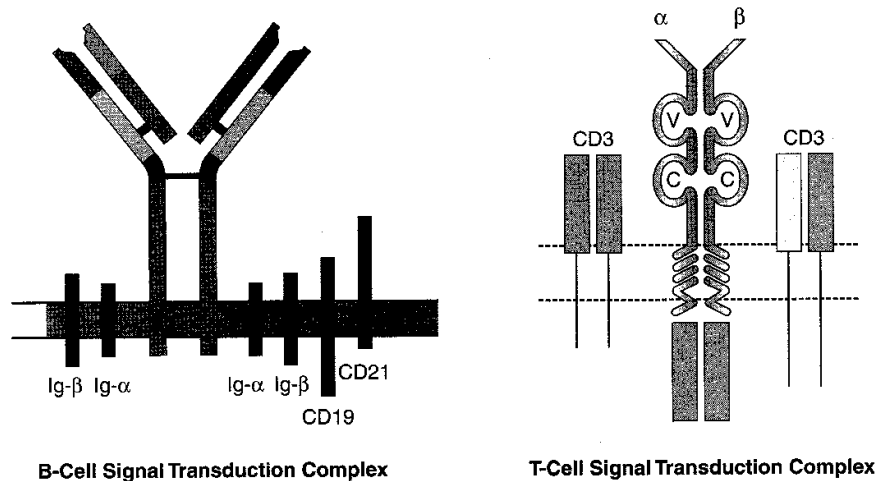


Figure II-2-4

Table II-2-3. Comparison of B- and T-Lymphocyte Antigen Receptors

Property	B-Cell Antigen Receptor	T-Cell Antigen Receptor
Molecules/Lymphocyte	100,000	100,000
Idiotypes/Lymphocyte	1	1
Isotypes/Lymphocyte	2 (IgM and IgD)	1 (α/β)
Is secretion possible?	Yes	No
Number of combining sites/molecule	2	1
Mobility	Flexible (hinge region)	Rigid
Signal-transduction molecules	Ig- α , Ig- β , CD19, CD21	CD3

THE GENERATION OF RECEPTOR DIVERSITY

Because the body requires the ability to respond specifically to all of the millions of potential harmful agents it may encounter in a lifetime, a mechanism must exist to generate the millions of idiotypes of antigen receptors necessary to meet this challenge. If each of these idiotypes were encoded separately in the germline DNA of lymphoid cells, it would require more DNA than is present in the entire cell. The generation of this necessary diversity is accomplished by a complex and unique set of rearrangements of DNA segments that takes place during the maturation of lymphoid cells.

In A Nutshell

- Millions of distinct idiotypes are generated by rearranging gene segments, which code for the variable domains of the B- or T-cell receptors.
- Three gene segments (V, D, and J) are combined to create the variable domain of the B cell heavy chain or the TCR β chain.

In the first place, it was discovered that individuals inherit a large number of different segments of DNA, which may be recombined and alternatively spliced to create unique amino acid sequences in the N-terminal ends (**variable domains**) of the chains that compose their antigen recognition sites. For example, to produce the **heavy chain variable domains** of their antigen receptor, B-lymphocyte progenitors select randomly and in the absence of stimulating antigen to recombine three gene segments designated variable (V), diversity (D), and joining (J) out of hundreds of germline-encoded possibilities to produce unique sequences of amino acids in the variable domains (VDJ recombination). An analogous random selection is made during the formation of the β chain of the TCR.

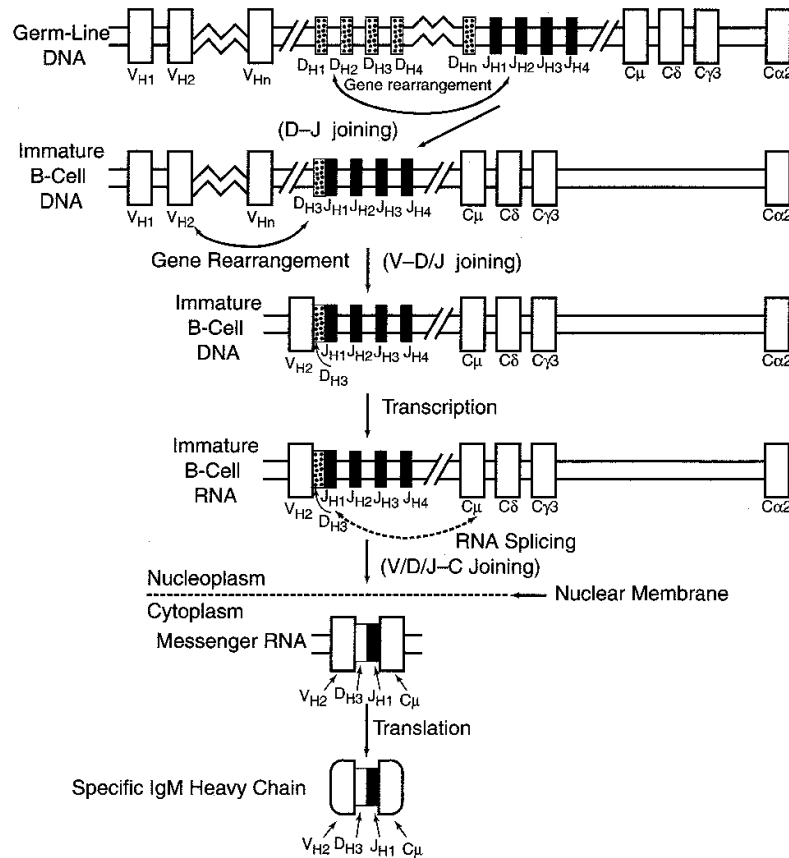


Figure II-2-5. Production of Heavy (B-Cell) or Beta (T-Cell) Chains of Lymphocyte Antigen Receptors

In A Nutshell

Two gene segments (V and J) are combined to create the B-cell light chain or the TCR- α chain of the T-cell receptor.

Next, the B-lymphocyte progenitor performs random rearrangements of two types of gene segments (V and J) to encode the **variable domain amino acids of the light chain**. An analogous random selection is made during the formation of the α chain of the TCR.

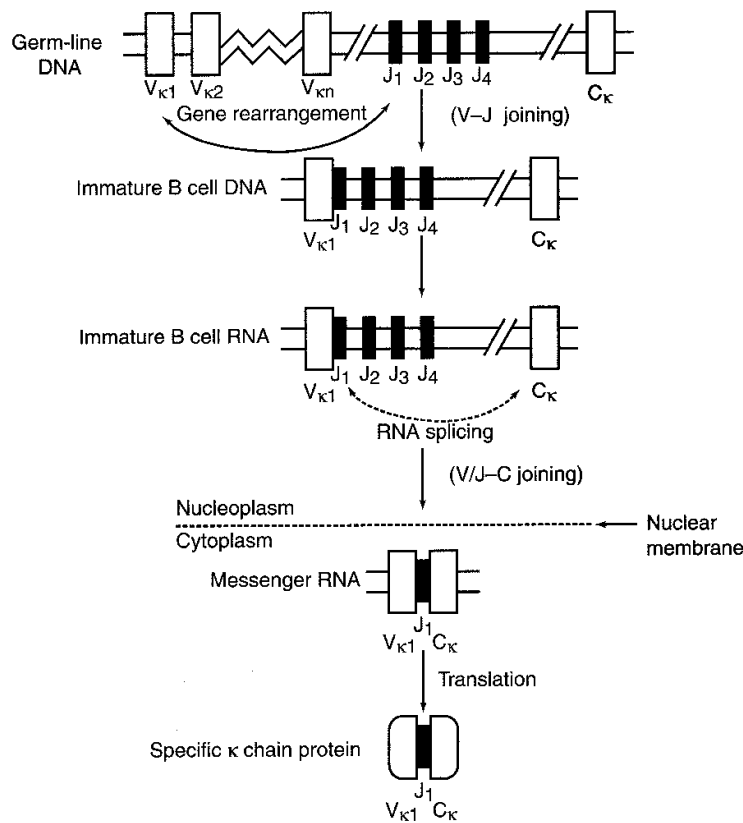


Figure II-2-6. Production of Light (B Cell) or Alpha (T Cell) Chain of a Lymphocyte Antigen Receptor

While heavy chain gene segments are undergoing recombination, the enzyme **terminal deoxynucleotidyl transferase** (Tdt) randomly inserts bases (without a template on the complementary strand) at the junctions of V, D, and J segments (N-nucleotide addition). When the light chains are rearranged later, Tdt is not active, but it is active during the rearrangement of all gene segments in the formation of the TCR. This generates even more diversity than the random combination of V, D, and J segments alone.

Needless to say, many of these gene segment rearrangements result in the production of truncated or nonfunctional proteins. When this occurs, the cell has a second chance to produce a functional strand by rearranging the gene segments of the homologous chromosome. If it fails to make a functional protein from rearrangement of segments on either chromosome, the cell is induced to undergo **apoptosis** or programmed cell death. In this way, the cell has two chances to produce a functional heavy (or β) chain. A similar process occurs with the light or α chain. Once a functional product has been achieved by one of these rearrangements, the cell shuts off the rearrangement and expression of the other allele on the homologous chromosome—a process known as **allelic exclusion**. This process ensures that B and T lymphocytes synthesize only **one specific antigen-receptor per cell**.

In A Nutshell

- The enzyme Tdt inserts bases randomly at the junctions of V, D, and J and creates more variability.
- Once a functional product has been made, the homologous chromosome is inactivated (allelic exclusion).

Clinical Correlate

Tdt is used as a marker for early stage T- and B-cell development in acute lymphoblastic leukemia. (See Chapter 15.)

In A Nutshell

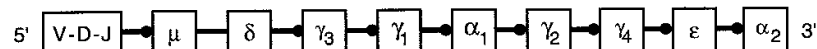
- The products of variable domain rearrangements are alternatively spliced at the RNA level to IgM or IgD constant domains to form the B-cell antigen receptor.
- IgM and IgD molecules with identical idiotypes are expressed on mature B cells.

Because any heavy (or β) chain can associate with any randomly generated light (or α) chain, one can multiply the number of different possible heavy chains by the number of different possible light chains to yield the total number of possible idiotypes that can be formed. This generates yet another level of diversity.

Table II-2-4. Summary of Mechanisms for Generating Receptor Diversity

Mechanism	Cell in Which It Is Expressed
Existence in genome of multiple V, D, J segments	B and T cells
VDJ recombination	B and T cells
N-nucleotide addition	B cells (only heavy chain) T cells (all chains)
Combinatorial association of heavy and light chains	B and T cells
Somatic hypermutation	B cells only, after antigen stimulation (<i>see</i> Chapter 7)

Downstream on the germline DNA from the coding segments, which have now been rearranged to yield the variable domain that will serve as the antigen-combining site of the molecule, are encoded in sequence, the amino acid sequences of all of the remaining domains of the chain. These domains tend to be similar within the classes or isotypes of immunoglobulin or TCR chains and are thus called constant domains. The first set of constant domains for the heavy chain of immunoglobulin that is transcribed is that of IgM and next, IgD. These two sets of domains are alternatively spliced to the variable domain product at the RNA level. There are only two isotypes of light chain constant domains, named κ and λ , and one will be combined with the product of light chain variable domain rearrangement to produce the other half of the final molecule. Thus, the B lymphocyte produces IgM and IgD molecules with identical idiotypes and inserts these into the membrane for antigen recognition.

**Figure II-2-7. Immunoglobulin Heavy Chain DNA**

Chapter Summary

- The cells of the immune system arise from a pluripotent stem cell in the bone marrow.
- The common lymphoid progenitor will give rise to B lymphocytes, T lymphocytes, and NK cells.
- The common myeloid progenitor will give rise to erythrocytes, platelets, basophils, mast cells, eosinophils, neutrophils, monocytes, macrophages, and dendritic cells.
- The phagocytic cells of the myeloid series include monocytes, macrophages, dendritic cells, neutrophils, and eosinophils.
- Basophils and mast cells are nonphagocytic cells, which mediate allergic responses.
- B lymphocytes secrete immunoglobulin; T cells may be helper or killer cells; and NK cells kill tumor or virus-infected target cells.
- Plasma cells are the end cells of B-lymphocyte differentiation and secrete antibody.
- The antigen receptor of the B lymphocyte is membrane-bound IgM and IgD and is designed to bind unprocessed antigens of almost any chemical composition.

The antigen receptor of the T lymphocyte is composed of two chains (α/β or γ/δ) and is designed to recognize cell-bound peptides.

- B-cell antigen receptors can be secreted, whereas T-cell receptors are always cell-bound.
- The antigen receptors of B and T cells are associated with signal transduction molecules: Ig α , Ig β , CD19, and CD21 for B cells and CD3 for T cells.
- The diversity of idiotypes of antigen-combining sites is generated by rearrangements of gene segments coding for variable domain amino acids and is assisted by the action of the enzyme terminal deoxynucleotidyl transferase.
- There are two major points when considering gene rearrangement:
 - 1) The difference between the heavy and light chains is the presence of the D region in the heavy chain, and
 - 2) Only rearranged genes can actually be expressed; therefore, only lymphocytes express antigen receptors.
- Allelic exclusion is the process by which one chromosome of a homologous pair will be inactivated, and it ensures that only one idioype of antigen-recognition molecule will be produced per cell.

Review Questions

1. A germline B lymphocyte possesses 200 distinct V region genes, 5 J region genes, and 2 isotypic possibilities to rearrange for its selection of light chain synthesis. Assuming no recombinational inaccuracies, how many distinct idiotypes could be produced by combining this coding sequence with one heavy chain?
 - (A) 10
 - (B) 205
 - (C) 400
 - (D) 1000
 - (E) 2000

2. Isotype switching during B-cell ontogeny dedicates mature B cells to production of a single heavy chain isotype, except in the case of IgM and IgD, which can be expressed concomitantly. How is this expression of both isotypes simultaneously possible?
 - (A) Allelic exclusion
 - (B) Allelic codominance
 - (C) Affinity maturation
 - (D) Alternative RNA splicing
 - (E) Somatic hypermutation

3. A 4-year-old Caucasian boy is brought to his pediatrician with complaints of abnormal bruising and repeated bacterial infections. A blood workup reveals thrombocytopenia and neutropenia and the presence of numerous small, dense lymphoblasts with scant cytoplasm. Immunophenotyping of the abnormal cells determines them to be extremely primitive B cells, which are CD19+, HLA-DR+, and Tdt+. Which of the following best describes the status of immunoglobulin chain synthesis most likely in these cells?
 - (A) IgM heavy chains inserted in the membrane
 - (B) IgM heavy chains present in the cytoplasm
 - (C) IgM monomers inserted in the membrane
 - (D) IgM monomers present in the cytoplasm
 - (E) No immunoglobulin chain synthesis present

4. A young woman with acute myeloblastic leukemia is treated with intensive chemotherapy and achieves remission of her symptoms. Because the prognosis for relapse is relatively high, a bone marrow transplant is undertaken in her first remission. Which of the following cytokines administered with the bone marrow cells would have the beneficial result of stimulating lymphoid-cell development from the grafted stem cells?
- (A) Interleukin (IL)-1
 - (B) IL-2
 - (C) IL-3
 - (D) IL-6
 - (E) IL-7
5. A 2-year-old boy is evaluated for a severe combined immunodeficiency disease. His bone marrow has normal cellularity. Radioactive tracer studies demonstrate a normal number of T-cell precursors entering the thymus, but no mature T lymphocytes are found in the blood or peripheral organs. Cells populating the thymus are found to lack CD3. Which of the following capabilities would his cells lack?
- (A) Ability to bind cell-bound peptides
 - (B) Ability to express CD4/CD8 coreceptors
 - (C) Ability to produce terminal deoxyribonucleotidyl transferase
 - (D) Ability to proliferate in response to specific antigen
 - (E) Ability to rearrange T-cell receptor gene segments

Answers and Explanations

1. **The correct answer is D.** The portion of the light chain that will be found within the antigen-combining site (idiotype) of an antibody molecule is formed by random rearrangement of V and J gene segments. Thus, given the numbers here, there are 200×5 different possible combinations. The isotypic (constant domain) possibilities do not play a part in the formation of the idiotype.

Choice A, 10, is not correct. If you selected this answer, you multiplied the number of J region genes times the number of isotypes. This is not a recombination that would produce the idiotype.

Choice B, 205, is not correct. If you selected this answer, you added the number of V region and J region genes together. Although you chose the correct gene segments to recombine, remember that the number of possible combinations of 200 choices and 5 choices requires that you *multiply*, not add, those figures.

Choice C, 400, is not correct. If you selected this answer, you multiplied the number of V region genes times the number of isotypic possibilities. This is not a recombination that would produce the idiotype.

Choice E, 2,000, is not correct. If you selected this answer, you multiplied the number of V region genes times the number of J region genes (to this point you were correct), but then further multiplied by the number of isotypic possibilities. The isotypic possibilities do not play a part in the formation of the idiotype.

2. **The correct answer is D.** Alternative RNA splicing allows a mature B cell to attach either δ or μ chains on a single idiotype that has been generated by germ-line DNA rearrangements.

Allelic exclusion (**choice A**) refers to the expression of products of either parental chromosome type, but not both. This allows lymphoid cells to express only one type of antigen receptor (one idiotype) per cell and is essential to cellular specificity of action.

Allelic codominance (**choice B**) refers to the expression of products of both parental chromosomes simultaneously. It is found in the expression of MHC class I and II products, but not in the expression of immunoglobulin gene products.

Affinity maturation (**choice C**) refers to the increase of affinity (binding strength) of a population of antibodies over time during the development of an immune response. Because the affinity of an antibody is dependent on the goodness-of-fit of its idiotype for its antigen, isotype switching does not affect the shape of the idiotype and does not change the affinity of the molecule.

Somatic hypermutation (**choice E**) is the phenomenon that allows affinity maturation to occur. It is the accelerated mutation of DNA coding within the hypervariable region that occurs during B-cell proliferation in response to antigenic stimulation. Again, the isotype of the antibody does not affect the shape of the idiotype, and this term refers to a process that changes the shape of the idiotype.

3. **The correct answer is E.** This child has acute lymphoblastic leukemia (ALL), and the malignant cells have the characteristics of early B-cell precursors. This leukemia has peak incidence at approximately 4 years of age, is twice as common in whites than in non-whites, and is slightly more frequent in boys than in girls. A leukemic cell that is positive for terminal deoxyribonucleotidyl transferase (Tdt) is in the process of rearranging the gene segments for synthesis of the heavy chain of immunoglobulin but will not yet have completed a functional product. Tdt is active for all heavy-domain gene segment rearrangements but is not used during light-chain gene segment rearrangements.

IgM heavy chains inserted in the membrane (**choice A**) would be found in cells that have completed the rearrangement of their heavy chain variable domain gene segments, and these may transiently be expressed on the surface of a cell in association with a surrogate light chain before light chain rearrangement is complete. These cells would not be using their Tdt any more.

IgM heavy chains in the cytoplasm (**choice B**) would be found in leukemic cells that are more highly differentiated than those described. Once the variable domain gene segments for the heavy chain have been successfully rearranged in a cell, μ chains can be found in the cytoplasm. In ALL, this is usually associated with a decreased expression of Tdt and appearance of CD10 (the common acute lymphoblastic leukemia antigen; CALLA) and CD20.

IgM monomers inserted in the membrane (**choice C**) would be found in leukemic cells that are at the mature B-cell stage. Such cells would have completed the rearrangements for both heavy and light chains and would lack Tdt as a marker. They would express surface HLA-DR, CD19, and CD20 in addition to surface immunoglobulin.

IgM monomers present in the cytoplasm (**choice D**) would be found in cells that have completed the rearrangement of their variable domain gene segments. They would no longer express Tdt.

4. **The correct answer is E.** The cytokine most strongly associated with stimulation of production of lymphoid cells from the bone marrow is interleukin (IL)-7.

IL-1 (**choice A**) is the endogenous pyrogen. It is produced by macrophages and acts on the hypothalamus to raise the temperature set point. It is associated with systemic inflammatory processes, but is not known to have an effect on lymphopoiesis.

IL-2 (**choice B**) is a product of T cells that stimulates proliferation of T cells in the periphery. It is not known to have an effect on lymphopoiesis.

IL-3 (**choice C**) is the cytokine that is most strongly associated with stimulation of myeloid cell precursors in the bone marrow.

IL-6 (**choice D**) is a second endogenous pyrogen. It causes production of acute-phase proteins from hepatocytes and acts on myeloid stem cells in the bone marrow to induce differentiation.

5. **The correct answer is D.** CD3 is the signal transduction complex in T lymphocytes. When specific antigen binding has occurred on the surface of the cell, this complex is responsible for transferring the message to the cytoplasm of the cell. This culminates in intracytoplasmic phosphorylation events, which activate the cell and induce its proliferation (cloning). A cell lacking CD3 would be capable of binding specific antigen, but incapable of activation and proliferation in response to that first signal.

Ability to bind cell-bound peptides (**choice A**) would not be affected by the absence of CD3. Binding to peptides presented by antigen-presenting cells is through interaction of the T-cell receptor with major histocompatibility antigens on the surface of other cells.

Ability to express coreceptors (**choice B**) would not be affected by the absence of CD3, although cells would not be able to complete their differentiation in the thymus and become fully committed T cells.

Ability to produce terminal deoxyribonucleotidyl transferase (**choice C**) would not be affected by the absence of the T-cell signal transduction complex. T-cell precursors rearrange their receptor gene segments (and use terminal deoxyribonucleotidyl transferase) in the absence of antigenic stimulation and before signal transduction through CD3 becomes critical.

Ability to rearrange T-cell receptor gene segments (**choice E**) would not be affected by the absence of the T-cell signal transduction complex. T-cell precursors rearrange their receptor gene segments in the absence of antigenic stimulation and before signal transduction through CD3 becomes critical.

The Selection of Lymphocytes

3

What The USMLE Requires You To Know

- The primary lymphoid organs: structure and function
- The ontogeny of T- and B-lymphocyte cell surface markers
- The structure and function of MHC gene products

As lymphoid progenitors develop in the bone marrow, we have seen that they make random rearrangements of their germline DNA to produce the unique idiotypes of antigen-recognition molecules that they will use throughout their lives. The bone marrow, therefore, is considered a **primary lymphoid organ** in humans because it supports and encourages these early developmental changes. B lymphocytes complete their entire formative period in the bone marrow and can be identified in their progress by the immunoglobulin chains they produce.

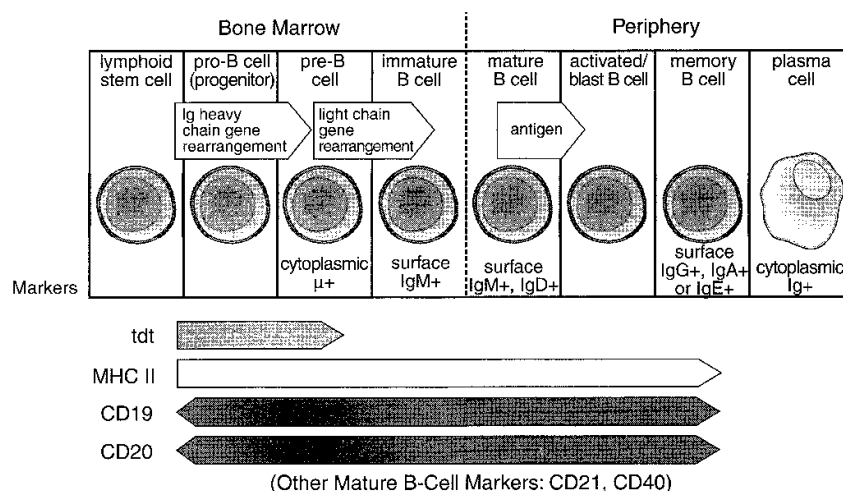


Figure II-3-1. B-Cell Differentiation

Because these gene segment rearrangements occur randomly and in the absence of stimulation with foreign antigen, it stands to reason that many of the idiotypes of receptors produced could have a binding attraction or **affinity** for normal body constituents. These cells, if allowed to develop further, could develop into self-reactive lymphocytes that could cause harm to the host.

In A Nutshell

Primary lymphoid organs are sites of lymphoid-cell development (lymphopoiesis).

- Bone marrow
- Thymus

Secondary lymphoid organs are sites of antigen exposure.

- Spleen
- Lymph nodes
- Mucosal-associated lymphoid tissues

In A Nutshell

Clonal anergy and clonal deletion produce self-tolerance.

In A Nutshell

T-cell precursors leave the bone marrow to undergo selection and maturation in the thymus.

Therefore, one of the key roles of the bone marrow stroma and interdigitating cells is to remove such potentially harmful products. Cells whose idiotype has too great an affinity for normal cellular molecules are either deleted in the bone marrow (**clonal deletion**) or inactivated in the periphery (**clonal anergy**). In such a way, only those cells that are **selectively unresponsive (tolerant)** to self-antigens are allowed to leave the bone marrow.

Immature lymphocytes destined to the T-cell lineage leave the bone marrow and proceed to the **thymus**, the second **primary lymphoid organ** dedicated to the maturation of T cells. The thymus is a bilobed structure located above the heart that consists of an outer **cortex** packed with immature T cells and an inner **medulla** into which cells pass as they mature. Both the cortex and medulla are laced with a network of epithelial cells, dendritic cells, and macrophages, which interact physically with the developing thymocytes.

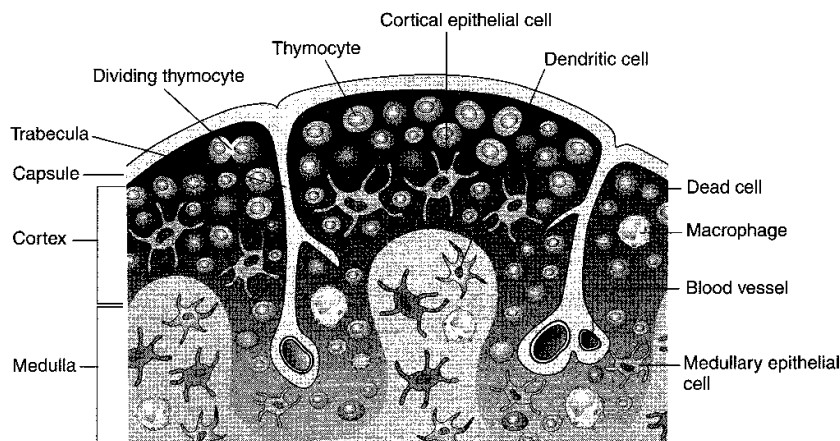


Figure II-3-2. The Structure of the Thymus

In A Nutshell

Thymocytes are exposed to MHC class I and II antigens.

As the developing thymocytes begin to express their TCRs, they are subjected to a rigorous two-step selection process. Because the TCR is designed to bind antigenic peptides presented on the surface of **antigen-presenting cells (APCs)** in the body, a selection process is necessary to remove those cells that would bind to normal self antigens and cause **autoimmunity**, as well as those that have no attraction whatsoever for the surfaces of APCs. This is accomplished by exposure of developing thymocytes to high levels of a unique group of membrane-bound molecules known as **major histocompatibility complex (MHC)** antigens.

The MHC is a collection of highly polymorphic genes on the short arm of chromosome 6 in the human. There are two classes of cell-bound MHC gene products (classes I and II). Both class I and class II molecules are expressed at high density on the surface of cells of the thymic stroma.

In A Nutshell

MHC Class I

- α chain plus β_2 -microglobulin
- Codominantly expressed
- All nucleated cells of the body

Table II-3-1. Class I and II Gene Products

Class I Gene Products			Class II Gene Products		
HLA-A	HLA-B	HLA-C	HLA-DP	HLA-DQ	HLA-DR

Class I molecules are expressed on all nucleated cells in the body, as well as platelets. They are expressed in codominant fashion, meaning that each cell expresses two A, two B, and two C products (one from each parent). The molecules (A, B, and C) consist of an α heavy chain with

three extracellular domains and an intracytoplasmic carboxy-terminus. A second light chain, β_2 -microglobulin, is not encoded within the MHC and functions in transport of the class I antigen to the cell surface. A groove between the first two extracellular domains of the α chain is designed to accommodate small peptides to be presented to the TCR.

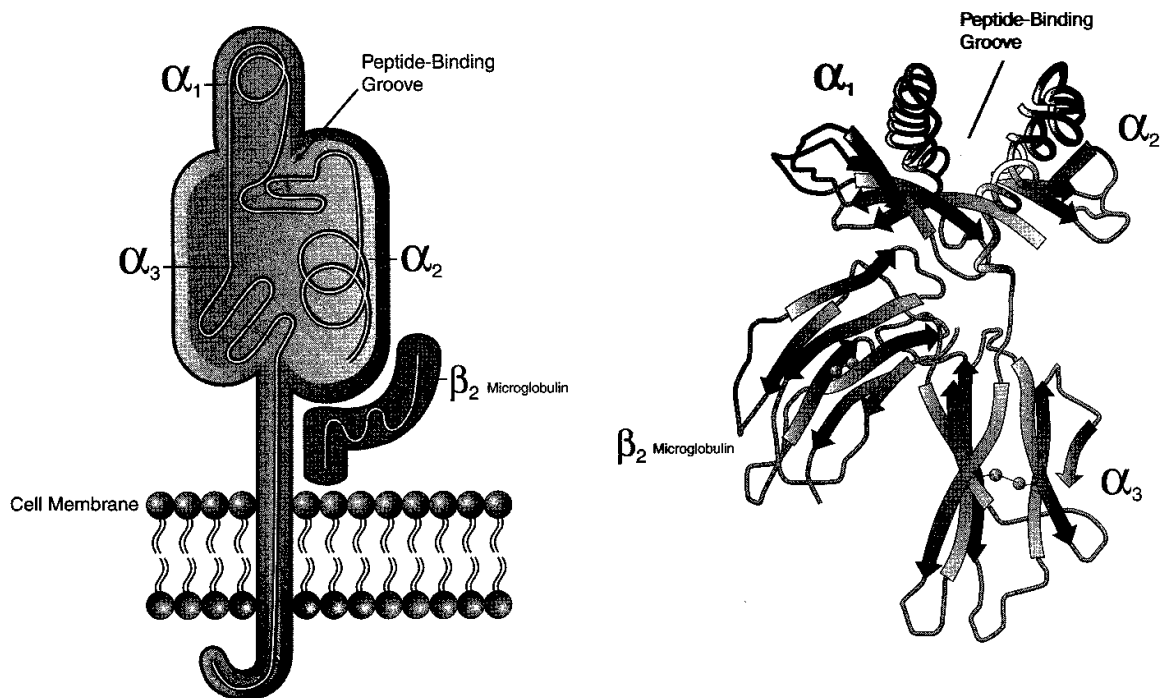


Figure II-3-3. Left) The Class I MHC Molecule, and right) X-Ray Crystallographic Image of Class I MHC Peptide-Binding Groove

In A Nutshell

Class II MHC

- α and β chains
- Expressed codominantly
- Present on APCs

Class II MHC molecules are expressed (also codominantly) on the antigen-presenting cells of the body (macrophages, B lymphocytes, dendritic cells, and Langerhans cells). The molecules are two chain structures of similar length, called α and β , and each possesses two extracellular domains and one intracytoplasmic domain. A groove that will accommodate peptides to be presented to the TCR is formed at the N-terminal end of both chains.

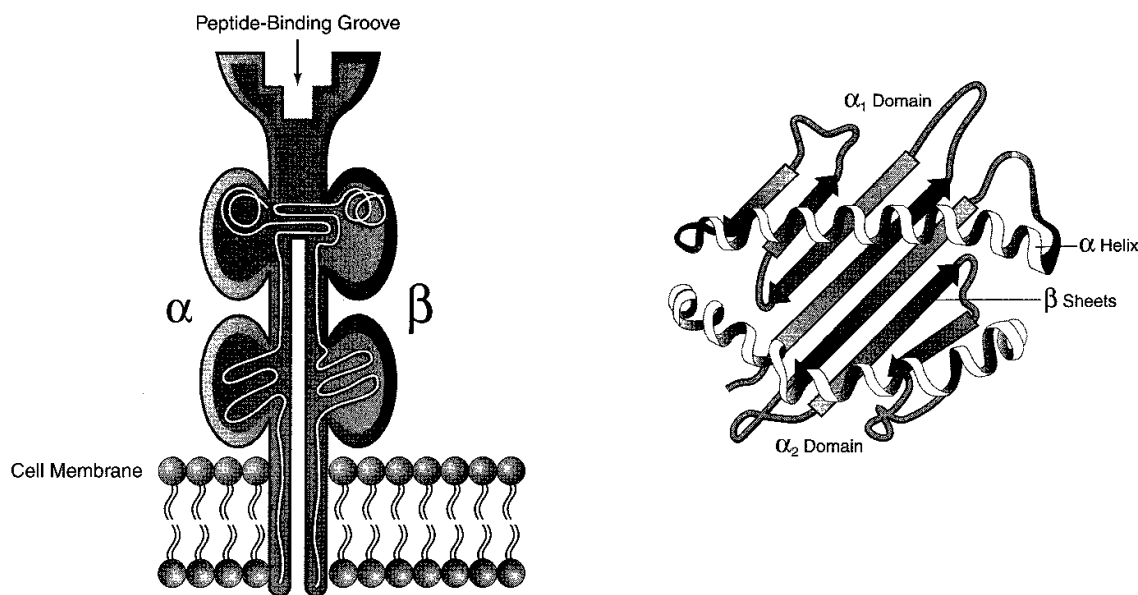


Figure II-3-4. *Left*) The Class II MHC Molecule, and *right*) X-ray Crystallographic Image of Class II MHC Peptide-Binding Groove

In A Nutshell

- Cells with "good" receptors receive positive selection.
- Cells with "useless" receptors receive no positive selection.
- Cells with "bad" receptors receive negative selection.
- CD4 stabilizes MHC II/TCR interaction.
- CD8 stabilizes MHC I/TCR interaction.

Within the thymus, each of these MHC products, loaded with normal self-peptides, is presented to the developing thymocytes. Those that have TCRs capable of binding with low affinity will receive a **positive selection** signal to divide and establish clones that will eventually mature in the medulla. Those that fail to recognize self-MHC at all will not be encouraged to mature (**failure of positive selection**). Those that bind too strongly to self MHC molecules will be induced to undergo apoptosis (**negative selection**) because these cells would have the potential to cause autoimmune disease. Although immature thymocytes express two accessory molecules on their surfaces designed to stabilize the interaction between MHC and TCR called **CD4** and **CD8**, as the affinity of the TCR for class I or class II MHC is "evaluated," the cells are directed to express only CD8 if their TCR binds class I molecules and only CD4 if their TCR binds class II molecules.

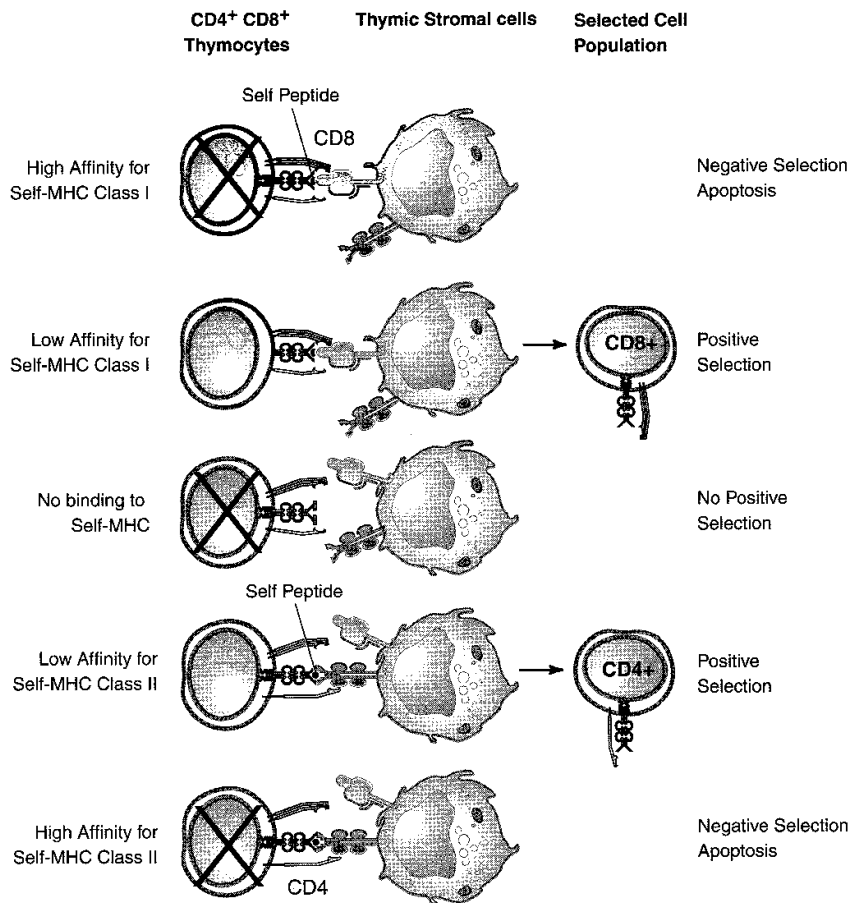


Figure II-3-5. T-Cell Selection in the Thymus

This selection process is an extraordinarily rigorous one. A total of 95 to 99% of all T-cell precursors entering the thymus are destined to die there. Only those with TCRs appropriate to protect the host from foreign invaders will be permitted to leave to the periphery: CD4⁺ cells that recognize class II MHC are destined to become “helper” T cells (TH), and CD8⁺ cells that recognize class I MHC are destined to become **cytotoxic T cells** (CTLs).

In A Nutshell

CD4⁺ cells that recognize class II MHC = TH cells.

CD8⁺ cells that recognize class I MHC = CTLs.

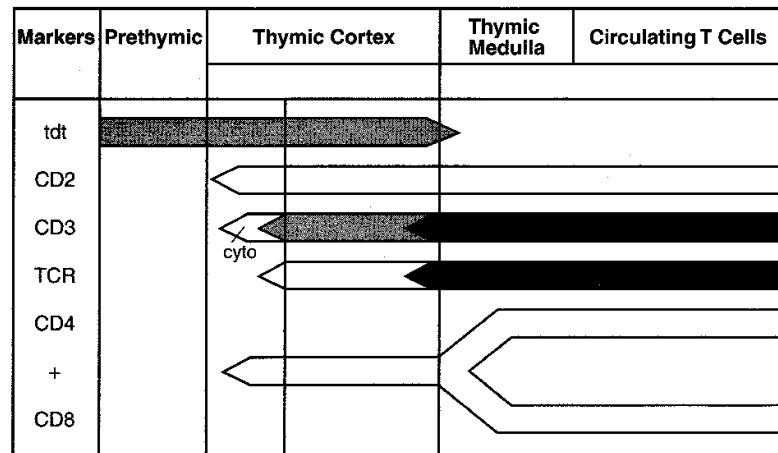


Figure II-3-6. Human T-Cell Differentiation

Chapter Summary

- The bone marrow and thymus are primary lymphoid organs in which the early development and selection of lymphocytes occurs (lymphopoiesis).
- Self-tolerance is induced by deletion of self-reactive cells in the bone marrow (clonal deletion) or inactivation of self-reactive cells in the periphery (clonal anergy).
- T-cell precursors move from the bone marrow to the thymus where they are selected for self-tolerance by exposure to major histocompatibility complex (MHC) antigens on stromal cells.
- Class I MHC products are two chain structures: the α chain is encoded within the MHC and β_2 -microglobulin is not.
- Class I MHC products are expressed on all nucleated cells of the body in a codominant fashion.
- Class II MHC products are two chain structures of which both α and β chains are encoded within the MHC.
- Class II MHC products are expressed on antigen-presenting cells in a codominant fashion.
- Thymocytes with antigen receptors that bind self-peptides presented in the groove of MHC I or II molecules will be induced to undergo apoptosis (negative selection).
- Thymocytes with antigen receptors that have no binding affinity whatsoever for classes I or II MHC are not directed to mature further (failure of positive selection).
- Thymocytes with antigen receptors that can recognize "altered" self are encouraged to clone themselves and mature (positive selection) and express CD4 molecules if their affinity is for MHC class II. These will become helper T cells.
- Thymocytes with antigen receptors that can recognize "altered" self are encouraged to clone themselves and mature (positive selection) and express CD8 molecules if their affinity is for MHC class I. These will become cytotoxic T cells.

Review Questions

1. An 8-year-old boy is diagnosed with acute lymphoblastic leukemia. Flow cytometry is used to determine the immunophenotype of the malignant cells. The patient's cells are evaluated with monoclonal antibodies for MHC class II, CD19, and CD34, and are found to have high levels of fluorescence with all of these markers. They also possess cytoplasmic μ heavy chains. What is the developmental stage of these cells?
 - (A) Immature B cell
 - (B) Lymphoid progenitor cell
 - (C) Mature B cell
 - (D) Pre-B cell
 - (E) Pro-B cell

2. The blood from an 8-year-old boy was analyzed by flow cytometry. The cells were treated with fluorescent-labeled antibodies to various cell surface markers before they were evaluated by flow cytometry. Which of the following markers would identify the B lymphocytes in the sample?
 - (A) CD3
 - (B) CD4
 - (C) CD8
 - (D) CD19
 - (E) CD56

3. An 18-year-old member of a college soccer team is seen by a physician because of chest tightness and dyspnea on exertion. A 15-cm mediastinal mass is detected radiographically. Eighty percent of the white blood cells in the peripheral blood are small, abnormal lymphocytes with lobulated nuclei and scant cytoplasm. Immunophenotyping of the abnormal cells shows them to be CD4+ and CD8+. Where would such cells normally be found in the body?
 - (A) Bone marrow
 - (B) Peripheral blood
 - (C) Thymic cortex
 - (D) Thymic medulla
 - (E) Splenic periarteriolar lymphoid sheaths

4. A 12-year-old child is diagnosed with a T-cell lymphoma. The phenotype of the malignant cell matches that of normal progenitor cells that leave the bone marrow to enter the thymus. What cell surface markers would you expect to find on the malignant cells?
- (A) CD4-, CD8-, TCR-
 - (B) CD4-, CD8-, TCR+
 - (C) CD4-, CD8+, TCR+
 - (D) CD4+, CD8-, TCR+
 - (E) CD4+, CD8+, TCR+
5. Herpes simplex viruses are extremely successful pathogens because they have a variety of immunologic evasion mechanisms. For example, both HSV I and II depress the expression of MHC class I molecules on the surface of infected cells. Which coreceptor's binding would be inhibited by this technique?
- (A) CD2
 - (B) CD4
 - (C) CD8
 - (D) CD16
 - (E) CD56

Answers

- The correct answer is D.** The leukemic cells are pre-B cells. They have rearranged their immunoglobulin genes to encode a μ heavy chain. MHC class II antigens are expressed beginning at the pro-B cell stage, as are CD19 and CD20. CD34 is a marker for early lymphohematopoietic stem and progenitor cells, and it functions as a cell–cell adhesion molecule. These cells would also have expressed CD10, the common acute lymphoblastic leukemia antigen (CALLA), which functions as a metalloendopeptidase.

Immature B cells (**choice A**) have accomplished both and heavy and light immunoglobulin chain rearrangements and therefore express IgM molecules on their cell surface. They would be Tdt-negative, CD19- and CD20-positive, MHC class II-positive, and CD34-negative.

Lymphoid progenitor cells (**choice B**) would not have completed any of the gene rearrangements necessary to create an immunoglobulin molecule. They would be Tdt-negative, MHC class II-negative, CD19- and CD20-negative, and CD34-positive.

Mature B cells (**choice C**) possess surface IgM and IgD molecules and are capable of responding to foreign antigen. They are Tdt-negative, MHC class II-positive, CD19- and CD20-positive, CD34-negative, and may express CD40.

Pro-B cells (**choice E**) are rearranging their immunoglobulin heavy chain gene segments but have not yet completed the process. Therefore, they have no completed chains either cytoplasmically or on their cell surfaces. They would be positive for Tdt, MHC class II, CD19, and CD20.
- The correct answer is D.** The best markers for identification of B lymphocytes are CD19, CD20, and CD21. CD19 and CD21 form a coreceptor complex during B-cell activation. The role of CD20 in B-cell activation is unclear, although it forms a calcium-ion channel. CD21 is also a receptor for the C3d component of complement and the Epstein-Barr virus.

CD3 (**choice A**) is the signal transduction complex of T cells. It is found on all T cells in association with the T-cell antigen receptor.

CD4 (**choice B**) is found on all helper T lymphocytes.

CD8 (**choice C**) is found on all cytotoxic T lymphocytes.

CD56 (**choice E**) is a marker for human natural killer cells.
- The correct answer is C.** This patient has a T-cell lymphoblastic lymphoma. In his case, the malignant cell is “double-positive”: it possesses both CD4 and CD8. In a normal individual, these would only be found as an early developmental stage in the cortex of the thymus. Once cells have rearranged their receptor genes and been subjected to positive and negative selection, the cells leaving the thymus will express one coreceptor or the other but never both.

Bone marrow (**choice A**) would contain T lymphocyte precursors that are double negative: They will lack both CD4 and CD8.

Peripheral blood (**choice B**) would have mature T cells that have differentiated into either helper (CD4+) or cytotoxic (CD8+) cells. There should be no double-positive T cells in the peripheral blood.

Thymic medulla (**choice D**) is the location of maturing T cells ready to circulate into the bloodstream and peripheral lymphoid organs. It would have only single-positive cells.

Splenic periarteriolar lymphoid sheaths (**choice E**) are the T-cell-dependent areas of the spleen. They would have fully committed helper (CD4+) or cytotoxic (CD8+) cells.
- The correct answer is A.** T-lymphocyte precursors that leave the bone marrow and move to the thymus have neither CD4 nor CD8 coreceptors, and they have not rearranged the DNA of the variable domains of their antigen receptor, the TCR.

CD4⁻, CD8⁻, and TCR⁺ (**choice B**) is not a possible T-cell phenotype. Once the TCR gene segments are rearranged and the TCR is expressed, the cells will bear both CD4 and CD8 coreceptors.

CD4⁻, CD8⁺, and TCR⁺ (**choice C**) is the phenotype of cytotoxic T cells that would be in the circulation, not in the thymus, unless it were immediately prior to their release into the circulation following the thymic selection process.

CD4⁺, CD8⁻, and TCR⁺ (**choice D**) is the phenotype of helper T cells that would be in the circulation, not in the thymus, unless it were immediately prior to their release into the circulation following thymic selection processes.

CD4⁺, CD8⁺, and TCR⁺ (**choice E**) is the phenotype of cells in the thymic cortex. These are the cells that have rearranged their receptor genes and bear both CD4 and CD8 coreceptors. As the specificity of their TCR is tested, they will be directed to express either CD4 (and become a helper T cell) or CD8 (and become a cytotoxic T cell).

5. **The correct answer is C.** The interaction between the TCR and MHC class I/peptide conjugates is stabilized by the CD8 coreceptor. By downregulating the expression of MHC class I antigens on the surface of infected cells, the virus protects the infected host cell from killing by cytotoxic T lymphocytes.

CD2 (**choice A**), also known as LFA-2, is a adhesion molecule within the immunoglobulin superfamily of genes. Its ligand is the integrin LFA-3. It is found on T cells and mediates attachment to other lymphocytes and antigen-presenting cells. It does not have a coreceptor role that would impact MHC class I-restricted killing.

CD4 (**choice B**) is the coreceptor that stabilizes the interaction between MHC class II antigens and the TCR. It is thus important for helper T cells, not cytotoxic T cells.

CD16 (**choice D**) is the Fc receptor involved in binding to immune complexes and promoting antibody-dependent cell-mediated cytotoxicity. It is not involved in the MHC class I-restricted killing by cytotoxic T cells.

CD56 (**choice E**) is a cell surface marker found on NK cells. Its function is unknown. However, since NK activity is enhanced in the absence of MHC class I antigen expression, the downregulation of these molecules by herpes simplex I and II actually makes infected cells more susceptible to the NK cell form of lysis.

Lymphocyte Recirculation and Homing

4

What The USMLE Requires You To Know

- The structure and function of the secondary lymphoid organs
- The areas in which B and T lymphocytes localize in the peripheral lymphoid organs
- The role of chemokines and adhesion molecules in lymphocyte trafficking

Lymphocytes of the B- and T-cell lineages that have completed their selection in the bone marrow and thymus respectively are now **mature, naive lymphocytes** ready to begin their role in the surveillance of the body against invaders. These mature, naive lymphocytes will begin the process of **recirculation** through the body, which is essential for ensuring that the limited number of cells with receptors for a specific antigen is enabled to search for that antigen throughout the body. Naive cells preferentially recirculate through the **peripheral (secondary) lymphoid organs**, the **lymph nodes**, **spleen**, and **mucosal-associated lymphoid tissue (MALT)** to maximize the chances of encounter with foreign antigen and thereby initiate specific immune responses.

Lymph nodes are the small nodular aggregates of secondary lymphoid tissue found along the lymphatic channels of the body and are designed to initiate immune responses to **tissue-borne antigens**. Each lymph node is surrounded by a fibrous capsule that is punctured by **afferent lymphatics**, which bring lymph into the **subcapsular sinus**. The fluid percolates through an outer **cortex** area that contains aggregates of cells called **follicles**. The lymph then passes into the inner **medulla** and the **medullary sinus** before leaving the node through the **hilum** in an efferent lymphatic vessel. Ultimately, lymph from throughout the body is collected into the **thoracic duct**, which empties into the vena cava and returns it to the blood.

In A Nutshell

Peripheral (Secondary)
Lymphoid organs

- Lymph nodes, spleen, and MALT
- Sites of foreign antigen exposure

In A Nutshell

- Lymph nodes filter tissue fluids.
- Outer cortex contains follicles (B-cell areas).
- Paracortex is a T-cell area.
- Inner medulla contains macrophages.

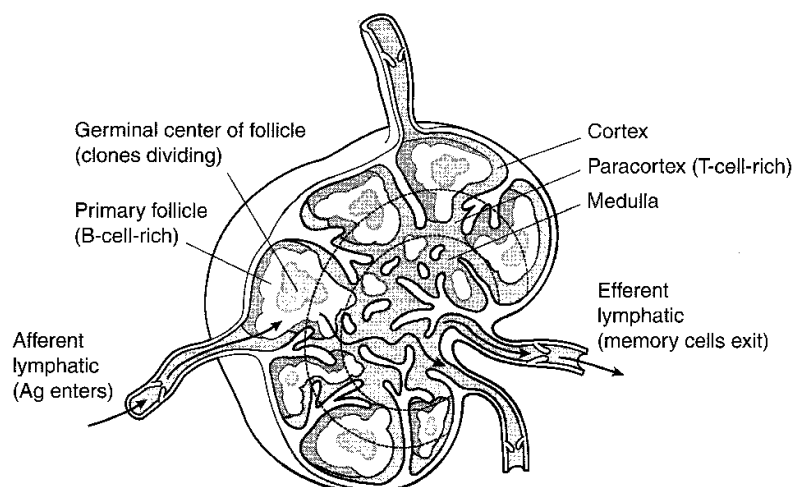


Figure II-4-1. Compartmentalization of a Lymph Node

In A Nutshell

- The spleen filters blood.
- PALS are T-cell areas.
- Follicles are B-cell areas.

The **spleen** is the secondary lymphoid organ designed to initiate immune responses to **blood-borne antigens**. A single splenic artery enters the capsule at the hilum and branches into arterioles, which become surrounded by cuffs of lymphocytes, the **periarteriolar lymphoid sheaths (PALS)**. **Lymphoid follicles** surrounded by a rim of lymphocytes and macrophages are attached nearby. This constitutes the **white pulp**. The arterioles ultimately end in vascular **sinusoids**, which make up the **red pulp**. From here, venules collect blood into the splenic vein, which empties into the portal circulation.

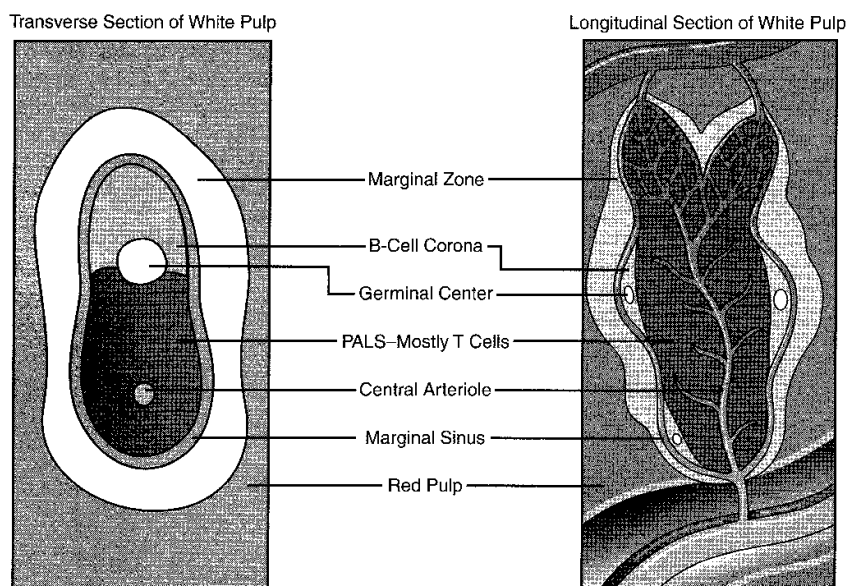


Figure II-4-2. The Structure of the Spleen

Naive lymphocytes enter the lymph nodes from the blood through **high endothelial venules** (HEVs). This migration involves a multistep sequence of **interactions between the lymphocyte and adhesion molecules on the endothelium**. The initial low-affinity **interaction is mediated by homing receptors, called L-selectins, present on the lymphocytes, which bind to addressins present on the HEVs**.

Naive T cells migrate into the **paracortical areas** under the influence of a **chemokine gradient**. Naive B cells are guided along a chemokine gradient into the **lymphoid follicles**.

Less is known about lymphocyte recirculation through the spleen. **Naive T cells home to the periarteriolar lymphoid sheaths (PALS)**, whereas naive B cells **home to the lymphoid follicles**.

The rate of recirculation of naive lymphocytes through the body **by this system is quite high**. It is estimated that each lymphocyte passes through each lymph node **in the body once a day** and through the spleen once every two days on average. This high rate of **trafficking maximizes** the possibility that the very small number of cells having a specific **antigen receptor have the maximum possibility of encountering that antigen when it is present**.

In A Nutshell

- Naive lymphocytes leave the blood through HEVs.
- L-selectins on lymphocytes bind to addressins on HEVs.
- Naive T cells home to the lymph node paracortex or the splenic PALS.
- Naive B cells home to lymphoid follicles.

Chapter Summary

- Mature naive lymphocytes recirculate through the peripheral lymphoid organs (lymph nodes and spleen) to search for antigen.
- Lymph nodes are designed to filter antigens from the tissue fluids.
- Lymph enters through afferent lymphatics and percolates through an outer cortex and inner medulla before leaving through the efferent lymphatic in the hilus.
- The spleen is designed to filter antigens from blood. Blood enters through a single splenic artery, which branches into arterioles that become surrounded by cuffs of lymphocytes (periarteriolar lymphoid sheaths) with follicles and macrophages nearby.
- Lymphocytes leave the bloodstream to enter lymph nodes through high endothelial venules.
- L-selectins on lymphocytes bind to addressins on the high endothelial venules, and chemokine receptors mediate the homing of specific cells to specific areas.
- T lymphocytes migrate into the lymph node paracortical areas.
- B lymphocytes migrate into the lymph node follicles.
- In the spleen, T lymphocytes are attracted to the periarteriolar lymphoid sheaths.
- B cells entering the spleen are attracted to the lymphoid follicles.
- Every lymphocyte passes through every lymph node once per day and through the spleen once every two days on average.

Review Questions

1. A lymph node biopsy of a 6-year-old boy shows markedly decreased numbers of lymphocytes in the paracortical areas. Analysis of his peripheral blood leukocytes is likely to show normal to elevated numbers of cells expressing surface
 - (A) CD2
 - (B) CD3
 - (C) CD4
 - (D) CD8
 - (E) CD19

2. A 65-year-old woman was involved in an automobile accident that necessitated the removal of her spleen. To which of the following pathogens would she have the most increased susceptibility?
 - (A) *Babesia microti*
 - (B) *Bordetella pertussis*
 - (C) *Corynebacterium diphtheriae*
 - (D) Enteroaggregative *Escherichia coli*
 - (E) Human papilloma virus

3. A 4-year-old boy is referred to a specialist for the diagnosis of a possible immunologic problem. The child has extremely elevated white blood cell counts, with a profound lymphocytosis. A biopsy performed on a cervical lymph node reveals extreme hypocellularity in both cortical and paracortical areas. Absence of which of the following leukocyte surface molecules could result in this clinical picture?
 - (A) Addressins
 - (B) Chemokines
 - (C) Immunoglobulin family cell adhesion molecules
 - (D) Integrins
 - (E) L-selectins

4. A 6-year-old child is taken to his pediatrician because the parents are alarmed about an indurated fluctuant mass on the posterior aspect of his neck. The mass is nontender and shows no signs of inflammation. The child is examined carefully, and no other masses are found. The pediatrician decides to submit a biopsy of this area to a pathologist. The pathologist reports back that the mass is a lymph node with markedly increased numbers of cells in the cortical area. Fluorescent antisera to which of the cell surface markers is most likely to bind to cells in this area?
- (A) CD2
 - (B) CD3
 - (C) CD4
 - (D) CD16
 - (E) CD19
5. A radioactive tracer dye is injected subcutaneously into the forearm of an experimental subject. What is the first area of the first draining lymph node that would develop significant radioactivity?
- (A) Cortex
 - (B) Medulla
 - (C) Paracortex
 - (D) Primary follicle
 - (E) Subcapsular sinus

Answers and Explanations

1. **The correct answer is E.** The paracortex of a lymph node is a T-cell–dependent area. If this area is lacking cellularity, then the patient has a deficiency of T lymphocytes. B-lymphocyte numbers could be normal or even elevated. The only B-cell marker on this list is CD19, the marker which is used clinically to enumerate B cells in the body.

CD2 (**choice A**), also known as LFA-2, is an adhesion molecule found on T cells, thymocytes, and NK cells. In a person with a T-cell deficiency, there would be decreased numbers of cells bearing this marker.

CD3 (**choice B**) is found on all T cells. It is also called the “pan-T” cell marker. In a person with a T-cell deficiency, there would be decreased numbers of cells bearing this marker.

CD4 (**choice C**) is found on all helper T lymphocytes. In a person with a T-cell deficiency, there would be decreased numbers of cells bearing this marker.

CD8 (**choice D**) is found on all cytotoxic T lymphocytes. In a person with a T-cell deficiency, there would be decreased numbers of cells bearing this marker.

2. **The correct answer is A.** The spleen is the secondary lymphoid organ that is responsible for primary surveillance against blood-borne antigens. *Babesia microti* is an intraerythrocytic parasite of humans, transmitted by the same vector tick as Lyme disease. Red blood cells (and their parasites) are filtered by the spleen, so splenectomy is a predisposing factor in development of serious disease with this parasite.

Bordetella pertussis (**choice B**) is a mucosal surface pathogen that attaches to the upper airways. Although its toxin becomes blood-borne, the organism itself is confined to the respiratory tree.

Corynebacterium diphtheriae (**choice C**) is a mucosal surface pathogen that attaches to the upper airways. Although its toxin becomes blood-borne, the organism itself is confined to the respiratory tree.

Enteroaggregative *Escherichia coli* (**choice D**) is an organism that causes diarrhea by producing a biofilm-like aggregation of organisms on the surface of the colonic mucosa, which impedes absorption. It is not likely to be a blood-borne pathogen.

Human papilloma virus (**choice E**) produces localized infections in epithelial cells where it is transferred by human-to-human or human-to-fomite contact. It is not likely to be a blood-borne pathogen.

3. **The correct answer is E.** L-selectins are the molecules found on the surfaces of lymphocytes that mediate their binding to the high endothelial venules of lymph nodes. This is the means by which lymphocytes enter lymph nodes; without these molecules, they would be unable to leave the blood. Thus, they would rise in the blood to extreme levels and be absent from their appropriate areas in the secondary lymphoid organs.

Addressins (**choice A**) are the molecules complementary to L-selectins, which are found on endothelial cells. Although their absence could also cause similar signs, the question asks about a leukocyte surface molecule, and addressins are not found on leukocytes.

Chemokines (**choice B**) are not cell surface molecules, but do play a role in the homing of lymphocytes to specific regions of the secondary lymphoid organs. An absence of chemokine receptors on leukocytes could have caused similar signs, but chemokines themselves are soluble substances, not surface molecules.

Immunoglobulin family cell adhesion molecules (**choice C**), or IgCAMs, are adhesion molecules found on various cells of the immune system. They mediate cell-to-cell interactions, as well as binding to the extracellular matrix. Their ligands are the integrins. If these molecules were absent, cell-cell interactions would be diminished, but movement of lymphocytes out of the circulation and into the secondary lymphoid organs would not be affected.

Integrins (**choice D**) are the ligands for the IgCAMs. They are involved in cell-to-cell interactions, as well as binding to the extracellular matrix. If these molecules were absent, cell-cell interactions would be diminished, but movement of lymphocytes out of the circulation and into the secondary lymphoid organs would not be affected.

4. **The correct answer is E.** The cortex of lymph nodes is a B-lymphocyte area. Thus, cells in this area would stain with fluorescent antibodies against CD19, the molecule that serves as a portion of the B-cell signal transduction complex. This molecule would be found on all B cells, but would be absent from T cells, macrophages, and NK cells.

CD2 (**choice A**) is a T-cell marker. T cells will be found in the paracortical areas of lymph nodes.

CD3 (**choice B**) is a T-cell marker. It is the signal transduction complex of the T cell and will be found on all T cells. T cells will be found in the paracortical areas of lymph nodes.

CD4 (**choice C**) is a marker for helper T cells. These cells would be found in the paracortical areas of lymph nodes.

CD16 (**choice D**) is the Fc receptor for IgG antibodies. It would be found on natural killer cells, which would not be numerous in the cortex of the lymph nodes.

5. **The correct answer is E.** Lymph nodes are designed to filter tissue fluids. Fluids entering the lymph nodes do so through the afferent lymphatics and are released into the subcapsular sinus. From there, fluids percolate through the cortex, into the medulla, through the medullary cords, and finally exit through the efferent lymphatics in the hilum.

The cortex (**choice A**) of the lymph node is directly beneath the subcapsular sinus. It would be the second region of the lymph node to be exposed to the radioactive tracer. The cortex is a B-lymphocyte-rich area.

The medulla (**choice B**) of the lymph node is rich in macrophages. It would not receive the radioactive fluid until it had passed through the cortex and paracortex.

The paracortex (**choice C**) of the lymph node is a T-cell area. It lies between the cortex and the medulla and thus would receive the radioactive fluid after the cortical areas.

Primary follicles (**choice D**) are found in the cortex of the lymph node. These are areas of active B-lymphocyte proliferation and cloning. They would receive the radioactivity after it left the subcapsular sinus.

